

Sharp tests of instrument validity for general outcome distributions

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Abstract

I combine novel identification and inference results to obtain the first test of instrument validity that can accommodate outcomes with unrestricted, potentially infinite, and multi-dimensional support, along with a large class of shape restrictions. The main result shows that, even if the outcome is continuous, the sharp testable implications of the exclusion restriction can be reduced to a finite set of linear inequality constraints on the model for selection into treatment. Hence, if the instrument and treatment are discrete, an infinite system of equations is reduced to an equivalent finite one. The resulting system involves non-differentiable functions of the data, for which existing inference procedures are not applicable. I therefore develop a procedure for joint inference on these parameters, which delivers a formal test of instrument validity. I apply the proposed test to the Oregon Health Experiment, where I reject the hypothesis that the intervention affected insurance coverage only on the extensive margin. Instead, I find evidence that individuals were incentivized to retain insurance for at least six months. Leveraging a shape constraint that models this empirical finding, I obtain estimates of the effect of insurance on health and health care utilization that are consistent with previous studies.

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1 Introduction

Instrumental variables (IV) is one of the most used methods in empirical economics. Following seminal work by [Imbens and Angrist \(1994\)](#), a common approach to IV is to leverage non-parametric shape restrictions (such as no-defiers or monotonicity) to identify causal parameters without restricting treatment effect heterogeneity. Furthermore, the assumptions that underpin IV models for heterogeneous treatment effects are equivalent to the hypothesis that a set of variables can fully account for the effect of an exogenous treatment in the context of mediation analysis. Existing tests for the validity of IV models for heterogeneous treatment effects either apply to and rely on specific shape restrictions, or require that the outcome be discrete or discretized. When the outcome is continuous, tests based on fixed discretizations neither exhaust the model’s testable implications nor use the entire empirical content of the data.

In this paper, I combine novel identification and inference results to propose the first test of instrument validity that can accommodate general outcome distributions and a large class of non-parametric shape restrictions. The test is sharp: it employs a set of conditions that is logically equivalent to the IV assumptions. The observable implications involve subdensity overlaps, a class of parameters for which no valid joint inference methods exist. Thus, I develop a novel method for inference on multiple density-overlaps based on non-parametric classification. Density and subdensity overlaps arise in econometric characterizations beyond IV validity, as well as in other disciplines such as medicine, political science, and biology. The proposed inferential method may therefore be of independent interest.

Finally, I propose a variant of the main test based on a novel micro-foundation for IV models with binary instruments and shape restrictions that prevent switches between specific pairs of treatments. This variant allows for simpler inference, and identifies the specific behaviors that exist in the population but contradict the model’s assumptions. Additionally, in contexts other than IV, it can test nonparametric assumptions about how an exogenous treatment affects the preferences and choices of a population of decision makers.

Identification: I consider a potential outcomes model with a discrete instrument Z , a discrete treatment X , and an outcome Y with unrestricted, and potentially multi-dimensional, support. The model is defined by potential treatments $X = X(Z)$ and potential outcomes $Y = Y(X, Z)$.

The key assumptions are: *i*) Z is independent from potential outcomes as if it was randomly assigned, *ii*) Z affects X without violating some pre-specified restrictions (e.g. no-defiers), and *iii*) an exclusion restriction on Z , which reduces potential outcomes to

$Y = Y(X)$. The goal is to obtain a sharp set observable implications for which a fully non-parametric test is feasible.

The model is not falsified if potential treatment and outcome functions exist that rationalize the observed conditional treatment probabilities:

$$\mathbb{P}[X = x|Z = z] = \mathbb{P}[X(z) = x|Z = z], \quad (1)$$

and conditional outcome probabilities:

$$\mathbb{P}[Y \in B_Y, X = x|Z = z] = \mathbb{P}[Y(x) \in B_Y, X(z) = x|Z = z] \quad (2)$$

for every x and z in the support of X and Z , and every measurable subset B_Y of the support of Y .

X and Z have finite support, thus, selection into treatment (equation 1) is described by a finite set of equations. But, if Y has infinite support, there are infinitely many measurable subsets B_Y and thus, infinitely many instances of equation 2. The insight that enables the test is that the sharp observable implications of the exclusion restriction combined with conditional outcome probabilities are equivalent to a finite set of linear restrictions on the selection part of the model. Thus, I reduce an infinite set of equations to a finite one.

The set of linear inequalities that expresses the sharp observable implications of the exclusion restriction involves a finite collection of subdensity overlaps. The overlap of n functions is the area that lies strictly below all of them, or, in other words, the integral of their point-wise minimum. In this paper, subdensities are the functions obtained by restricting the joint densities of the outcome and treatment conditional on the instrument to a unique treatment value. Thus, the target parameters are integrals of the point-wise minimum of the functions $f_{Y,X|Z=z}(y, x)$ across all possible values of z and some fixed x .

Inference: Estimation and inference on sub-density overlaps is a non-standard problem for which no previous, valid, fully non-parametric method exists. The non-standard nature of the problem arises from the region of the support of the outcome in which two or more subdensities are equal. This region is referred to as the contact set. The main inferential challenge follows from the fact that the minimizer is not unique in this region.

Ruling the contact set out is problematic. First, it would implicitly eliminate the possibility that the instrument has no effect on treatment adoption (i.e. instrument irrelevance), which, in the context of testing, might not be an *ex-ante* plausible assumption. Second, it also eliminates the possibility that the instrument affects just certain subsets of the population (e.g. the less healthy members of a population). Finally, the assumption that

the contact set has zero measure is fundamentally untestable. Thus, a test robust to this possibility is more credible.

Furthermore, the contact set is the reason why discretization based tests are not sharp. Discretization is equivalent to integration with a coarse grid, and it is precisely around the points where subdensities cross that coarse grids overestimate the integral of the minimum. Hence, discretization is only sharp if the contact set is known or estimated with high precision. An alternative solution is to use tests based on discretizations that become finer as the sample grows. Such tests rely on estimators that converge at a rate strictly slower than $n^{1/2}$, which reduces their power relative to tests based on root- n -consistent estimators. Moreover, in finite samples, these tests still use a set of non-sharp observable implications. Thus, to obtain a sharp, efficient test, the problems associated with the contact set cannot be avoided.

The contact set is problematic because non-differentiability of the minimum impedes using the delta method, while the non-parametric nature of the problem implies there are no guarantees that extensions of the delta method for directionally differentiable functions such as [Fang and Santos \(2019\)](#) work. Furthermore, if the minimum of two densities is not unique in some non-negligible subset of the support of the outcome, the asymptotic behavior of plug-in estimators is non standard. Any finite-sample sub-density estimator will have a statistical noise component that shifts the estimator away from the true value of the target function. If two sub-densities are estimated from samples that are approximately similar in size, the magnitude of the noise component of both estimators will be of the same order, and its ratio will remain constant even as the sample size grows. Hence, for any given sample size, the identity of in-sample minimizers inside the contact set is determined by statistical noise. As a consequence, the limit of the estimated minimizer, and thus of a plug-in overlap estimator, might be a random variable rather than constant.

To circumvent these problems, I note that inference on sub-density overlaps is equivalent to inference on the loss of an asymmetric classifier: the goal is not to learn the identity of the maximum/minimum across the support of Y , but the aggregate frequency with which the observed values of X and Y come from the sub-density associated with the least likely instrument value. The main advantage of this approach is that, precisely in the set in which sub-densities are equal, any classification decision attains the exact same loss. In other words, the estimated classifier might be random (even in the limit), but its loss, which corresponds to the target parameter, is deterministic.

I propose a consistent, asymptotically normal, and asymptotically unbiased estimator for the overlap of two or more sub-densities. To derive its asymptotic properties, I combine the classification framework with uniform rates of convergence for kernel density estimators

(Giné and Guillou (2002)) and a margin condition that controls the magnitude of classification errors in the proximity of the contact set. The estimator uses sample splitting. One fraction of the data is used to train a non-parametric classifier based on kernel density estimators, and the rest to evaluate its loss. Sample splitting ensures that the empirical loss of the trained classifier is unbiased for its theoretical loss.

In Monte-Carlo simulations, I show that the overlap estimators have negligible bias, and that my proposed confidence intervals achieve coverage close to the nominal level in large samples. Given this valid method for inference on sub-density overlaps, testing instrument validity reduces to verifying that a linear system of equations admits at least one solution. This is a well studied problem for which inference procedures exist. Using the methodology from Cox and Shi (2023) I formally show that the resulting test has asymptotically correct size. A second set of Monte-Carlo simulations supports the theoretical properties of the test.

The main overlap estimator I propose has two caveats: First, it uses sample splitting which reduces the effective sample size to $n/2$ as opposed to n . Second, it has no regular limit and asymptotic normality is only achieved after pre-multiplying by the right normalization sequence. This last caveat implies that the estimator is not compatible with some inference methods for linear systems that rely on the bootstrap such as Fang et al. (2023) or Goff and Mbakop (2025).

To address these caveats, I propose an alternative estimator based on a classifier that follows a pre-specified deterministic rule that only changes if the true minimizer is learned from the data with high confidence. This estimator has a fixed, non-random limit and thus the bootstrap is valid. Moreover, it enables K-fold cross fitting which restores n as the effective sample size. This estimator, however, requires one additional tuning parameter, and has potentially larger bias as a result of using a sub-optimal classifier. In simulations, I find no substantive difference in terms of bias between the two estimators, but confidence interval coverage is anti-conservative for the cross-fitted estimator in small and medium sized samples.

Generalized no-defiers and empirical application: One of the most used approaches to identification in IV models with heterogeneous treatment effects is to assume that some treatment switches cannot occur as the instrument changes (e.g. the no defiers assumption). I call restrictions in this class *generalized no-defiers assumptions*. For models with binary instruments that impose *generalized no-defiers assumptions*, I propose a specialized, simpler test based on a microeconomic interpretation of such restrictions.

Generalized no-defiers restrictions specify that, as the instrument increases, some treat-

ments are weakly promoted (at least in relative terms) with respect to others. Thus, if the assumption holds, adoption of the promoted treatments should weakly increase. I formalize the notion of *promoted treatments* in terms of restrictions on how the instrument affects the preferences of units in the population, and characterize how adoption of the promoted treatments *weakly increases* using a generalization of first order stochastic dominance (GFOSD) to arbitrary binary relations.

Additionally, the sharp testable implications of any *generalized no-defiers restriction* combined with the exclusion restriction are still a set of inequalities with a GFOSD interpretation. The difference is that the exclusion restriction imposes an upper bound on the probabilities associated with some of the promoted treatments, which results in a *tightening* of the GFOSD inequalities relative to the inequalities associated with the *generalized no-defiers restriction* alone.

These results provide a microfoundation for *generalized no-defiers assumptions*, an interpretable set of sharp testable implications, and a framework to formulate plausible restrictions that are consistent with the observed data.

I illustrate the proposed framework in an empirical application to the Oregon Health Experiment, a state-level lottery that randomized eligibility to the Medicaid program among low income households (Finkelstein et al. (2012)). Specifically, I test and reject the assumption that the intervention only affected insurance on the extensive margin proposed by Rose and Shem-Tov (2024). Instead, I find that the treatment also affected the duration of insurance for households that would otherwise have acquired insurance for less than six months. That is, the intervention also had an effect on the intensive margin. Leveraging a shape restriction that models my empirical findings, I estimate the effect of increased insurance periods on hospital utilization and health outcomes. My empirical results are consistent with previous results by Finkelstein et al. (2012) and Rose and Shem-Tov (2024).

Related literature: This paper contributes to the literature on testing the validity of instrumental variables models for heterogeneous treatment effects. The first to consider instrument validity for heterogeneous treatment effects where Balke and Pearl (1997), but without a formal inference procedure and only for binary outcomes. Subsequently, Kitagawa (2015), Huber and Mellace (2015), and Mourifié and Wan (2017) developed tests for the binary instrument and binary treatment case under a monotonicity assumption. Kédagni and Mourifié (2020) proposed a joint test for the exclusion restriction and random assignment but their results are only sharp for a binary outcome and do not allow for shape restrictions. Subsequent work has derived sharp testable implications for IV designs on a case-by-case basis (Bai et al., 2024c), or proposed general tests that are not sharp (Sun, 2023). More

recently, [Kwon and Roth \(2024\)](#) and [Kaido and Ponomarev \(2025\)](#) provided the first sharp characterizations for general IV models, but their proposed inference procedures require that the outcome be either discrete, or discretized at the cost of losing sharpness.

The observable implications I propose apply to a range of IV methods for heterogeneous treatment effects. [Imbens and Angrist \(1994\)](#) showed how to use non-parametric restrictions on instrument-response heterogeneity to identify the LATE parameter. Subsequent work has extended their approach to a wide range of settings and other parameters of interest, such as: re-interpretations of monotonicity ([Heckman and Pinto \(2018\)](#), [Navjeevan and Pinto \(2022\)](#)); alternative restrictions on how the instrument affects treatment take-up ([Richardson and Robins \(2010\)](#), [De Chaisemartin \(2017\)](#), [Dahl et al. \(2023\)](#), [Goff \(2024\)](#)); and generalizations to settings with multiple instruments and treatments ([Lee and Salanié \(2020\)](#), [Goff \(2020\)](#), [Mogstad et al. \(2021\)](#), [van't Hoff et al. \(2023\)](#), [Kazemi and Olsen \(2024\)](#), [Fusejima \(2024\)](#), [Bai et al. \(2024a\)](#)). These methodological advances have been applied in diverse empirical contexts, including education and labor market studies ([Kline and Walters \(2016\)](#), [Kirkeboen et al. \(2016\)](#), [Pinto \(2021\)](#), [Mountjoy \(2022\)](#)). The testing procedures I propose apply to all the aforementioned papers.

The GFOSD characterization and its relation to non-parametric random utility models contributes to the literature that investigates the empirical content of response type restrictions ([Goff \(2024\)](#) and [Bai et al. \(2024b\)](#)) and their microeconomic foundations ([Vytlacil \(2002\)](#), [Heckman and Pinto \(2018\)](#), [Navjeevan and Pinto \(2022\)](#)).

To obtain the GFOSD inequalities I show that finding a distribution over potential treatments and potential outcomes that satisfies the sharp instrument validity conditions amounts to solving a well known transportation problem over a directed graph. This technique, and its equivalence to the maximum flow/minimum cut problems, was first used to characterize incomplete econometric models by [Ekeland et al. \(2010\)](#) and [Galichon and Henry \(2011\)](#). My approach differs in two ways: first, instead of considering transportation problems from the space of observed variables to the domain of latent unobservables, my transportation problem is defined between observed treatment probabilities conditional on different instrument values. Second, to incorporate the exclusion restriction, I extend their minimum-cut characterization method to a broader class of capacity constraints.

My optimal transport approach is related to contemporaneous and independent work by [Kaido and Ponomarev \(2025\)](#) who study the falsification of shape constraints and exclusion restrictions using graph theory. The approaches are not nested. The main advantage of the approach from [Kaido and Ponomarev \(2025\)](#) is its generality since it is sharp for the multiple instruments case and applies to other econometric models beyond IV. The main advantage of my approach is that it does not rely on partitioning the support of

the outcome (discretization), obtaining a finite characterization even when the outcome has infinite support. A second advantage is that my approach can accommodate a wider class of restrictions on treatment adoption, for example, the *more compliers that defiers* assumption from [De Chaisemartin \(2017\)](#).

All the testable implications I provide are equivalent to the requirement that a particular linear system of inequalities admits at least one solution. Inference on the feasibility of linear systems and linear programs is an active area of research ([Bai et al. \(2022\)](#), [Fang et al. \(2023\)](#), [Cho and Russell \(2024\)](#), [Goff and Mbakop \(2025\)](#)) that is tightly related to inference methods for models characterized by moment inequalities such as [Cox and Shi \(2023\)](#) or [Andrews et al. \(2023\)](#).

To test IV validity using inferential methods for linear systems I first require a method for non-parametric joint inference on sub-density overlaps. This is a particular instance of the more general problem of estimating the integral of non-differentiable transformations of a collection of functions or densities. Two closely related papers are [Semenova \(2025\)](#) and [Levis et al. \(2025\)](#) who consider inference on the integral of the point-wise minimum/maximum of smooth functions, and its applications to IV models. Neither of these papers studies testing instrument validity. For inference, [Semenova \(2025\)](#) assumes that the minimum is unique almost-everywhere, which eliminates the non-regularity issue. [Levis et al. \(2025\)](#) propose a similar solution, and also consider the non-negligible contact set case. Their solution is to restore regularity using a continuous approximation to the minimum function. While the soft-min approach would be valid in the context of this paper, the continuous approximation to the minimum shifts the target parameter. Consequently, the resulting inequalities would be a set of necessary conditions for IV validity, but not the sharp testable implications

To the best of my knowledge, only two papers have considered exact inference on parameters defined by the integral of the minimum/maximum of a collection of functions that is robust to non-negligible contact sets. [Anderson et al. \(2012\)](#) consider a plug-in approach for inference on a single overlap parameter. Unfortunately, their variance estimator is not compatible with existing inference methods for linear systems. [Luedtke and Van Der Laan \(2016\)](#) propose using online estimation to conduct inference on the mean outcome under a non-unique optimal policy. While similar non-regularity issues arise in both problems, in subdensity overlap estimation, realized and potential outcomes enter the target parameter through their distribution rather than directly. Thus, the online estimation solution is not applicable without substantial modifications.

The key argument sustaining my inference procedure is that, in the contact set, the loss of a non-parametric classifier is invariant, and hence, an estimator based on the empirical

loss of a classifier is robust to the erratic behavior of the minimum inside the contact set. This insight reduces the inference problem to controlling the behavior of the classifier out of the contact set. To do so, I combine sample splitting, a margin assumption that controls the misclassification rate near the boundary of the contact set (Tsybakov (2004), Audibert and Tsybakov (2007)), and classical results on the rates of uniform convergence for kernel-density estimators by Giné and Guillou (2002).

The alternative estimator I propose to enable K-fold cross fitting relies on estimating a set that is known to contain the contact set with high probability. This idea was first proposed in the context of a test for first order stochastic dominance by Linton et al. (2010). In the context of inference on overlaps, it was also used by Anderson et al. (2012).

My method for inference on density and sub-density overlaps might be of independent interest. First, other IV identification methods such as Kitagawa (2010) or Kédagni et al. (2024) obtain identification results that involve or are related to the vector of sub-density overlaps from this paper, but do not have inference methods that allow for continuous outcomes. Second, similar parameters appear in other econometric settings such as Borusyak (2015) or Meango et al. (2025). Finally, density overlaps are a relevant non-parametric measure of similarity between populations in other disciplines such as political science, ecology, and medicine (see Dey et al. (2025) for a recent review).

The remainder of this paper is organized as follows: In section 2 I introduce the class of models and restrictions considered. In section 3 I derive the sharp testable implications of the exclusion restriction and provide the finite sharp characterization for IV models. In section 4 I specialize the characterization from section 2 to instrument-response-type restrictions and link them to FOSD and random utility models using optimal transport. In section 5 I describe my method for joint estimation and inference on subdensity overlaps and prove its asymptotic properties. Then, I discuss how the estimators can be used to construct a sharp test for instrument validity. In section 7 I illustrate the performance of both the estimators and the test. In section 8, I use my inference method and the GFOSD characterization to study the effect of the Oregon health experiment on the duration of insurance and its effect on health and hospital utilization. In section 9 I conclude.

2 IV models for heterogeneous treatment effects

The model has three observable random variables: an outcome Y , a treatment X and an instrument Z with joint distribution $F_{Y,X,Z}$. Both X and Z have discrete finite supports denoted $\mathcal{X} = \{x_0, x_1, \dots, x_{L-1}\}$ and $\mathcal{Z} = \{z_0, z_1, \dots, z_{L-1}\}$ respectively. The support of Y

which I denote as $\mathcal{Y}$ has unrestricted cardinality and dimension. The only requirement is that a measure μ_Y over $\mathcal{Y}$ exists.

I adopt a potential outcomes framework. The joint distribution of (Y, X, Z) is determined by the potential treatment and potential outcomes functions $X(z) : \mathcal{Z} \mapsto \mathcal{X}$ and $Y(x, z) : \mathcal{X} \times \mathcal{Z} \mapsto \mathcal{Y}$. These functions are random variables and their joint distribution fully determines the model.

Formally, for any measurable subset of the support of the outcome $B_Y \subset \mathcal{Y}$, and any fixed treatment and instrument values $x_l \in \mathcal{X}$ and $z_k \in \mathcal{Z}$, the joint distribution of (Y, X, Z) satisfies:

$$\mathbb{P}[Y \in B_Y, X = x_l, Z = z_k] = \mathbb{P}[Y(x_l, z_k) \in B_Y, X(z_k) = x_l, Z = z_k] \quad (3)$$

I assume that all other relevant covariates are exogenous. Since they play no role in the characterizations I provide, I assume that all the identification arguments are conditional on such covariates and omit them from the formal proofs and results.

Unless otherwise noted, I will present and discuss all the results using the IV notation and terminology. All results would also hold for the sharp null of full mediation from [Kwon and Roth \(2024\)](#) after appropriately relabeling the values of X as mediators and the values of Z as treatments.

I denote the joint density of X and Y conditional on $Z = z_k$ induced by $F_{Y,X,Z}$ as $f_{Y,X|Z=z_k}(y, x)$.

2.1 Instrument validity and response types

For Z to be a valid instrument, two assumptions are typically invoked, instrument independence, and an exclusion restriction:

Assumption 2.1 (Instrument validity).

1. *Instrument independence:*

$$\mathbb{P}[Y(x_{l'}, z_{k'}) \in B_Y, X(z_{k''}) = x_l | Z = z_k] = \mathbb{P}[Y(x_{l'}, z_{k'}) \in B_Y, X(z_{k''}) = x_l]. \quad (4)$$

For every $x_l, x_{l'} \in \mathcal{X}$, $z_k, z_{k'}, z_{k''} \in \mathcal{Z}$ and any measurable $B_Y \subset \mathcal{Y}$.

2. *Exclusion restriction:*

$$Y(x_l, z_k) = Y(x_l) \quad (5)$$

For every $x_l \in \mathcal{X}$ and $z_k \in \mathcal{Z}$

Put in words, the instrument must be independent from potential outcomes and treatments as if it was randomly assigned, and it must have no effect on outcomes after accounting for treatment.

Assumption 2.1 implies that the observed joint distribution of (X, Y, Z) is fully determined by the observed distribution of Z , and the unobserved distribution of the potential outcomes and potential treatments functions. This motivates the following definitions:

Definition 2.1.

1. An instrument-response type $t^Z \in \mathcal{T}_Z$ is a function $t^Z : \mathcal{Z} \rightarrow \mathcal{X}$ that maps instrument values to treatment values.
2. A treatment-response type $t^X \in \mathcal{T}_X$ is a function $t^X : \mathcal{X} \rightarrow \mathcal{Y}$ that maps treatment values to outcome values.
3. A response type t is a pair $t = (t_z, t_x) \in \mathcal{T} = \mathcal{T}_Z \times \mathcal{T}_X$.

An important feature of response types is that by construction they satisfy the exclusion restriction. The reason is that for any $x_l \in \mathcal{X}$, a response type $t = (t^Z, t^X)$ assigns a unique outcome value $t^X(x_l)$ when $X = x_l$ regardless of the value that the instrument takes. Thus, under assumption 2.1, the model can be fully specified in terms of response types.

Since $\mathcal{Y}$ is μ_Y -measurable, and $\mathcal{X}$ is finite, the set $\mathcal{T}$ is measurable with respect to the product measure induced by μ_Y and the count measures on $\mathcal{X}$. I denote the resulting measure space as $(\mathcal{T}, \mathcal{B}_T)$.

Definition 2.2.

1. A distribution over response types is a probability measure $g : \mathcal{B}_T \rightarrow [0, 1]$ over the measure space $(\mathcal{T}, \mathcal{B}_T)$.
2. A distribution over instrument-response types is a function $p : \mathcal{T}_Z \rightarrow [0, 1]$ such that $p(t^Z) \geq 0$ for all $t^Z \in \mathcal{T}_Z$, and $\sum_{t^Z \in \mathcal{T}_Z} p(t^Z) = 1$.

Remark 2.1. Any distribution over response types g defines a unique distribution over instrument-response types f_g according to the following equation:

$$p_g(t'^Z) := g\left(\{t = (t^Z, t^X) \in \mathcal{T} : t^Z = t'^Z, t^X \in \mathcal{Y}^L\}\right), \quad \forall t'^Z \in \mathcal{T}_Z, \quad (6)$$

$$= g\left(\{t = (t^Z, t^X) \in \mathcal{T} : t^Z = t'^Z\}\right), \quad \forall t'^Z \in \mathcal{T}_Z. \quad (7)$$

If assumption 2.1 holds, then an unobserved true distribution over response types g_0 must exist that generates the observed distribution of (Y, X, Z) .

Definition 2.3 (Consistency). *A distribution over response types $g \in \Delta(\mathcal{T})$ is consistent with observed conditional outcome and treatment probabilities if:*

$$\mathbb{P}[Y \in B_Y, X = x_l | Z = z_k] = g_0\left(\{(t^Z, t^X) \in \mathcal{T} : (t^Z)_k = x_l, (t^X)_l \in B_Y\}\right), \quad (8)$$

$$\forall x_l \in \mathcal{X}, z_k \in \mathcal{Z}, \text{ and } B_Y \in \mathcal{B}_Y.$$

In the previous equation, the observed conditional probability on the left hand side must correspond to the mass that g_0 assigns to response types whose potential treatment is equal to x_l when the instrument takes the value z_k and realize outcomes in B_Y when treatment is equal to x_l .

Let $\Delta(\mathcal{T})$ be the set of all distributions over response types, assumption 2.1 is falsified if no $g \in \Delta(\mathcal{T})$ exists that satisfies equation 8.

2.2 Shape restrictions

IV models that allow for heterogeneous treatment effects, typically rely on additional shape restrictions that discipline how units might select into different treatments as the instrument changes. The most prominent example is monotonicity in the binary instrument and treatment case, which, as Imbens and Angrist (1994) showed, allows interpreting the OLS estimand as a local average treatment effect. Subsequent work has explored relaxations and extensions of this assumption to settings with multiple instruments and multiple, ordered and unordered, treatments (Heckman and Pinto (2018), Goff (2020), Mogstad et al. (2021), van't Hoff et al. (2023), Lee and Salanié (2020), Bai et al. (2024c)), these generalizations typically rely on ruling out particular instrument-response types.

Alternatively, researchers may impose inequality restrictions on the prevalence or relative frequency of certain instrument-response types—for example, requiring that the proportion of compliers exceeds that of defiers as De Chaisemartin (2017), or imposing upper bounds on the share of defiers for sensitivity analysis as Huber (2014), Noack (2021) or Yap (2025)).

The class of shape restrictions I consider covers all the aforementioned assumptions on their own or combined, as well as multiple potential extensions. More specifically, my characterization allows for any restriction than can be expressed as a set of linear constraints on the set of admissible distributions over instrument-response types.

To formalize the shape restrictions allowed, note that the total number of possible instrument-response types is $|\mathcal{T}_Z| := |\mathcal{X}|^{|\mathcal{Z}|} = (L)^K$. Therefore, a distribution p over $\mathcal{T}_Z$ can be represented as a vector $\mathbf{p}$ in $\mathbb{R}_+^{|\mathcal{T}_Z|}$, where each coordinate corresponds to the probability

that $p \in \Delta(\mathcal{T}_Z)$ assigns to a different $t \in \mathcal{T}_Z$.¹

I consider any shape restriction that takes the form $A_Z \mathbf{p} \leq \mathbf{r}$ where A_Z is an M by $|\mathcal{T}_Z|$ matrix with known coefficients for some fixed $M \geq 1$, and $\mathbf{r}$ is an M dimensional vector with entries that are either known or point identified from the data. I use the subscript Z to emphasize that the matrix A_Z encodes restrictions that pertain to treatment assignment as a function of the instrument.

Assumption 2.2. *Let $M \geq$ be a fixed integer, A_Z a known fixed matrix $A_Z \in \mathbb{R}^{M \times |\mathcal{T}_Z|}$, and $\mathbf{r}_Z \in \mathbb{R}^M$ a point identified vector that depends on the joint distribution of (Y, X, Z) .*

A distribution over response types $g_0 \in \Delta(\mathcal{T})$ exists such that if $\mathbf{p}_{g_0}$ denotes the vector representation of p_{g_0} , the following system of inequalities holds:

$$A_Z \mathbf{p}_{g_0} \leq \mathbf{r}_Z. \quad (9)$$

Note that strict equality restrictions can be incorporated by representing them as pairs of inequality constraints. In particular, if the k -th entry of $\mathbf{p}$ corresponds to the instrument-response type $t^Z \in \mathcal{T}_Z$, the restriction that t^Z occurs with zero probability can be imposed by including the canonical basis vectors e_k and $-e_k$ as rows of A_Z , and setting the corresponding entries of $\mathbf{r}_Z$ to zero.² Thus, assumption 2.2 can accommodate restrictions that rule-out specific response types. In appendix D I provide examples of models and restrictions encompassed by assumption 2.2.

To conclude the discussion of shape restrictions, note that if no distribution over instrument-response types $p \in \Delta(\mathcal{T}_Z)$ exists such that $\mathbf{p}$ satisfies equation 9 and is consistent with observed conditional treatment probabilities, then, the restrictions on instrument-response heterogeneity imposed by assumption 2.2 are falsified.

Proposition 2.1 (Sharp observable implications of assumptions 2.1 and 2.2). *A distribution over instrument-response types $p \in \Delta(\mathcal{T}_Z)$ is consistent with assumption 2.2 and observed conditional treatment probabilities, if and only if it satisfies the following system of linear equalities and inequalities:*

$$A_Z \mathbf{p} \leq \mathbf{r}_Z, \quad (10)$$

$$A_X \mathbf{p} = \mathbf{r}_X. \quad (11)$$

¹To formally define $\mathbf{p}$ it suffices to note that the set of response types is finite and thus can be well ordered. Then, the n -th coordinate of $\mathbf{p}$ represents the probability that the distribution $p \in \Delta(\mathcal{T}_Z)$ assigns to the n -th response type. The specific ordering of response types is not relevant as long as it is well defined.

² e_k denotes the vector that takes the value 0 in every entry except the k -th which is equal to 1.

Where A_Z and $\mathbf{r}_Z$ are defined as in assumption 2.2 and the system $A_X \mathbf{p} = \mathbf{r}_X$ stacks the following set of equations:

$$\mathbb{P}[X_i = x_l | Z_i = z_k] = \sum_{t^Z \in \mathcal{T}_Z: t^Z(z_k) = x_l} p(t^Z), \quad \forall z_l \in \mathcal{Z}, x_k \in \mathcal{X}. \quad (12)$$

The proof of proposition 2.1 is straightforward and follows from the previous discussion.

3 Sharp testable implications

3.1 Binary instruments

In this section I show that, in the binary instrument case, the restrictions implied by the observed conditional distribution of the outcome combined with the exclusion restriction can be expressed as a finite set of linear restrictions on instrument-response probabilities. This result, combined with proposition 2.1 transforms the potentially infinite set of linear inequalities that describe the original model into an equivalent finite one.

In the binary instrument case, the sharp testable implications are just a collection of upper bounds on the share of units that realize the exact same treatment at every instrument value. To define these bounds, I first introduce additional notation.

For any $x_l \in \mathcal{X}$, define the population frequency of always- x_l -takers as $\mathbb{P}[X(z_0) = X(z_1) = x_l]$, this is, the share of the population that realizes treatment x_l at all possible instrument values. Let $p(AT_{x_l})$ denote the probability that a distribution over response types $p \in \Delta(\mathcal{T}_Z)$ assigns to always- x_l takers. Formally:

$$p(AT_{x_l}) := \sum_{t^Z \in \mathcal{T}_Z} p(t^Z) \mathbf{I}_{\{t^Z(z_0) = t^Z(z_1) = x_l\}}, \quad (13)$$

where $\mathbf{I}_{\{t^Z(z_0) = t^Z(z_1)\}}$ is an indicator that response type t^Z realizes treatment x_l at both values of Z . Note that this equation is linear in $\mathbf{p}$ with known, deterministic coefficients.

Let $f_{Y,X|Z=z_k}(y, x_l)$ denote the joint distribution of Y and X conditional on $Z = z_k$. Then, for any fixed $x_l \in \mathcal{X}$, define the function $\Psi_{x_l} : \mathcal{Y} \rightarrow \mathbb{R}_+$ as follows:

$$\psi_{x_l}(y) = \min_{k=0,1} \{ [f_{Y,X|Z=z_k}(y, x_l)] \}, \quad (14)$$

and the quantity:

$$\Psi_{x_l} = \int_{\mathcal{Y}} \psi_{x_l}(y) d\mu_Y. \quad (15)$$

The goal is to express the set of inequalities given by $p(AT_{x_l}) \leq \Psi_{x_l}$ for every $x_l \in \mathcal{X}$ in matrix form. To this end, Let t_j^Z denote the j -th instrument-response type according to some fixed, arbitrary, pre-specified order over instrument-response types. Let A_Y be a $L \times |\mathcal{T}_Z|$ matrix with entries $(A_Y)_{l,j} = \mathbf{I}_{\{t_j^Z(z_0)=t_j^Z(z_1)=x_l\}}$ and let Ψ denote the vector that stacks the parameters $\Psi_{x_0}, \dots, \Psi_{x_{K-1}}$ and has Ψ_{x_l} in its l -th coordinate. As the next theorem shows, the sharp observable implications of the exclusion restriction can be expressed as $A_Y \mathbf{p} \leq \Psi$.

Theorem 3.1. *[Sharp observable implications of binary instruments]*

The following statements are equivalent:

1. *A distribution over response types $g \in \Delta(\mathcal{T})$ that satisfies, assumption 2.1 (instrument validity), assumption 2.2 (shape restrictions) and definition 2.3 (consistency with observed probabilities) exists.*
2. *A distribution over instrument-response types $p \in \Delta(\mathcal{T}_Z)$ exists such that p satisfies:*

$$A_Z \mathbf{p} \leq \mathbf{r}_Z, \quad (16)$$

$$A_X \mathbf{p} = \mathbf{r}_X, \quad (17)$$

$$A_Y \mathbf{p} \leq \Psi. \quad (18)$$

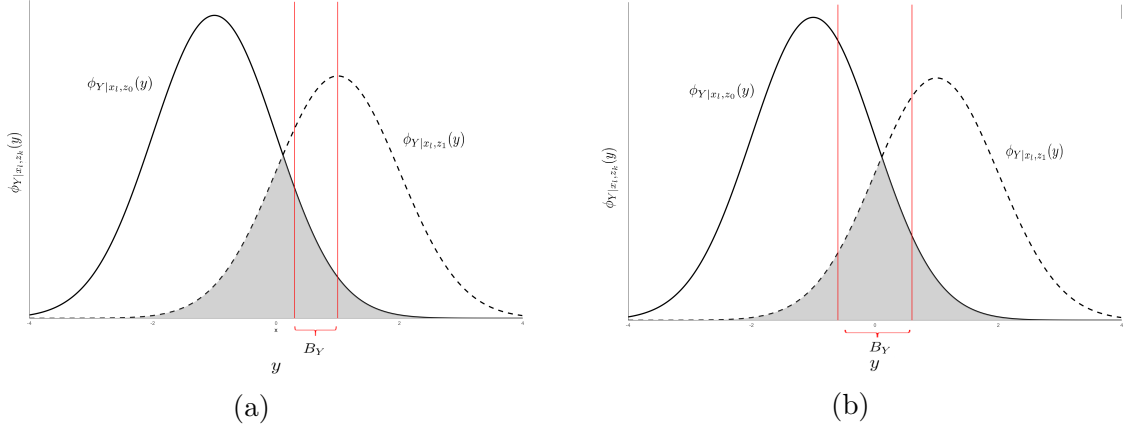
Where $A_Z, A_X, \mathbf{r}_Z$ and $\mathbf{r}_X$ are finite dimensional matrices as defined in assumption 2.2 and proposition 2.1.

The formal proof of theorem 3.1 is given in appendix A. Here, I provide a graphical sketch of the proof and compare it to existing falsification procedures.

To see why the inequality $p(AT_{x_l}) \leq \Psi_{x_l}$ must hold for every $x_l \in \mathcal{X}$ note that always- x_l takers are a potentially strict subset of all the units that realize treatment x_l both when $Z = z_0$ and when $Z = z_1$. Therefore, if the exclusion restriction holds, the group of always- x_l takers that realize outcomes in any measurable set $B_Y \subset \mathcal{B}_Y$ must also be a (potentially strict) subset of the units that realize outcomes in B_Y when $X = x_l$ and $Z = Z_k$ for $z_k \in \{z_0, z_1\}$. Thus, the fraction of always takers with outcomes in B_Y must be weakly dominated by $\psi_{x_l}(y)$ in B_Y (see figure 1 for a graphical representation). Setting $B_Y = \mathcal{Y}$ yields the desired result.

To see the sufficiency of the restrictions, suppose that some $p \in \mathcal{T}_Z$ is consistent with observed conditional treatment probabilities and satisfies $p(AT_{x_l}) \leq \Psi_{x_l}$ for every $x_l \in \mathcal{X}$. To construct a distribution over response types consistent with the observed joint distribution of (Y, X, Z) , the unobserved marginal distribution of potential outcomes for always- x_l takers when $X = x_l$ can be set to $\left[\frac{p(AT_{x_l})}{\Psi_{x_l}}\right]\psi_{x_l}(y)$ so that always- x_l takers account for a fraction $\frac{p(AT_{x_l})}{\Psi_{x_l}}$ of the mass represented by Ψ_{x_l} . This construction guarantees that the unconditional sub-density associated to the outcomes of always- x_l -takers is dominated by $f_{Y,X|Z=z_0}(y, x_l)$ and $f_{Y,X|Z=z_1}(y, x_l)$. Moreover, this construction ensures that the distribution of the outcome for always takers is invariant to instrument changes, which ensures the exclusion restriction is satisfied.

Figure 1: Upper bounds on always takers.

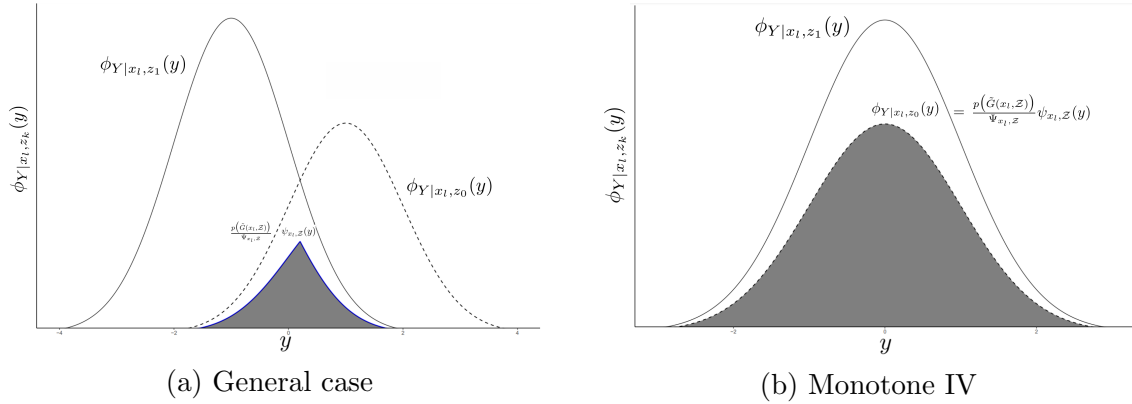


Notes: In both sub-figures the solid and dashed black lines respectively represent hypothetical sub-densities $f_{Y,X|Z=z_0}(y, x_l)$ and $f_{Y,X|Z=z_1}(y, x_l)$. The constant Ψ_{x_l} is defined as the integral over $\mathcal{Y}$ of the point-wise minimum of these two sub-densities, thus, its magnitude corresponds to the gray colored area. The two vertical red lines delimit a measurable interval $B_Y \in \mathcal{B}_Y$.

The exclusion restriction implies that the distribution of outcomes for the group AT_{x_l} of subjects that realize treatment x_l at both instrument values (z_0 and z_1) must be the same conditional on $Z_i = z_0$ or $Z_i = z_1$. Moreover, the units in AT_{x_l} whose outcomes belong to some measurable $B_Y \in \mathcal{B}_Y$ are a (potentially strict) subset of all the units that would realize treatment x_l and outcomes in B_Y . As a consequence, conditional on $Z_i = z_k$, the probability that the outcomes of such group belong to any measurable set $B_Y \in \mathcal{B}_Y$ must be lower than the probability that the events $\{Y_i \in B_Y\}$ and $\{X_i = x_l\}$ are jointly realized. For instance, in subfigure 1a, the fraction of units that realize treatment x_l and outcomes in B_Y conditional on $Z = z_0$ strictly coincides with the intersection of the gray area and the rectangle delimited by the red lines. On the other hand, conditional on $Z = z_1$, there is an additional group of units (whose mass corresponds to the area delimited by the red lines, the dashed black line, and the contour of the gray area) who realize outcomes in B_Y . This additional group cannot belong to AT_{x_l} , otherwise, their outcomes would change without their treatment changing, which would be a violation of the exclusion restriction. In subfigure 1b none of the two sub-densities delimits the upper contour of the gray area, and therefore, the integral of the point-wise minimum yields a tighter bound on always takers that realize outcomes in B_Y than the minimum of the integrals of $f_{Y,X|Z=z_0}(y, x_l)$ and $f_{Y,X|Z=z_1}(y, x_l)$ over B_Y . This argument is formalized in appendix ??.

It remains to distribute the fraction $1 - \frac{p(AT_{x_l})}{\Psi_{x_l}}$ of the area Ψ_{x_l} as well as any mass strictly above at least one sub-density across response types that realize x_l at just one instrument value (i.e. x_l -compliers and x_l -defiers). Assigning potential outcomes to these units is simple because –since they realize treatment x_l at just one instrument value– the exclusion restriction does not bind their potential outcomes. Figure 2a provides a visual illustration of this procedure. The formal construction is described in appendix A.

Figure 2: Sufficiency of the necessary conditions



Notes: In both sub-figures the solid and dashed black lines represent hypothetical sub-densities $f_{Y,X|Z=z_0}(y, x_l)$ and $f_{Y,X|Z=z_1}(y, x_l)$ respectively. The function $\psi_{x_l}(y)$ is defined as the point-wise minimum $\psi_{x_l}(y) := \min\{f_{Y,X|Z=z_0}(y, x_l), f_{Y,X|Z=z_1}(y, x_l)\}$, and the constant Ψ_{x_l} as its integral over $\mathcal{Y}$, thus, its magnitude corresponds to the area that lies strictly below both sub-densities. In this figure, the gray area equals the quantity $p(AT_{x_l})$. The blue solid line represents the sub-density $\frac{p(AT_{x_l})}{\Psi_{x_l}}\psi_{x_l}(y)$ which is proportional to the distribution of potential outcomes for always- x_l -takers. Then, the distribution of potential outcomes for x_l -compliers and x_l -defiers can be defined to be proportional to the sub-density $f_{Y,X|Z=z_k}(y, x_l) - \frac{p(AT_{x_l})}{\Psi_{x_l}}\psi_{x_l}(y)$ for $k = 0, 1$ respectively. Sub-figure (a) corresponds to the general case without monotonicity. It also illustrates a situation in which a valid distribution over instrument response types $p \in \Delta(\mathcal{T}_Z)$ is such that $p(AT_{x_l}) < \Psi_{x_l}$. The monotone case is illustrated in sub-figure (b). In this case, monotonicity enforces that $p(AT_{x_l}) = \Psi_{x_l}$, furthermore, the distribution of potential outcomes for always- x_l -takers is identified and is proportional to $\psi_{x_l}(y)$.

A benchmark case is when assumption 2.2 imposes that treatment x_l can only increase (or decrease) as the instrument changes. This case corresponds to the classical monotone binary instrument and treatment model from Imbens and Angrist (1994), Kitagawa (2015) and Mourifié and Wan (2017). It also corresponds to maximal elements in the ordered or unordered case as in Sun (2023).

Suppose that adoption of x_l weakly increases as Z changes from z_0 to z_1 . Note that in this special case, the function $f_{Y,X|Z=z_1}(y, x_l)$ weakly dominates the function $f_{Y,X|Z=z_0}(y, x_l)$, therefore, $\psi_{x_l}(y)$ is equal to $f_{Y,X|Z=z_0}(y, x_l)$. Furthermore, the fraction

of always x_l takers is identified and coincides with Ψ_{x_l} and $\mathbb{P}[X = x_l|Z = z_0]$. This particular case is illustrated in figure 2b.

Figure 2b illustrates why, while seemingly simpler, my approach is equivalent to Kitagawa (2015) in the binary monotone case. The integral of $\psi_{x_l}(y)$ over $\mathcal{Y}$ constrains the frequency of always takers, this statement is true even if monotonicity does not hold. The test of Kitagawa (2015) takes the function $f_{Y,X|Z=z_0}(y, x_l)$ and asks if, indeed, it coincides with ψ_{x_l} almost everywhere. Instead, I first compute the integral of $\psi_{x_l}(y)$ without invoking monotonicity. Then, I note that if violations of monotonicity occur with positive probability, $\psi_{x_l}(y)$ will be strictly smaller than $f_{Y,X|Z=z_0}(y, x_l)$ in some positive measure set. Thus, the two integrals are equal if and only if random assignment, exclusion, and monotonicity are not violated.

For the general, non-monotone case, Kwon and Roth (2024) note that the population frequency of x_l -defiers (units that realize $X_i = x_l$ only when $Z_i = z_0$) is bounded below by the supremum over measurable sets $B_Y \subset \mathcal{Y}$ of the difference $\mathbb{P}[Y_i \in B_Y, X_i = x_l|Z_i = z_0] - \mathbb{P}[Y_i \in B_Y, X_i = x_l|Z_i = z_1]$. This supremum is attained in the set $\{y \in \mathcal{Y} : f_{Y,X|Z=z_0}(y, x_l) \geq f_{Y,X|Z=z_1}(y, x_l)\}$ and its value coincides with the quantity $\mathbb{P}[X_i = x_l, Z_i = z_0] - \Psi_{x_l}$. Since at $Z = z_0$ only x_l -defiers and x_l -always takers can realize x_l , the upper bound on always takers I derive implies and is implied by the lower bound on compliers from Kwon and Roth (2024). The difference is that instead of taking a supremum over a potentially infinite collection of measurable sets, I obtain an equivalent bound by integrating the point-wise minimum of two sub-densities.

3.2 Multi-valued instruments

4 Generalized no-defiers restrictions

Now I specialize theorem 3.1 to assumptions that rule out specific treatment switches from the population, such as ordered and unordered monotonicity (Imbens and Angrist (1994), Angrist and Imbens (1995), Heckman and Pinto (2018)). I refer to these type of restriction as *generalized no-defiers restrictions*. The specialized characterization is a set of sharp inequalities with a clear microeconomic interpretation.

The essence of *generalized no-defiers restrictions* is that as the instrument changes, some treatments are promoted or relative to others. This notion can be formalized

as follows: If alternative x_1 is preferred over x_0 absent any exogenous incentive, it must, must also be preferred over x_0 when the instrument promotes x_1 . Hence, no switches from x_1 to x_0 are observed.

If *Generalized no-defiers restriction* assume that the instrument promotes some treatments relative to others, then, adoption of the promoted treatments should increase in some way as the instrument changes. While this idea is natural, exactly how should this increase happen is not obvious. Given a specific *generalized no-defiers restriction*, GFOSD describes the exact way in which adoption of promoted treatments increases.

Furthermore, I show that the sharp testable implications of any *generalized no-defiers restriction* combined with the exclusion restriction from assumption 2.1 is also a set of inequalities with a GFOSD interpretation. The reason is that the exclusion restriction on its own does not prevent specific treatment switches. But, combined with the observed distribution of the outcome, it imposes upper bounds on the prevalence of always-takers, which in turn imposes upper bounds on the total share of units that can realize certain treatments.

The GFOSD characterization is a set of inequalities rather than a system of linear equations. Having a characterization in terms of inequalities has two advantages, first, it eliminates nuisance parameters which allows for simpler inference, second, every inequality is linked to a set of behaviors that the imposed shape restriction (on its own or combined with the exclusion restriction) rules out. Hence, if an inequality is violated, it reveals a set of behaviors that exist in the population but are inconsistent with the model's assumptions.

4.1 Sharp testable implications

First, I formally define *generalized no-defiers restrictions* and note that any such restriction can be represented as a binary relation.

Assumption 4.1 (Generalized no-defiers).

Let $R \subset \mathcal{T}_Z = \mathcal{X} \times \mathcal{X}$ be a fixed and known collection of instrument-response types. The underlying distribution over response types $g_0 \in \Delta(\mathcal{T})$ is such that $p_{g_0}(t^Z) = 0$ for all $t^Z \in R$.

Definition 4.1 (Binary relation associated to $R \subset \mathcal{T}_Z$).

Let $R \subset \mathcal{T}_Z$ be a collection of response types. Define the binary relation $\geq^R \subset \mathcal{X} \times \mathcal{X}$

as follows:

$$x_l \geq^R x_{l'} \iff (x_l, x_{l'}) \in R \quad (19)$$

Put in words, a *generalized no-defiers restrictions* simply imposes that instrument-response types contained in the set R occur with zero probability. Moreover, x_l is related to $x_{l'}$ according to $\geq^R$ if switches from x_l to $x_{l'}$ when the instrument goes from z_0 to z_1 are ruled out by the *generalized no-defiers restrictions* associated with R .

Now consider a situation in which a natural ordering over treatments exist (e.g. the number of months a household had insurance in a given period), and suppose that an exogenous intervention promotes the adoption of weakly higher treatments (e.g. having insurance for longer periods). Then, as a consequence of the intervention, no unit should realize a strictly weaker treatment, or, put formally, no unit should realize a treatment in the lower contour of its initial one.

The generalization of the previous idea is that, for any given initial set of treatments S , there is a unique second set of treatments $L_{\geq^R}(S)$ that units with pre-intervention treatments in S cannot realize. For a *generalized no-defiers restrictions* associated with a binary relation R , the set $L_{\geq^R}(S)$ is the *common lower contour* of the set S according to $\geq^R$.

Definition 4.2 (Common lower contour of $\geq^R$).

Let $\geq^R$ be a binary relation over $\mathcal{X}$. For any $S \subset \mathcal{X}$, the common lower contour of S according to $\geq^R$ is:

$$L_{\geq^R}(S) = \{x_k \in \mathcal{X} : s \geq^R x_k, \forall s \in S\} \quad (20)$$

Having defined common lower contours, I state the sharp observable implications of *generalized no-defiers restrictions* and binary instruments. The sharp testable implications can be summarized as follows: For every $S \subset \mathcal{X}$, no unit can switch its treatment from S to $L_{\geq^R}(S)$.

Theorem 4.1.

A distribution over instrument-response types $p \in \Delta(\mathcal{T}_Z)$ that satisfies assumption 2.1 (instrument validity), assumption 4.1 (instrument-response type restriction), and definition 2.3 (consistency) exists, if and only if:

1. **Shape restriction:** The following inequality holds for every $S \subset \mathcal{X}$.

$$\mathbb{P}[X_i \in S | Z_i = z_0] \leq \mathbb{P}[X_i \in L_{\geq R}(S)^C | Z_i = z_1] \quad (21)$$

2. **Exclusion restriction:** For any $S \subset \mathcal{X}$ and any partition (Λ_S, Λ'_S) of $L_{\geq R}(S)^C$ satisfying (i) $\emptyset \neq \Lambda'_S \subset S$, (ii) $L_{\geq R}(x_{l'})^C \setminus \{x_{l'}\} \subset \Lambda_S$ for every $x_{l'} \in \Lambda'_S$, and (iii) $L_{\geq R}(S \setminus \Lambda'_S)^C \subset \Lambda_S$:

$$\mathbb{P}[X_i \in S | Z = z_0] \leq \mathbb{P}[X_i \in \Lambda_S | Z = z_1] + \sum_{x_{l'} \in \Lambda'_S} \Psi_{x_{l'}} \quad (22)$$

Where Ψ_{x_l} is defined in equation 15.

The formal proof of theorem 4.1 is given in appendix B. The key idea is to represent the instrument validity conditions as a flow problem: If the observed conditional treatment probability $\mathbb{P}[X_i = x_l | Z_i = z_0]$ represents the share of the population that realizes treatment x_l at instrument value z_0 , then, finding a valid distribution over response types is equivalent to finding a valid set of flows that, for every $x_l \in \mathcal{X}$, takes the share $\mathbb{P}[X = x_l | Z = z_0]$ of the population that starts with treatment x_l and distributes it across potential counterfactual treatments in a way that matches the observed treatment probabilities conditional on $Z = z_1$ without recurring to the treatment switches or flows that the *generalized no-defiers restriction* forbids.

This flow representation of the falsification problem is equivalent to a pair of well known combinatorial problems: the maximum flow and minimum cut problems.³ The minimum cut formulation is particularly useful to derive the sharp observable implications of incomplete models in terms of inequalities that only depend on observable quantities as first noted by Ekeland et al. (2010) and Galichon and Henry (2011).

Another interpretation for the first part of theorem 4.1 (i.e. falsification of *generalized no-defiers restrictions* without an outcome and an exclusion restriction), is that mutually exclusive events cannot occur in the population with probability greater than one (the events $\{X_i(0) \in S\}$ and $X_i(1) \in L_{\geq R}(S)$ are mutually exclusive). A

³The equivalence is also proven in appendix B. The entire reasoning would also apply to more general restrictions such as those considered in assumption 2.2. The only difference would be that more general restrictions cannot necessarily be stated as capacity constraints on the edges of the graph. This impossibility would complicate using the equivalence with minimum-cut problem to obtain observable inequalities that do not depend on unobservables.

generalization of a logically equivalent idea beyond the binary instrument case was recently developed by [Kaido and Ponomarev \(2025\)](#).

To incorporate an outcome along with the exclusion restriction, [Kaido and Ponomarev \(2025\)](#) expand the graph representation of the model to the space of treatment-outcome pairs (i.e. $(x, y) \in \mathcal{X} \times \mathcal{Y}$). For discrete outcomes, this expanded space substantially increases the total number of inequalities to verify, and for continuous outcomes results in an infinite number of inequalities making statistical inference infeasible. With my approach, the number of inequalities remains the same when an outcome and an exclusion restriction are added.

The second part of theorem [4.1](#) incorporate the exclusion restriction which, as I show in theorem [3.1](#), binds the prevalence of always- x_l takers. In some cases, these bounds on always takers *tighten* the GFOSD inequalities by further limiting the maximum of units that could potentially realize treatments in certain upper contours.

The fact that the exclusion restriction *tightens* the GFOSD inequalities by replacing some conditional treatment probabilities $\mathbb{P}[X_i = x_l | Z = z_1]$ with the weakly smaller quantities Ψ_{x_l} allows to distinguish violations from the exclusion restriction from violations of instrument-response restrictions. In particular, one of the following three cases must occur:

1. All inequalities are satisfied. Hence, neither the exclusion restriction nor the response-type restriction is falsified.
2. Only inequalities from part 2 are violated. Hence, the response type restriction is not falsified on its own, but only when combined with the exclusion restriction.
3. At least one inequality from part 1 is violated. Hence, the response type restriction is falsified

Additionally, every inequality is linked to exactly one set S . Thus, if the inequalities associated with S are not satisfied, it must be that the assumed restriction on how the instrument affects the potential treatments (or preferences over treatments) of units that realize treatments in S when $Z_i = z_0$ is falsified, while restrictions associated to other treatment switches might not.

To illustrate its usefulness, I use theorem [4.1](#) to characterize the binary monotone instrument IV model from [Angrist and Imbens \(1995\)](#).

Corollary 4.1 (Sharp observable implications of binary monotone instruments). *Suppose that $\mathcal{X} = \{x_0, x_1, \dots, x_L\}$, with $x_l < x_{l'}$ for every $l < l'$. A distribution over response types exists that satisfies assumptions 2.1 and:*

$$\mathbb{P}[X(z_0) \leq X(z_1)] = 0 \quad (23)$$

If and only if:

$$\mathbb{P}[X \geq x_l | Z = z_0] \leq \Psi_{x_l} + \mathbb{P}[X > x_{l+1} | Z = z_1] \quad \forall x_l \in \mathcal{X} \quad (24)$$

Note that in the ordered monotone case, the set $L_{\geq}(x_l)^C$ corresponds to the standard upper contour of x_l (i.e. $\{x_l, x_{l+1}, \dots, x_L\}$), thus, the sharp inequalities reduce to FOSD but *tightened* by the exclusion restriction in the sense that the probability $\mathbb{P}[X = x_l | Z = z_1]$ associated to the minimal element of every $L_{\geq}(x_l)^C$ (i.e. x_l) is replaced by the weakly smaller quantity Ψ_{x_l} .⁴

4.2 A submonotone interpretation

I conclude this section by formalizing the way in which *generalized no-defiers restrictions* can be derived from non-parametric restrictions on how the instrument affects the preferences or utility of units in the population. In particular, I assume that selection into treatment follows a utility maximization model, and, rather than restricting potential treatment choices, I restrict potential preferences over the set of treatments $\mathcal{X}$.

Assumption 4.2 ($\geq^*$ -Submonotonicity). *Let $\geq^*$ be an irreflexive⁵ binary relation over $\mathcal{X}$. Suppose that:*

1. *Units are endowed with strict preference profiles over $\mathcal{X}$ that depend on Z_i denoted $\succ_i^{(z_k)}$ for $k = 0, 1$ where the function $\succ_i^{(z_k)}$ maps $\mathcal{Z}$ to strict preference profiles over $\mathcal{X}$. In other words, potential preferences.*

⁴As a consequence of this result, it is easy to see that the conditions from Sun (2023) do not incorporate the effect that the exclusion restriction has on upper contours associated to non minimal/maximal elements. This illustrates why the observable implications from Sun (2023) are not sharp for monotone instruments, and why they are nested by the ones proposed in this paper.

⁵The extension to reflexive binary relations is conceptually simple. The reasoning remains the same but requires special treatment of response type restrictions that rule out “always takers” of one particular treatment. In the interest of brevity this case is addressed in appendix B.2.

2. Units are rational in the sense that $X_i(z_k) = x_l$ implies $x_l \succ_i^{(z)} x_{l'}$ almost surely for all $l, l' \in \{0, 1, \dots, L\}$ such that $l \neq l'$.
3. There are no preference reversals against $\geq^*$ when the instrument changes from z_0 to z_1 . That is, if $x_l \geq^* x_{l'}$ and $x_l \succ_i^{(z_0)} x_{l'}$, then, $x_l \succ_i^{(z_1)} x_{l'}$ almost surely.

The first two points of assumption 4.2 simply posit that response types arise from latent utility maximization. The third point captures the notion that if $x_l \geq^* x_{l'}$, then alternative x_l is being weakly incentivized or promoted by the instrument. Note that this promotion of x_l is not necessarily relative to all other treatments, but relative to $x_{l'}$.

I call assumption 4.2 submonotonicity because it encodes a notion weaker than monotonicity; it rules out two-way flows between certain pairs of treatments, without necessarily involving all alternatives or treatments. In this sense, any submonotonicity assumption can always be extended to a full monotonicity condition with respect to a total order (or the linear extension of a total order) by incorporating additional restrictions. For instance, both ordered and unordered monotonicity correspond to special cases of submonotonicity where the binary relation is transitive. The notion of preventing “two-way flows” from Navjeevan and Pinto (2022) corresponds to assuming that the binary relation in question is asymmetric.

Theorem 4.2. *Let $\geq^*$ be an irreflexive binary relation over $\mathcal{X}$ and suppose that $R \subset \mathcal{T}$ is such that $\geq^R = \geq^*$. Then:*

1. *If $\geq^*$ -Submonotonicity holds, then, $p_{g_0}(t^Z) = 0$ for every $t^Z \notin R$.*
2. *R is the smallest cardinality subset of $\mathcal{T}_Z$ such that if $p_{g_0}(t) = 0$ for all $t \notin R$, then, $\geq^*$ -Submonotonicity cannot be falsified.*

The assumption that $\geq^*$ is irreflexive is not necessary, but it substantially simplifies exposition. The proof of theorem 4.2 is straightforward and can be found in appendix B.2 where the general case is covered.

Theorem 4.2 shows that $\geq^R$ -submonotonicity is sufficient to guarantee that instrument-response types outside of R occur with zero probability. Moreover, R is the minimal (in the sense of set inclusion) restriction over instrument-response types, such that $\geq^R$ -submonotonicity is not falsified. Thus, the set of inequalities from 21 comprises the sharp observable implications of $\geq^R$ -submonotonicity.

In Appendix E I explore in depth the interpretation of different assumptions within the submonotonicity class and provide multiple motivating examples. These examples include extensions of unordered monotonicity (Heckman and Pinto (2018)); assumptions that draw from both ordered and unordered monotonicity; specific cases of targeting and encouragement designs as well as some possible relaxations and generalizations (Lee and Salanié (2020), Bai et al. (2022)); and designs in which the instrument encourages variety and experimentation, or its effect depends on *status-quo* ($Z = z_0$) choices.

In the empirical section of this paper I use the framework from this section to assess the *compliers in the extensive margin* assumption from Rose and Shem-Tov (2024) in the Oregon Health Experiment, and formulate alternative, more plausible restrictions.

5 Inference

In this section I propose a non-parametric inference method for the vector of sub-density overlaps Ψ defined in section 3 along with conditional treatment probabilities. With this method in hand I propose a valid test for the null hypothesis of IV validity.

I assume that the data is an *i.i.d.* random sample $\{(Y_i, X_i, Z_i)\}_{i=1}^n$ drawn from the distribution $F_{Y,X,Z}$. In what follows, I assume that the outcome is a real-valued vector of finite dimension d so that $\mathcal{Y} = \mathbb{R}^d$.

In this section I consider the two instruments case, but the results of this section extend naturally to estimation of the overlap any finite number of densities or subdensities.

5.1 Contact set, non-differentiability, and plug-in estimators

The main challenge for inference on the vector of sub-density overlaps is the possibility that the contact set might have positive measure. When this is the case, non-differentiability of the minimum implies that plug-in estimators are biased and have a non-standard limit. These problems are exacerbated if the goal is to have a fully non-parametric method.

Before describing the technical challenges, I argue that in the context of testing IV validity, robustness to the possibility that the contact set has positive measure is

desirable. The first and simplest reason is that an IV validity test should be robust to the possibility that the instrument has no effect on realized treatments, in other words, it should allow for instrument irrelevance. If the instrument has no effect on realized treatments and the exclusion restriction holds, sub-densities will not change with the instrument and therefore the contact set is equal to the entire support of the outcome.

A second, related but subtler reason has to do with the quasi-experimental nature of IV models. Consider a randomized intervention that seeks to induce individuals in a population to acquire medical insurance with the goal of estimating its effect on health outcomes. If the intervention had a 100% compliance rate, the setup would correspond to a perfectly randomized experiment; it is precisely because of non-compliance that an IV method is needed.

But non-compliance is likely to be correlated with the outcome. The healthiest subjects in the population might not get insurance even if it is promoted. If healthier subjects do not respond to the instrument, their treatment and outcomes do not change, and thus, the joint distribution of the outcome and treatment is the same at different instrument values which results in a non-negligible contact set.

Finally, unless the relevant subdensities do not cross, finite sample estimators cannot determine if the contact set has positive measure. The reason is that, for any given sample size, around the true crossing points there will be a positive measure set in which the difference between densities will be small and statistically indistinguishable from zero. Hence, equality on a non-negligible set cannot be rejected. If the existence of a positive measure contact set is fundamentally untestable, robustness to this possibility becomes a desirable property for both overlap estimators and instrument validity tests.

To illustrate the challenges that arise from non-differentiability, recall that the target parameter is a vector whose entries are:

$$\Psi_{x_l} = \int_{\mathbb{R}^d} \min\{f_{Y,X|Z=z_0}(y, x_l), f_{Y,X|Z=z_1}(y, x_l)\} dy \quad (25)$$

For every $x_l \in \mathcal{X}$.

Then, a plug-in estimator would replace the conditional densities $f_{Y,X|Z=z_k}(y, x_l)$ for $k = 0, 1$ in equation 25 with non-parametric density estimators $\hat{f}_{Y,X|Z=z_k}(y, x_l)$ and calculate the integral of the function $\hat{\psi}_{x_l}(y) := \min\{\hat{f}_{Y,X|Z=z_0}(\tilde{y}, x_l), \hat{f}_{Y,X|Z=z_1}(\tilde{y}, x_l)\}$.

These density estimators can be decomposed into a signal $f_{Y,X|Z=z_k}(y, x_l)$ and noise $\hat{f}_{Y,X|Z=z_k}(y, x_l) - f_{Y,X|Z=z_k}(y, x_l)$ components. Let $\mathcal{C}_{x_l} = \{y \in \mathbb{R}^d : f_{Y,X|Z=z_0}(y, x_l) = f_{Y,X|Z=z_1}(y, x_l)\}$ denote the contact set. In $\mathcal{C}_{x_l}$, the signal components are the same. Thus, for any given sample size, the minimum is entirely determined by statistical noise.

If the minimum is determined by statistical noise regardless of the sample size, then, for any fixed $y \in \mathcal{C}_{x_l}$, the minimizer can switch back and forth from z_0 to z_1 as the sample size increases. Consequently, $\hat{\psi}_{x_l}(y)$ need not converge in probability to any fixed limit. Then, the asymptotic distribution of a plug-in estimator depends on the random process $\{\hat{\psi}_{x_l}(y) : y \in \mathbb{R}^d\}$. Characterizing the variance-covariance of this process and understanding how it affects plug-in estimators is technically challenging.

Moreover, applying the minimum operator in the contact set creates a bias problem. Instead of averaging out statistical noise, the minimum consistently picks negative signals which results in a downward biased estimator.

Finally observe that, even if the minimum is a directionally differentiable function, non-parametric density estimators, and more specifically kernel density estimators, are not tight in general which is a necessary condition for validity of the *m out of n* bootstrap method proposed by [Fang and Santos \(2019\)](#).

To circumvent these technical challenges, I recast sub-density overlaps as the loss of an asymmetric classifier. This alternative representation of the target parameters has the advantage of eliminating most of the randomness that arises from the contact set.

5.2 A classification-based estimator

Given the functions $f_{Y,X|Z=z_0}(y, x_l)$ and $f_{Y,X|Z=z_1}(y, x_l)$, calculating Ψ_{x_l} amounts to identifying the regions in which each of these two functions is dominated by the other, and then computing their integrals over such sets. Thus, the core problem is to identify the points in $\mathbb{R}^d$ that belong to the set $\{y \in \mathbb{R}^n : f_{Y,X|Z=z_1}(y, x_l) \geq f_{Y,X|Z=z_0}(y, x_l)\}$. This is a classification problem.

Consider the random vector (Y, X, Z) . The classification problem consists on predicting Z given Y and X . A classifier is a function $\eta : \mathcal{Y} \times \mathcal{X} \rightarrow \{0, 1\}$ that maps pairs (y, x) to an instrument label $k = 0, 1$. A classification mistake occurs on the event $\{\eta(Y, X) \neq k, Z = z_k\}$ for $k = 0, 1$.

The loss of a symmetric classifier would be the expected missclassification probability $\mathbb{E}[\mathbf{1}_{\{\eta(Y,X) \neq k, Z=z_k\}}]$, but, classification mistakes can be penalized differently depending on the value of Z . A more general loss function could penalize differently the events $\{\eta(Y, X) \neq 0, Z = z_0\}$ and $\{\eta(Y, X) \neq 1, Z = z_1\}$ and even let the penalties depend on the values of X . This results in a loss function:

$$L(\eta, x_l; c) = c_0(X) \mathbb{E}[\mathbf{1}_{\{\eta(Y, X) \neq 0, Z = z_0\}}] + c_1(X) \mathbb{E}[\mathbf{1}_{\{\eta(Y, X) \neq 1, Z = z_1\}}]. \quad (26)$$

Proposition 5.1. *Define the classifier $\eta^*(y, x_l) := \mathbf{1}_{\{f_{Y,X|Z=z_1}(y, x_l) \geq f_{Y,X|Z=z_0}(y, x_l)\}}$ and the loss function $L^*(\eta^*, x_l) = L^*(\eta^*, x_l, c^*)$ with missclassification penalties $c_k^*(X) = \frac{\mathbf{1}_{\{X=x_k\}}}{\mathbb{P}[Z=z_k]}$ for $k = 0, 1$. Then:*

$$\Psi_{x_l} = L^*(\eta^*, x_l) = \sum_{k=0,1} \mathbb{P}[\eta^*(Y, X) \neq k, Z = z_k, X = x_l | Z = z_k]. \quad (27)$$

Furthermore, for any classifier $\eta'(y, x_l)$ such that $\eta'(y, x_l) = \eta^*(y, x_l)$ for every $y \in \mathbb{R}^n \setminus \mathcal{C}_{k_l}$:

$$L^*(\eta^*, x_l) = L^*(\eta', x_l). \quad (28)$$

Proposition 5.1 shows that the overlap parameter can be expressed as the loss of the classifier $L^*(\eta^*, x_l)$. Furthermore, it shows that any classifier that only differs from $L^*(\eta^*, x_l)$ in the contact set attains the same loss as $L^*(\eta^*, x_l)$, resulting in an alternative valid expression for Ψ_{x_l} . Thus, at the population level, the classification approach is invariant to the behavior of classifiers in the contact set. This invariance property makes analyzing the asymptotic behavior of a classification-based estimator tractable.

The estimator is constructed as follows: first, the sample is split in two sub-samples S_1 and S_2 with n_1 and n_2 observations respectively and such that $n_1/n_2 \rightarrow \tau \in (0, 1)$. In the first split the classifier η^* is estimated. Then, the empirical loss of the estimated classifier is evaluated in the second split. The resulting empirical loss is an estimator for Ψ_{x_l} .

To formalize the estimator, let n_1 and n_2 be the number of observations in the first and second split respectively, $K(\cdot)$ a kernel function, and h_n a bandwidth sequence

so that $K_n(y - Y_i) := K\left(\frac{y - Y_i}{h_n}\right)$. The sub-density estimators are:

$$f_{Y,X|Z=z_k,n}(y, x_l) := P_{k,n_1}^{-1} \left[\frac{1}{n_1 h_n^d} \sum_{i=1}^{n_1} \mathbf{1}\{X_i = x_l, Z_i = z_k\} K_n(y - Y_i) \right], \quad (29)$$

where the sum is over all units in the first split, $n_{k,1} := \sum_{i=1}^{n_1} \mathbf{1}\{Z_i = z_k\}$ and $P_{k,n_1} := n_{k,1}/n_1$.

Then, define $\Delta_{n_1}(y, x_l) := f_{Y,X|Z=z_1,n}(y, x_l) - f_{Y,X|Z=z_0,n}(y, x_l)$ and $\eta_{n_1}^*(y, x_l) := \mathbf{1}\{\Delta_{n_1}(y, x_l) \geq 0\}$. The empirical loss of the estimated classifier $\eta_{n_1}^*(y, x_l)$ evaluated in the second split is the empirical analogue of the last expression in (27):

$$L_{n_2}(\eta_{n_1}^*, x_l) := \frac{1}{n_2} \sum_{i=1}^{n_2} \left(\frac{\mathbf{1}\{\eta_{n_1}^*(Y_i, X_i) = 1, Z_i = z_0, X_i = x_l\}}{P_{0,n_2}} + \frac{\mathbf{1}\{\eta_{n_1}^*(Y_i, X_i) = 0, Z_i = z_1, X_i = x_l\}}{P_{1,n_2}} \right), \quad (30)$$

where the sum is over all observations that belong to the second split and P_{k,n_2} denotes the sample probability that Z is equal to z_k in the second split. The estimator is $L_{n_2}(\eta_{n_1}^*, x_l)$.

Note that by construction the estimated classifier $\eta_{n_1}^*$ will a higher loss than the oracle classifier η^* , thus, the estimator is upward biased. But, as I show in the next sub-section, this bias is asymptotically zero.⁶

Recall that the IV validity conditions involve the vector of overlaps Ψ along with conditional treatment probabilities $\mathbb{P}[X = x_l | Z = z_k]$. Thus, in the evaluation split I also compute conditional treatment probability estimators:

$$P_{x_l|z_k,n_2} = \frac{\sum_{i=1}^{n_2} \mathbf{1}\{X_i = x_l, Z_i = z_k\}}{\sum_{i=1}^{n_2} \mathbf{1}\{Z_i = z_k\}}. \quad (31)$$

Conditional treatment probabilities are perfectly collinear, to avoid degenerate asymptotic distributions I restrict the target parameter to all but one observed treatment values. $\mathcal{X}_k = \{l : \mathbb{P}[X = x_l | Z = z_k] > 0\}$ denote the set of treatment indices that occur with strictly positive probability conditional on $Z = z_k$. Define $\mathbf{P}_{\mathcal{X}_k,n}$

⁶In simulations I find that this bias is negligible even in small and medium samples.

as the vector that stacks all the estimators $P_{x_0|z_k, n_2}$ for every $l \in \mathcal{X}_k$ except the one associated with the highest treatment index in $\mathcal{X}_k$. Let $\mathbf{P}_{\mathcal{X}_k}$ be the vector that stacks the associated target conditional treatment probabilities.⁷

5.3 Asymptotic properties

To analyze the asymptotic properties of the estimator, I decompose its scaled difference with respect to the target parameter into two terms. One that corresponds to the theoretical difference between the target and the trained classifiers, and another that corresponds to the difference between the empirical and theoretical loss of the trained classifier.

$$\sqrt{n_2} \left[L_{n_2}(\eta_{n_1}^*, x_l) - \Psi_{x_l} \right] = \sqrt{n_2} \left[L_{n_2}(\eta_{n_1}^*, x_l) - L(\eta_{n_1}^*, x_l) \right] + \sqrt{n_2} \left[L(\eta_{n_1}^*, x_l) - L(\eta^*, x_l) \right] \quad (32)$$

To obtain valid inference, I note that the asymptotic behavior of the first term depends on neither the distribution of the classification decision at particular values of $y \in \mathbb{R}^d$, nor the correlation between classification decisions at different values of $y \in \mathbb{R}^d$. It only depends on aggregate missclassification rates, which substantially simplifies analyzing and computing its variance. Combining this insight with a set of necessary conditions that make the second term $o_p(1)$ I prove that the estimator is asymptotically normal and propose a valid variance estimator.

To make the second term in equation 32 $o_p(1)$ I first use the fact that the population level loss is invariant to the behavior of $\eta_{n_1}^*$ in the contact set (see proposition 5.1).

It remains to analyze the loss of $\eta_{n_1}^*$ out of the contact set. The key insight is that the rate of uniform convergence for KDEs is $r_n = \sqrt{\frac{nh_n^d}{\log(h_n^{-1})}}$. Therefore, in the region of $\mathbb{R}^d$ in which the absolute difference $|\Delta_{x_l}(y)| := |f_{Y,X|Z=z_1}(y, x_l) - f_{Y,X|Z=z_0}(y, x_l)|$ is greater than r_n , classification mistakes occur with very low probability. Consequently, while the additional loss in which $\eta_{n_1}^*$ might incur relative to η^* in this region can be large, the expected additional loss is negligible.

The only remaining region is the set in which $0 < |\Delta_{x_l}(y)| \leq r_n$. Since r_n tends to zero, both the measure of this region and the magnitude of the extra loss that

⁷I restrict the conditional probability estimators to treatment values in $\mathcal{X}_k$ to ensure that none of the estimators has a degenerate limit distribution.

results from classification mistakes in this region tend to zero as the sample grows. This, however, is not enough to guarantee that the additional misclassification loss is negligible at the rate $n^{-1/2}$. Thus, I make an additional assumption that forces the region $\{y \in \mathbb{R}^d : 0 < |\Delta_{x_l}(y)| \leq r_n\}$ to shrink at a sufficiently fast rate, or, in other words, I impose a margin condition (see assumption 5.1.3).

Now I formalize the assumptions I make on the functions $f_{Y,X|Z=z_k}$, the kernel function, and bandwidth sequence:

Assumption 5.1.

1. **Non-zero instrument probabilities:** A constant $C \in (0, 1)$ exists such that:

$$\mathbb{P}[Z = z_k] > C \quad \forall z_k \in \mathcal{Z}. \quad (33)$$

2. **Bounded continuous Hessian :** For every $x_l \in \mathcal{X}$ and $z_k \in \mathcal{Z}$, the function $f_{Y,X|Z=z_k}(y, x_l)$ is twice continuously differentiable in y with Hessian matrix $\nabla^2 f_{Y|X=x_l, Z=z_k}$. Furthermore, fixed constants $M > 0$ and $M' > 0$ exist such that:

$$\sup_{y \in \mathbb{R}^d} f_{Y,X|Z=z_k}(y, x_l) \leq M, \quad (34)$$

and

$$\sup_{\|u\|=1} \|(\nabla^2 f_{Y|X=x_l, Z=z_k})u\| \leq M' \quad (35)$$

3. **Margin condition:** For every $x_l \in \mathcal{X}$ and $z_k \in \mathcal{Z}$, positive constants B, α , and $\delta > 0$ exist such that $\forall t \in (0, \delta)$

$$\mu\{y \in \mathcal{Y} : 0 < |f_{Y,X|Z=z_1}(y, x_l) - f_{Y,X|Z=z_0}(y, x_l)| \leq t\} \leq Bt^\alpha \quad (36)$$

Where μ denotes the Lebesgue measure over $\mathbb{R}^d$.

Assumption 5.1 imposes some mild conditions on the data generating process. 5.1.1 simply requires that all considered instrument values occur with positive probability in the population, which is guaranteed to occur in most empirical settings. 5.1.2 requires that the distribution of the outcome be bounded, and twice continuously differentiable with bounded Hessian. I combine this assumption with standard

properties of kernel density estimators (KDE) to argue that the contribution of density estimation bias is negligible for classification mistakes. The margin condition from 5.1.3 is typically imposed in the non-parametric classification literature to obtain fast rates of convergence for estimated classifiers (Tsybakov (2004)).⁸ In the one-dimensional outcome case, to satisfy the margin condition, values of α as low as 3/5 suffice, depending on the bandwidth choice.

Assumption 5.2.

1. **Kernel regularity:** The kernel function $K : \mathcal{Y} \mapsto \mathbb{R}$ is positive, symmetric around zero, bounded, and square integrable. Furthermore, it satisfies assumption (K_1) from Giné and Guillou (2002).⁹
2. **Bandwidth sequence restrictions:** The bandwidth sequence h_n satisfies:
 - (a) $h_n = n^{-\gamma}$ for some $\gamma > \frac{1}{d+4}$.
 - (b) $nh_n^d/\ln(n) \rightarrow \infty$.
 - (c) The constant α from assumption 5.1.3 and γ satisfy: $\alpha > \frac{\gamma}{1-\gamma}$.

Assumption 5.2 is a set of common requirements for kernel functions and bandwidth sequences. 5.2.1 is required to control bias and to use the results from Giné and Guillou (2002) to bound the probability that KDEs take extreme values and thus incur in classification mistakes. Typical kernel choices such as gaussian, triangular, uniform or Epanechnikov satisfy this assumption. The first part of 5.2.2 simply requires some degree of undersmoothing so that KDEs bias is negligible, the second part requires that the bandwidth decreases at a rate slower than n^{-d} . The third part of 5.2.2 is necessary to invoke the margin condition from assumption 5.1.3 and guarantee that the bias caused by classification mistakes in the proximity of the contact set is asymptotically negligible.

Assumption 5.3.

⁸It is also used in optimal policy and DML (add citations!!!!).

⁹The assumption requires that K lies in the linear span of functions $\{g : \mathcal{Y} \mapsto \mathbb{R}_+\}$ such that the set $\{(s, u) : g(s) \geq u\}$ can be represented as a finite number of Boolean operations among sets of the form $\{(s, u) : p(s, u) \geq \phi(u)\}$ where p is a polynomial and ϕ an arbitrary real function. See Giné and Guillou (2002) for a discussion of this assumption.

1. **Strictly positive-overlaps:** For every $x_l \in \mathcal{X}$,

$$\Psi_{x_l} > \epsilon. \quad (37)$$

2. **Non-trivial classification problem** A constant $\epsilon > 0$ exist such that for every $x_l \in \mathcal{X}$ and every $z_k \in \mathcal{Z}$:

$$\lim_{n \rightarrow \infty} \int_{C_k} f_{Y,X|Z=z_k}(y) dy > \epsilon \quad (38)$$

Where $C_k = \mathbb{R}^d \setminus \{y \in \mathbb{R}^d : \eta_{n_1}^*(y, x_l) \neq k\}$.

Assumption 5.3 is not required for inference on sub-density overlaps, just for testing IV validity. The condition $\Psi_{x_l} > 0$ from assumption 5.3.1 simply ensures the limit distribution of the overlap estimator is non-degenerate asymptotically. This condition is testable using the proposed inference framework on the overlap parameters alone.

Assumption 5.3.2 rules out the possibility that, as the sample grows, the classifier estimated in the first split always selects the same instrument value. This condition guarantees that the overlap estimator is not perfectly correlated with conditional treatment probability estimators.

Assumption 5.3.2 can only be violated in two cases, the first is when the contact set is equal to $\mathbb{R}^d$ and yet, even as the sample size tends to infinity, the estimated classifier picks the same fixed instrument value z_k for every $y \in \mathbb{R}^d$. Since this first case is implausible, I assume it away. The second case is when one subdensity strictly dominates the other almost everywhere. This second condition is testable and, if verified, both the classification and overlap estimation problems become trivial with a sufficiently large sample.¹⁰

Under these assumptions, I obtain valid joint inference on conditional treatment probabilities and subdensity overlaps.

Theorem 5.1. Suppose assumptions 5.1 to 5.3 hold.

Define the vectors $\theta_n^T = (P_{\mathcal{X}_0, n}^T, P_{\mathcal{X}_1, n}^T, \Psi_n^T)$ and $\mu_\theta^T = (P_{\mathcal{X}_0}^T, P_{\mathcal{X}_1}^T, \Psi^T)$ and let m denote their dimension. Let Ω_n denote the symmetric positive definite matrix $3m \times 3m$ matrix $n_2 \text{Var}(\theta_n | S_1)$.

¹⁰For the alternative estimator based on a semi-deterministic classifier that I introduce in the next section, assumption 5.3.2 can be guaranteed to hold by choosing an appropriate default deterministic classification rule.

Then:

$$\sqrt{n_2}\Omega_n^{-1/2}(\boldsymbol{\theta}_n - \boldsymbol{\mu}_\theta) \xrightarrow{d} \mathcal{N}(0, \mathbf{I}_m) \quad (39)$$

Furthermore, the formulas from lemma C.8 in appendix C show how to construct an estimator Σ_n for the matrix Ω_n . The matrix Σ_n satisfies:

$$\|\Sigma_n - \Omega_n\| \xrightarrow{p} 0. \quad (40)$$

Furthermore, in lemma C.5 from appendix C I further show that sample splitting implies that the expectation of the empirical loss $L_{n_2}(\eta_{n_1}^* m, x_l)$ is equal to the theoretical loss $L(\eta_{n_1}^* m, x_l)$. Hence, since $L(\eta_{n_1}^* m, x_l)$ converges to Ψ_{x_l} , the overlap estimators are asymptotically unbiased.

Note that theorem 5.1 does not yield asymptotic normality for $\boldsymbol{\theta}_n$ but for the normalized sequence $\Omega_n^{-1/2}\boldsymbol{\theta}_n$. The reason is that, even if the overlap estimator Ψ_{x_l} converges to a fixed constant, it is based on using the second split to evaluate the empirical loss of the classifier trained in the first split $\eta_{n_1}^*$. Therefore, its variance conditional on the first split depends on $\eta_{n_1}^*$. If the contact set has positive measure, the estimated classifier $\eta_{n_1}^*$ has no stable limit and thus $\sqrt{n_2}(\Psi_{x_l, n} - \Psi_{x_l})$ has no stable asymptotic distribution.

In particular, the conditional variance of the empirical loss in the second split is given by:

$$n_2 \text{Var}(\Psi_{x_l, n} - \Psi_{x_l} | S_1) = \sum_{k=0,1} \frac{P_{\xi, x_l | z_k, n_2}(1 - P_{\xi, x_l | z_k, n_2})}{\mathbb{P}[Z = z_k]} \quad (41)$$

Where $P_{\xi, x_l | z_k, n_2} := \mathbb{P}[\eta_{n_1}^*(Y, X, Z; z_k, x_l) \neq k, X = x_l | Z = z_k]$.

If classification decisions in the contact set are dictated by statistical noise, then, $\eta_{n_1}^*$ randomly re-classifies subsets of the contact set from one instrument value to another. Furthermore, this re-classification does not stop as the sample size tends to infinity. Hence, $P_{\xi, x_l | z_k, n_2}$ does not converge to a fixed constant and as a consequence the limit of the scaled variance is not a constant. The same phenomenon occurs with the covariance between overlap estimators associated with different treatment values, and their covariance with conditional treatment probabilities.

Variance instability does not prevent inference on the vector of overlap parameters,

it only implies that it has no fixed limit distribution. That is why normalization by $\Omega_n^{-1/2}$ is required to obtain asymptotic normality.

5.4 Semi-deterministic classifiers

Now I propose a slight modification of the estimator, the modification restores a fixed limit distribution which makes the estimator compatible with a larger class of methods for inference on linear systems including Fang et al. (2023) and Goff and Mbakop (2025). Moreover, the modified estimator becomes asymptotically linear and thus allows for K-fold cross fitting. The modification is to use a sub-optimal estimated classifier that has the advantage of converging to a fixed function.

The key modification is to define a default pre-determined, arbitrary classification rule as well as a deterministic sequence d_n that dominates the rate of uniform convergence for KDEs r_n . Then, the classifier differs from the pre-determined rule only when the estimated difference $\Delta_{n1}(y, x_l)$ is larger than d_n .

The modified classifier is asymptotically optimal in the sense that it converges to a classifier that only follows the pre-determined rule in the contact set, where, as proposition 5.1 shows, the loss is invariant to classification decisions.

Assumption 5.4. [*Deterministic decision rate:*] The sequence d_n satisfies:

1. $\lim_{n \rightarrow \infty} \frac{d_n}{r_n} = \infty$.
2. $\lim_{n \rightarrow \infty} d_n = 0$.
3. A constant $\tilde{\gamma} > 0$ exists such that $\lim_{n \rightarrow \infty} \frac{d_n}{n^{-\tilde{\gamma}}} = 0$ and $\alpha > \frac{\tilde{\gamma}}{1-\tilde{\gamma}}$.

The first part of assumption 5.4 simply requires that d_n dominates r_n so that the classifier only changes when KDE-based classifiers are unlikely to incur in classification mistakes. The second part requires that d_n tends to zero so that, asymptotically, the classifier only coincides with the pre-specified deterministic rule inside the contact set. The last condition is required to apply the margin condition and prove that the loss from classification mistakes near the boundary of the contact set is negligible. Note that this last condition is strictly stronger than its analogue in assumption 5.2.2.

For a sample of size n and some fixed integer $\tilde{K} \geq 2$, a $\tilde{K}$ -fold partition randomly distributes all observations into $\tilde{K}$ sub-samples or folds denoted $(J_\kappa)_{\kappa=1}^{\tilde{K}}$ of size $\lfloor n/\tilde{K} \rfloor$ each. Let, J_κ^C denote the set of all observations except those in fold κ .

Let $\eta_{\kappa,n}^d(y, x_l)$ be the classifier trained in J_κ^C according to the following semi-deterministic rule:

$$\eta_{\kappa,n}^d(y, x_l) := \begin{cases} \mathbf{1}_{\{\Delta_{\kappa,n}(y, x_l) \geq 0\}} & \text{if } |\Delta_{\kappa,n}(y, x_l)| \geq d_n \\ \mathbf{1}_{\{y \in D\}} & \text{otherwise} \end{cases} \quad (42)$$

Where D is a fixed, pre-specified measurable subset of $\mathbb{R}^n$ and $\Delta_{\kappa,n}(y, x_l)$ is the difference between the kernel density estimators for $f_{Y,X|Z=z_1}$ and $f_{Y,X|Z=z_0}$ trained in J_κ^C . Note that $\eta_{\kappa,n}^d$ converges to the classifier η^d that coincides with $\mathbf{1}_{\{\Delta_{\kappa,n}(y, x_l) > 0\}}$ everywhere except in $\mathcal{C}_{x_l}$ where it is equal to $\mathbf{1}_{\{y \in D\}}$.

Let $L_\kappa(\eta_{\kappa,n}^d, x_l)$ denote the empirical loss of the classifier $\eta_{\kappa,n}^d$ evaluated in J_κ . Define the estimator $\Psi_{x_l, \kappa, n}^d := L_\kappa(\eta_{\kappa,n}^d, x_l)$.

In appendix C I prove that the difference $L_\kappa(\eta_{\kappa,n}^d, x_l) - \Psi_{x_l}$ admits the following decomposition:

$$\begin{aligned} L_\kappa(\eta_{\kappa,n}^d, x_l) - \Psi_{x_l} = & \frac{1}{\tilde{n}} \sum_{i \in J_\kappa} \left[\sum_{k=0,1} \frac{P_{z_k} - \mathbf{1}\{Z_i = z_k\}}{P_{z_k}^2} \mu_{\eta^d}(z_k, x_l) \right. \\ & \left. + \frac{1}{P_{z_k}} \xi_{\eta^d}(Y, X, Z; z_k, x_l) \right] + o_p(n^{-1/2}) \end{aligned} \quad (43)$$

Where $\mu_{\eta^d}(z_k, x_l)$ denotes the misclassification probability $\mathbb{P}[\eta^d(Y, X) \neq k, X = x_l, Z = z_k]$. That is, the estimator is asymptotically linear with influence function given by the sum in brackets. Thus, the estimator based on a semi-deterministic classifier is asymptotically normal and its asymptotic distribution does not depend on the trained classifier. This enables K-fold cross fitting.

Theorem 5.2. Set $\Psi_{x_l, \kappa, n}^d := L_\kappa(\eta_{\kappa,n}^d, x_l)$ and define:

$$P_{x_l|z_k, \kappa, n} = \frac{\sum_{i \in J_\kappa} \mathbf{1}\{X_i = x_l, Z_i = z_k\}}{\sum_{i=1}^n \mathbf{1}\{Z_i = z_k\}}. \quad (44)$$

Let $\mathbf{P}_{\mathcal{X}_1, n}^T$ and $\mathbf{\Psi}_n^T$ stack the estimators $P_{x_l|z_k, \kappa, n}$ associated with treatment values in $\mathcal{X}_k$ except the one with the highest index.

Define the random vector:

$$\boldsymbol{\theta}_{\kappa,n}^T = (\mathbf{P}_{\mathbf{x}_0,n}^T, \mathbf{P}_{\mathbf{x}_1,n}^T, \Psi_{x_0,\kappa,n}^d, \dots, \Psi_{x_{L-1},\kappa,n}^d) \quad (45)$$

With expectation $\boldsymbol{\mu}_\theta^T = (\mathbf{P}_{\mathbf{x}_0}^T, \mathbf{P}_{\mathbf{x}_1}^T, P_{x_{L-1}|z_1}, \Psi_{x_0}, \dots, \Psi_{x_{L-1}})$.

Furthermore, define:

$$\boldsymbol{\theta}_n^T = \frac{1}{\tilde{K}} \sum_{\kappa=1}^{\tilde{K}} \boldsymbol{\theta}_{\kappa,n}^T. \quad (46)$$

Then, a symmetric positive definite matrix $\boldsymbol{\Omega}$ exist such that:

$$n\text{Var}(\boldsymbol{\theta}_n^T) \xrightarrow{p} \boldsymbol{\Omega} \quad (47)$$

And:

$$\sqrt{n}(\boldsymbol{\theta}_n - \boldsymbol{\mu}_\theta) \xrightarrow{d} \mathcal{N}(0, \boldsymbol{\Omega}) \quad (48)$$

Furthermore, a consistent estimator $\boldsymbol{\Sigma}_n$ for the matrix $\boldsymbol{\Omega}$ can be constructed using the formulas from lemma C.8 in appendix C.

The proof of theorem 5.2 is in appendix C.

Notice that the estimator from theorem 5.2 has a fixed asymptotic variance. Hence, without imposing assumption 5.3, or including conditional treatment probabilities that are equal to zero, the limit would still be asymptotically normal, but potentially degenerate.

5.5 Formal sharp tests of IV validity

Now I discuss how to implement a test of instrument validity based on the estimators from subsections 5.1 and 5.4 and the characterizations from theorems 3.1 and 4.1. In particular, I show that the testing problem satisfies the assumptions from Cox and Shi (2023) and thus their method is applicable.

Let $\boldsymbol{\theta}$ denote the vector of non-redundant conditional treatment probabilities and sub-density overlaps. The null hypothesis that assumptions 2.1 and 2.2 are jointly

satisfied is equivalent to the assumption that the linear system:

$$\mathbf{H}_0 : \exists \delta \text{ s.t. } B_0 \delta + B_1 \boldsymbol{\theta} \leq \mathbf{0} \quad (49)$$

Where, B_0 and B_1 are deterministic matrices with known coefficients and, in the case of *generalized no-defiers restrictions* from section 4 the matrix B_0 is zero.

Cox and Shi (2023) show that the null hypothesis can be recast as

$$\mathbf{H}_0 : T := \min_{\mathbf{b}: A\mathbf{b} \leq \mathbf{0}} (\boldsymbol{\theta} - \mathbf{b})^T \boldsymbol{\Omega}^{-1} (\boldsymbol{\theta} - \mathbf{b}) = 0, \quad (50)$$

for some fixed matrix A with known coefficients and a positive-definite matrix $\boldsymbol{\Omega}$. Furthermore, they propose a testing procedure based on comparing the sample analogue of T to the critical values of a χ -squared distribution with degrees of freedom equal to the rank of the rows of A associated with the inequalities that bind at the optimal $\mathbf{b}$.

Definition 5.1 (Admissible nulls). *Let $\mathcal{F}$ be the set of all distributions over $\mathbb{R}^d \times \mathcal{X} \times \mathcal{Z}$ such that:*

1. *F satisfies assumptions 5.1 and 5.3 for some fixed values of C , M , M' , α , B and ϵ .*
2. *The set of instrument-response types $p \in \Delta(\mathcal{T}_Z)$ such that p satisfies equations 16 to 18 from theorem 3.1 with respect to the probabilities induced by $F_{Y,X,Z} \in \mathcal{F}$ is non-empty.*

Theorem 5.3. *Suppose that $\boldsymbol{\theta}_n$ is the estimator from either theorem 5.1 or theorem 5.2 with variance estimator $\boldsymbol{\Sigma}_n$ and let A be the matrix from equation (50).*

Define:

$$T_n := \min_{\mathbf{b}: A\mathbf{b} \leq \mathbf{0}} (\boldsymbol{\theta}_n - \mathbf{b})^T \boldsymbol{\Sigma}_n^{-1} (\boldsymbol{\theta}_n - \mathbf{b}), \quad (51)$$

and let $\hat{r}$ be the rank of the matrix comprised by the rows of A associated with the inequalities that bind at the optimal $\mathbf{b}$ in equation 51.

For $\alpha \in [0, 1]$, let $\chi_{m, 1-\alpha}^2$ be the $1 - \alpha$ -quantile of a χ -squared distribution with m degrees of freedom. Define $\phi_n^{CC}(\alpha) := \mathbf{1}_{\{T_n > \max\{\chi_{\hat{r}, 1-\alpha}^2, \text{tol}\}\}}$ and some tolerance $\text{tol} > 0$.

Then:

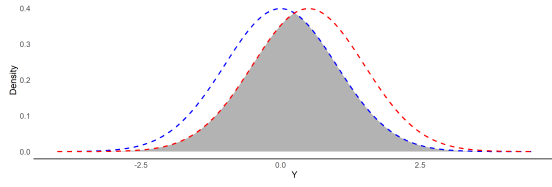
$$\limsup_{n \rightarrow \infty} \sup_{F_{Y,X,Z} \in \mathcal{F}} \mathbb{P}_{F_{Y,X,Z}} [\phi_n^{CC}(\alpha) = 1] \leq \alpha \quad (52)$$

To prove theorem 5.3 I verify that every sequence F_n in $\mathcal{F}$ is such that the centered estimator $\sqrt{n}(\boldsymbol{\theta}_n - \boldsymbol{\theta})$ and its variance estimator $\boldsymbol{\Sigma}_n^{-1/2}$ satisfy assumption 3(a) from Cox and Shi (2023). The proof is in appendix C. The only requirement is that, as definition 5.1 imposes, assumptions 5.1 and 5.3 must hold for the same fixed, strictly positive constants for all admissible nulls. This condition ensures that all admissible distributions yield non-degenerate variance-covariance matrices and therefore the test statistic is well defined. This restriction rules out sequences of null hypothesis with conditional treatment probabilities $\mathbb{P}[X = x_l | Z = z_k]$ or overlaps Ψ_{x_l} that are strictly positive for every sample size n , but decrease with n and reach zero in the limit .

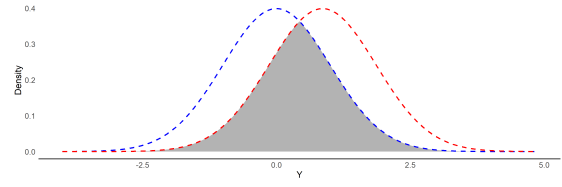
An alternative proof for the estimator based on the semi-deterministic classifier would be to use the asymptotically linear expression in equation (43) to verify assumption 1 from Cox and Shi (2023). Asymptotic linearity implies that other inference methods for linear systems such as Fang and Santos (2019) or Goff and Mbakop (2025) are valid too.

6 Simulations

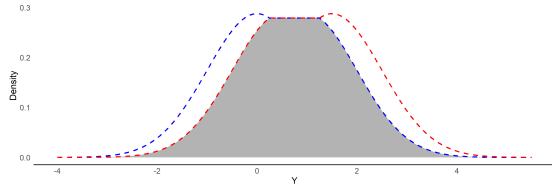
Figure 3: Outcome distribution under DGP_1



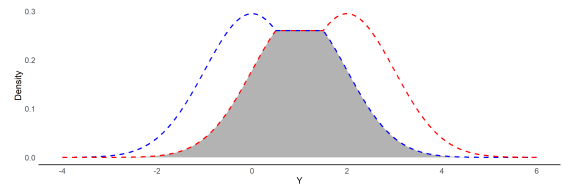
(a) $X = x_2, \beta_{IV} = \frac{\tau}{2}, \Phi_C = 0, |\Delta_B| = 0.31$



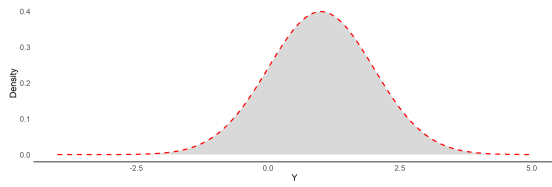
(b) $X = x_2, \beta_{IV} = \tau, \Phi_C = 0, |\Delta_B| = 0.31$



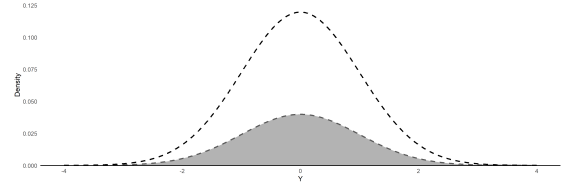
(c) $X = x_2, \beta_{IV} = \frac{\tau}{2}, \Phi_C = .28, |\Delta_B| = 0.16$



(d) $X = x_2, \beta_{IV} = \tau, \Phi_C = .27, |\Delta_B| = 0.31$



(e) $X = x_2, \beta_{IV} = 0, \Phi_C = 27$

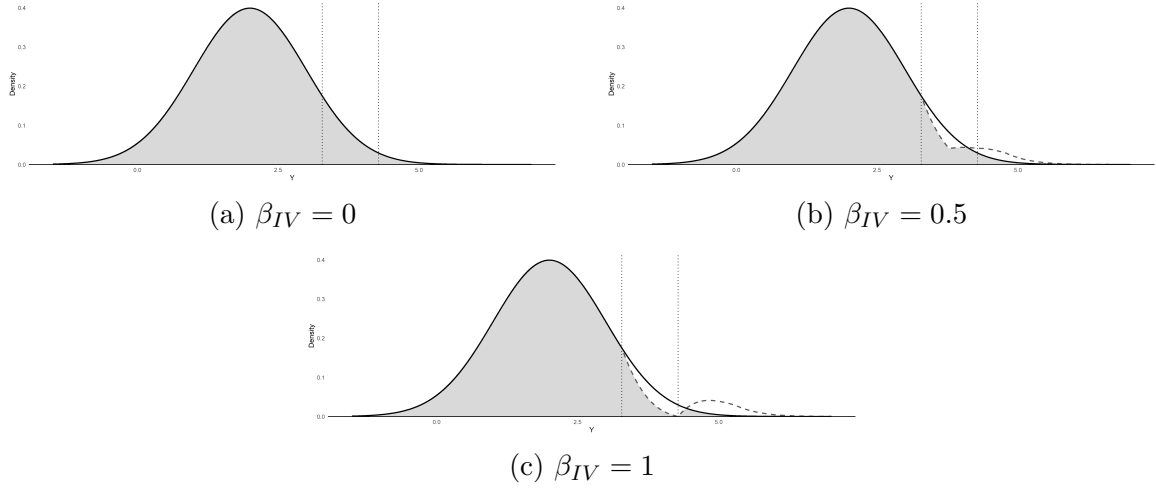


(f) $X = x_1, x_3$

Table 1: Bias, overlap coverage, and joint overlaps and CTPs coverage for sample splitting estimators.

			$h_n = n^{-1/3}$			$h_n = n^{-1/4}$			$h_n = n^{-1/5}$			
$n = 1000$ $\mathbb{E}n_{z_0, x_l} = 165, \mathbb{E}n_{z_1, x_l} = 330$			$C_{\frac{2}{3}}$	$C_{\frac{1}{2}}$	$C_{\frac{1}{3}}$	$C_{\frac{2}{3}}$	$C_{\frac{1}{2}}$	$C_{\frac{1}{3}}$	$C_{\frac{2}{3}}$	$C_{\frac{1}{2}}$	$C_{\frac{1}{3}}$	
$w = 0$	$\mu_2 = 0$	$\Psi_{x_0, n} - \Psi_{x_0}$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
		$\Psi_{x_0} \in C_{0.95}$	0.95	0.95	0.95	0.96	0.97	0.93	0.96	0.97	0.94	
		$\theta_0 \in E_{0.95}$	0.94	0.94	0.93	0.94	0.95	0.92	0.94	0.95	0.93	
	$\mu_2 = 0.5$	$\Psi_{x_0, n} - \Psi_{x_0}$	0.02	0.02	0.03	0.01	0.02	0.02	0.01	0.01	0.02	
		$\Psi_{x_0} \in C_{0.95}$	0.91	0.90	0.84	0.93	0.92	0.88	0.95	0.93	0.90	
		$\theta_0 \in E_{0.95}$	0.88	0.84	0.75	0.90	0.86	0.80	0.93	0.87	0.82	
	$\mu_2 = \tau$	$\Psi_{x_0, n} - \Psi_{x_0}$	0.01	0.01	0.01	0.00	0.00	0.00	0.00	0.00	0.00	
		$\Psi_{x_0} \in C_{0.95}$	0.94	0.95	0.95	0.94	0.94	0.95	0.94	0.95	0.94	
		$\theta_0 \in E_{0.95}$	0.91	0.93	0.91	0.91	0.93	0.92	0.92	0.94	0.91	
$w = 1$	$\mu_2 = 0.5$	$\Psi_{x_0, n} - \Psi_{x_0}$	0.02	0.02	0.02	0.01	0.01	0.02	0.01	0.01	0.01	
		$\Psi_{x_0} \in C_{0.95}$	0.94	0.92	0.90	0.94	0.92	0.90	0.94	0.94	0.92	
		$\theta_0 \in E_{0.95}$	0.89	0.88	0.81	0.89	0.90	0.81	0.90	0.87	0.81	
	$\mu_2 = \tau$	$\Psi_{x_0, n} - \Psi_{x_0}$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
		$\Psi_{x_0} \in C_{0.95}$	0.93	0.95	0.95	0.94	0.96	0.95	0.93	0.93	0.96	
		$\theta_0 \in E_{0.95}$	0.93	0.95	0.93	0.93	0.96	0.94	0.94	0.95	0.93	
	$n = 2000$ $\mathbb{E}n_{z_0, x_l} = 330, \mathbb{E}n_{z_1, x_l} = 660$			$C_{\frac{2}{3}}$	$C_{\frac{1}{2}}$	$C_{\frac{1}{3}}$	$C_{\frac{2}{3}}$	$C_{\frac{1}{2}}$	$C_{\frac{1}{3}}$	$C_{\frac{2}{3}}$	$C_{\frac{1}{2}}$	$C_{\frac{1}{3}}$
	$w = 0$	$\mu_2 = 0$	$\Psi_{x_0, n} - \Psi_{x_0}$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
			$\Psi_{x_0} \in C_{0.95}$	0.96	0.97	0.94	0.95	0.96	0.93	0.95	0.95	0.94
$\theta_0 \in E_{0.95}$			0.95	0.97	0.94	0.94	0.97	0.95	0.94	0.95	0.95	
$\mu_2 = 0.5$		$\Psi_{x_0, n} - \Psi_{x_0}$	0.01	0.01	0.02	0.01	0.01	0.01	0.01	0.01	0.01	
		$\Psi_{x_0} \in C_{0.95}$	0.91	0.92	0.88	0.93	0.92	0.91	0.94	0.93	0.92	
		$\theta_0 \in E_{0.95}$	0.90	0.89	0.79	0.92	0.91	0.87	0.92	0.90	0.89	
$\mu_2 = \tau$		$\Psi_{x_0, n} - \Psi_{x_0}$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
		$\Psi_{x_0} \in C_{0.95}$	0.95	0.94	0.96	0.95	0.95	0.94	0.95	0.94	0.94	
		$\theta_0 \in E_{0.95}$	0.93	0.91	0.95	0.92	0.92	0.94	0.93	0.93	0.94	
$w = 1$	$\mu_2 = 0.5$	$\Psi_{x_0, n} - \Psi_{x_0}$	0.01	0.01	0.01	0.01	0.01	0.01	0.00	0.01	0.01	
		$\Psi_{x_0} \in C_{0.95}$	0.93	0.93	0.91	0.94	0.94	0.92	0.93	0.93	0.92	
		$\theta_0 \in E_{0.95}$	0.92	0.90	0.86	0.93	0.90	0.87	0.93	0.91	0.88	
	$\mu_2 = \tau$	$\Psi_{x_0, n} - \Psi_{x_0}$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
		$\Psi_{x_0} \in C_{0.95}$	0.96	0.97	0.95	0.94	0.96	0.95	0.94	0.94	0.95	
		$\theta_0 \in E_{0.95}$	0.95	0.95	0.96	0.93	0.94	0.95	0.93	0.94	0.96	
	$n = 4000$ $\mathbb{E}n_{z_0, x_l} = 660, \mathbb{E}n_{z_1, x_l} = 1320$			$C_{\frac{2}{3}}$	$C_{\frac{1}{2}}$	$C_{\frac{1}{3}}$	$C_{\frac{2}{3}}$	$C_{\frac{1}{2}}$	$C_{\frac{1}{3}}$	$C_{\frac{2}{3}}$	$C_{\frac{1}{2}}$	$C_{\frac{1}{3}}$
	$w = 0$	$\mu_2 = 0$	$\Psi_{x_0, n} - \Psi_{x_0}$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
			$\Psi_{x_0} \in C_{0.95}$	0.95	0.95	0.95	0.95	0.96	0.95	0.95	0.95	0.94
$\theta_0 \in E_{0.95}$			0.95	0.95	0.94	0.96	0.93	0.95	0.95	0.94	0.95	
$\mu_2 = 0.5$		$\Psi_{x_0, n} - \Psi_{x_0}$	0.00	0.01	0.01	0.00	0.00	0.00	0.00	0.00	0.00	
		$\Psi_{x_0} \in C_{0.95}$	0.93	0.94	0.90	0.95	0.95	0.95	0.94	0.95	0.96	
		$\theta_0 \in E_{0.95}$	0.91	0.88	0.83	0.94	0.92	0.89	0.94	0.93	0.90	
$\mu_2 = \tau$		$\Psi_{x_0, n} - \Psi_{x_0}$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
		$\Psi_{x_0} \in C_{0.95}$	0.95	0.95	0.96	0.95	0.94	0.95	0.94	0.95	0.96	
		$\theta_0 \in E_{0.95}$	0.94	0.94	0.94	0.94	0.95	0.95	0.94	0.94	0.94	
$w = 1$	$\mu_2 = 0.5$	$\Psi_{x_0, n} - \Psi_{x_0}$	0.00	0.00	0.01	0.00	0.00	0.00	0.00	0.00	0.00	
		$\Psi_{x_0} \in C_{0.95}$	0.93	0.93	0.91	0.94	0.95	0.92	0.94	0.93	0.94	
		$\theta_0 \in E_{0.95}$	0.93	0.90	0.88	0.94	0.92	0.93	0.92	0.92	0.94	
	$\mu_2 = \tau$	$\Psi_{x_0, n} - \Psi_{x_0}$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
		$\Psi_{x_0} \in C_{0.95}$	0.96	0.96	0.94	0.96	0.96	0.95	0.96	0.96	0.94	
		$\theta_0 \in E_{0.95}$	0.93	0.93	0.94	0.93	0.94	0.94	0.95	0.94	0.94	

Figure 4: Outcome distribution at x_3 under DGP_3



Kitagawa (2010) Kaido and Ponomarev (2025) Mourifié and Wan (2017) Kwon and Roth (2024)

Table 2: Null hypothesis of IV validity rejection rates.

Panel A: Three treatments, instrument effect at x_2															
DGP	KDE overlaps				Finite grid overlaps				KR Discretization						
	$c_{\frac{1}{2}}$	$c_{\frac{2}{3}}$	$c_{\frac{1}{2}}$	$c_{\frac{2}{3}}$	$c_{\frac{1}{2}}$				Constant b_n				Increasing b_n		
	$h_n = n^{-1/4}$		$h_n = n^{-1/5}$		b_5	b_{15}	b_{25}	b_{50}	b_5	b_{15}	b_{25}	b_{50}	$b_{\frac{n}{100}}$	$b_{\frac{n}{50}}$	$b_{\frac{n}{25}}$
$n = 1500$ $\mathbb{E}n_{z_k, x_l} = 250$															
$\beta_{IV} = 0.9\tau$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.91	0.72	0.00	0.00	0.79	0.72	0.14
$\beta_{IV} = 1\tau$	0.01	0.02	0.01	0.02	0.00	0.00	0.00	0.01	0.76	0.50	0.00	0.00	0.54	0.50	0.11
$\beta_{IV} = 1.1\tau$	0.06	0.05	0.06	0.05	0.01	0.02	0.01	0.04	0.49	0.27	0.02	0.00	0.32	0.27	0.10
$\beta_{IV} = 1.2\tau$	0.16	0.13	0.17	0.14	0.04	0.04	0.05	0.12	0.29	0.22	0.07	0.00	0.33	0.22	0.20
$n = 3000$ $\mathbb{E}n_{z_k, x_l} = 500$															
$\beta_{IV} = 0.9\tau$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.98	0.87	0.00	0.00	0.87	0.21	0.00
$\beta_{IV} = 1\tau$	0.02	0.02	0.01	0.02	0.00	0.00	0.00	0.01	0.85	0.56	0.00	0.00	0.56	0.11	0.00
$\beta_{IV} = 1.1\tau$	0.10	0.08	0.11	0.08	0.00	0.00	0.00	0.02	0.53	0.29	0.03	0.00	0.29	0.14	0.01
$\beta_{IV} = 1.2\tau$	0.31	0.25	0.33	0.27	0.01	0.01	0.02	0.06	0.29	0.34	0.19	0.03	0.34	0.44	0.07
$n = 6000$ $\mathbb{E}n_{z_k, x_l} = 1000$															
$\beta_{IV} = 0.9\tau$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.00	0.97	0.00	0.00	0.23	0.00	0.00
$\beta_{IV} = 1\tau$	0.01	0.02	0.01	0.02	0.00	0.00	0.00	0.01	0.95	0.65	0.00	0.00	0.12	0.00	0.00
$\beta_{IV} = 1.1\tau$	0.17	0.13	0.17	0.14	0.00	0.01	0.01	0.06	0.64	0.27	0.08	0.01	0.28	0.03	0.00
$\beta_{IV} = 1.2\tau$	0.53	0.44	0.57	0.46	0.06	0.04	0.06	0.30	0.25	0.53	0.45	0.12	0.77	0.26	0.03
Panel B: Two treatments, instrument effect on right tail															
DGP	KDE overlaps				Finite grid overlaps				KR Discretization						
	$c_{\frac{1}{2}}$	$c_{\frac{2}{3}}$	$c_{\frac{1}{2}}$	$c_{\frac{2}{3}}$	$c_{\frac{1}{2}}$				Constant b_n				Increasing b_n		
	$h_n = n^{-1/4}$		$h_n = n^{-1/5}$		b_5	b_{15}	b_{25}	b_{50}	b_5	b_{15}	b_{25}	b_{50}	$b_{\frac{n}{100}}$	$b_{\frac{n}{50}}$	$b_{\frac{n}{25}}$
$n = 1000$ $\mathbb{E}n_{z_k, x_l} = 250$															
$\beta_{IV} = 0$	0.06	0.05	0.06	0.07	0.05	0.05	0.05	0.05	0.07	0.05	0.02	0.00	0.07	0.06	0.05
$\beta_{IV} = 0.5$	0.10	0.07	0.07	0.12	0.05	0.06	0.05	0.06	0.07	0.05	0.03	0.01	0.07	0.06	0.05
$\beta_{IV} = 1$	0.14	0.10	0.10	0.15	0.07	0.07	0.07	0.07	0.07	0.05	0.03	0.01	0.07	0.06	0.05
$n = 2000$ $\mathbb{E}n_{z_k, x_l} = 500$															
$\beta_{IV} = 0$	0.03	0.05	0.05	0.04	0.06	0.06	0.06	0.06	0.04	0.03	0.03	0.00	0.05	0.03	0.01
$\beta_{IV} = 0.5$	0.11	0.10	0.10	0.12	0.06	0.06	0.06	0.06	0.04	0.03	0.02	0.00	0.05	0.03	0.01
$\beta_{IV} = 1$	0.19	0.13	0.11	0.16	0.06	0.06	0.09	0.08	0.04	0.03	0.02	0.00	0.05	0.03	0.01
$n = 4000$ $\mathbb{E}n_{z_k, x_l} = 1000$															
$\beta_{IV} = 0$	0.05	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.03	0.02	0.00	0.03	0.01	0.00
$\beta_{IV} = 0.5$	0.20	0.15	0.13	0.20	0.04	0.05	0.06	0.06	0.04	0.03	0.02	0.01	0.03	0.01	0.05
$\beta_{IV} = 1$	0.31	0.25	0.18	0.31	0.04	0.05	0.13	0.12	0.04	0.03	0.02	0.01	0.03	0.01	0.05

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A Proofs for section 3

Proof of proposition ??.

First observe that for every measurable $B_Y \subset \mathcal{Y}$, every non-empty $\tilde{Z} \subset \mathcal{Z}$ and $x_l \in \mathcal{X}$, if $z_k \in \tilde{Z}$, then:

$$\mathbb{P}[Y \in B_Y, X = x_l | Z = z_k] \geq \mathbb{P}[Y(x_l) \in B_Y, X(\tilde{z}) = x_l \ \forall \tilde{z} \in \tilde{Z} | Z = z_k] \quad (53)$$

$$\implies \mathbb{P}[Y \in B_Y, X = x_l | Z = z_k] \geq \mathbb{P}[Y(x_l) \in B_Y, X(\tilde{z}) = x_l \ \forall \tilde{z} \in \tilde{Z}] \quad (54)$$

$$\implies \min_{z_k \in \tilde{Z}} \{\mathbb{P}[Y \in B_Y, X = x_l | Z = z_k]\} \geq \mathbb{P}[Y(x_l) \in B_Y, X(\tilde{z}) = x_l \ \forall \tilde{z} \in \tilde{Z}] \quad (55)$$

$$\implies \min_{z_k \in \tilde{Z}} \left\{ \int_{B_Y} f_{Y,X|Z=z_k}(y, x_l) d\mu_y \right\} \geq \mathbb{P}[Y(x_l) \in B_Y, X(0) = x, X(\tilde{z}) = x_l \ \forall \tilde{z} \in \tilde{Z}] \quad (56)$$

The initial inequality is true because outcomes in B_Y might be realized by response types other than the sufficient-takers associated to $\tilde{Z}$. The first implication follows from exogeneity of potential outcomes, the second is true because the inequality must hold for every individual $z_k \in \tilde{Z}$. The last simply applies the definition of $f_{Y,X|Z=z_k}(y, x_l)$.

Note that equation 56 must hold for every measurable set $B_Y \in \mathcal{B}_Y$. The next step is to transform this collection of equations into a single restriction.

To this end, consider the set of all orderings over $\tilde{Z}$ denoted $\mathcal{O}(\tilde{Z}) = \{\kappa : \{0, 1, \dots, |\tilde{Z}| - 1\} \longrightarrow \tilde{Z} : \exists \kappa^{-1}\}$. For compactness, for every $\pi \in \mathcal{O}(\tilde{Z})$ and every $k \in \{0, 1, \dots, |\tilde{Z}| - 1\}$ I denote the $k + 1$ -th element of $\tilde{Z}$ according to κ as κ_k (instead of $\kappa(k)$).

Then, for every x_l and $\kappa \in \mathcal{O}(\tilde{Z})$, define the set $Y_{x_l}^\pi$ as follows:

$$Y_{x_l}^\kappa = \left\{ y \in \mathcal{Y} : f_{Y,X|Z=\kappa_0}(y, x_l) \leq \min_{k=0,2,\dots,|\tilde{Z}|-1-1} \{f_{Y,X|Z=\kappa_k}(y, x_l)\} \right\} \quad (57)$$

$$\bigcap \left\{ y \in \mathcal{Y} : f_{Y,X|Z=\kappa_1}(y, x_l) \leq \min_{k=1,\dots,|\tilde{Z}|-1-1} \{f_{Y,X|Z=\kappa_k}(y, x_l)\} \right\} \quad (58)$$

$$\dots \bigcap \left\{ y \in \mathcal{Y} : f_{Y,X|Z=\kappa_m}(y, x_l) \leq \min_{k=m,\dots,|\tilde{Z}|-1} \{f_{Y,X|Z=\kappa_k}(y, x_l)\} \right\} \quad (59)$$

$$\dots \bigcap \left\{ y \in \mathcal{Y} : f_{Y,X|Z=\kappa_{K-2}}(y, x_l) \leq \min_{k=|\tilde{Z}|-1-2, |\tilde{Z}|-1-1} \{f_{Y,X|Z=\kappa_k}(y, x_l)\} \right\} \quad (60)$$

This is, the set in which the values that the functions $f_{Y,X|Z=z_k}(y, x_l)$ take are sorted by magnitude in exactly in the same order that κ sorts the instrument values. Note that since the functions $f_{Y,X|Z=z_k}(y, x_l)$ are point identified, the sets $Y_{x_l}^\kappa$ are identified too. Moreover, the sets $Y_{x_l}^\kappa$ are the intersection of sets defined by inequalities that depend on measurable functions, thus, they are measurable themselves.

Then:

$$\mathbb{P}[Y \in B_Y, X(\tilde{z}) = x_l \ \forall \tilde{z} \in \tilde{Z} | Z = z] = \sum_{\kappa \in \mathcal{O}(\tilde{Z})} \mathbb{P}[Y \in Y_{x_l}^\kappa \cap B_Y, X(\tilde{z}) = x_l \ \forall \tilde{z} \in \tilde{Z} | Z = z] \quad (61)$$

$$\leq \sum_{\kappa \in \mathcal{O}(\tilde{Z})} \min_{z_k \in \tilde{Z}} \left\{ \int_{Y_{x_l}^\kappa \cap B_Y} f_{Y,X|Z=z_k}(y, x_l) d\mu_Y \right\} \quad (62)$$

$$= \sum_{\kappa \in \mathcal{O}(\tilde{Z})} \int_{Y_{x_l}^\kappa \cap B_Y} f_{Y,X|Z=\kappa_0}(y, x_l)(y) d\mu_Y \quad (63)$$

$$= \sum_{\kappa \in \mathcal{O}(\tilde{Z})} \int_{Y_{x_l}^\kappa \cap B_Y} \psi_{x_l, \tilde{Z}}(y) d\mu_Y \quad (64)$$

The first line follows from the fact that the sets $Y_{x_l}^\kappa$ comprise a partition of $\mathcal{Y}$. The inequality in the second line follows from equation 56. The equality in the third line is crucial, it is true because the sets $Y_{x_l}^\kappa$ are constructed so that $f_{Y,X|Z=\kappa_0}(y, x_l)(y) \leq f_{Y,X|Z=z_k}(y, x_l)$ for every $y \in Y_{x_l}^\kappa$ and every $z_k \in \tilde{Z}$. Moreover, in $Y_{x_l}^\kappa$ it holds that $\psi_{x_l, \tilde{Z}}(y) f_{Y,X|Z=\kappa_0}(y, x_l)$ which proves the third line. Using again the fact that the sets

$Y_{x_l}^\kappa$ partition $\mathcal{Y}$ yields the last equality. $\square$

Proof of theorem 3.1.

1 $\implies$ 2:

The proof follows immediately from proposition ??.

2 $\implies$ 1:

Let $\mathbf{p}$ be a distribution over instrument response types such that:

$$A_Z \mathbf{p} \leq \mathbf{r}_Z \quad (65)$$

$$A_X \mathbf{p} = \mathbf{r}_X \quad (66)$$

$$A_Y \mathbf{p} \leq \Psi \quad (67)$$

The goal is to define a distribution over response types that satisfies equation 8 and proposition 2.1. To this end, it is necessary to specify potential treatments and outcomes for every possible response types, including potential outcomes for types that never realize certain treatments and thus never realize the potential outcome associated to such treatment.

To simplify exposition, let $\tilde{p}_{x_l}$ denote the quantity $p(AT_{x_l})$.

Let $t \in \Delta(\mathcal{T})$ be a response type. Define the set of treatments that response type $t = (t^Z, t^X)$ realizes as $R(t) = \{x_l \in \mathcal{X} : (t^Z)_{k'} = x_l \text{ for some } k' = 1, 2\}$. Note that by construction $|R(t)| \leq 2$

Let λ be an arbitrary fixed distribution over $\mathcal{Y}$. Then, define $g' : \Delta(\mathcal{T}) \longrightarrow \mathbb{R}$ as follows:

$$g'(t^Z, t^X) = \begin{cases} \left[\frac{1}{\Psi_{x_l, \mathcal{Z}}} \psi_{x_l, \mathcal{Z}}((t^X)_l) \right] \left[\prod_{l': x_{l'} \notin R(t)} \lambda((t^X)_{l'}) \right], & \text{if } R(t) = \{x_l\} \text{ and } \Psi_{x_l} > 0 \\ \left[\prod_{l, k: x_l = (t^Z)_k} \frac{1}{P_{x_l|z_k} - \tilde{p}_{x_l}} \xi_{x_l}((t^X)_l) \right] \left[\prod_{l': x_{l'} \notin R(t)} \lambda((t^X)_{l'}) \right], & \text{if } |R(t)| = 2 \text{ and } \Psi_{x_l} - \tilde{p}_{x_l} > 0 \\ 0, & \text{otherwise} \end{cases} \quad (68)$$

Where $\xi_{x_l}((t^X)_l) = f_{Y, X|Z=z_k}((t^X)_l, x_l) - \frac{\tilde{p}_{x_l}}{\Psi_{x_l, \mathcal{Z}}} \psi_{x_l, \mathcal{Z}}((t^X)_l)$ and $P_{x_l|z_k} = \mathbb{P}[X = x_l | Z = z_k]$.

Integrating the function g' allows to construct the desired distribution over response types. To this end, recall the following result from measure theory:

Lemma A.1. *Any measurable set of response types $B_{\mathcal{T}} \in \mathcal{B}_{\mathcal{T}}$ can be partitioned into slices as follows:*

$$B_{\mathcal{T}} = \bigcup_{t^Z \in \mathcal{T}_Z} \{t^Z\} \times B_{T_X}^{t^Z} \quad (69)$$

Where $B_{T_X}^{t^Z} = B_{Y_0}^{t^Z} \times B_{Y_1}^{t^Z} \times \dots \times B_{Y_{L-1}}^{t^Z} \in \mathcal{Y}^L$ is potentially empty and measurable with respect to the product measure over the product of the L spaces that correspond to the potential outcomes at each different treatment value $x_l \in \mathcal{X}$.

Moreover, each individual set $= B_{Y_l}^{t^Z}$ for $l = 0, 1, \dots, L-1$ is measurable with respect to μ_Y .

Then, using the lemma, define:

$$g(B_{\mathcal{T}}) = \sum_{t^Z \in \mathcal{T}_Z} \left[p(t^Z) \int_{B_{T_X}^{t^Z}} g'(t^Z, t^X) d\mu_{T^X} \right] \quad (70)$$

Where μ_{T^X} denotes the product measure $\mu_{T^X} = \mu_Y \otimes \mu_Y \otimes \dots \otimes \mu_Y$ over the space of potential treatment-responses.

$$\int_{B_{T_X}^{t^Z}} g'(t^Z, t^X) d\mu_{T^X} = \int_{B_{Y_0}^{t^Z}} \int_{B_{Y_1}^{t^Z}} \dots \int_{B_{Y_{L-1}}^{t^Z}} g'(t^Z, t^X) d\mu_Y \mu_Y \dots \mu_Y \quad (71)$$

It remains to prove that $g(B_{\mathcal{T}})$ is a valid distribution over response types, this is, it is non-negative, adds up to 1, satisfies countable additivity, and is consistent with the observed joint distribution of outcomes, treatments and instruments..

When $|R(t)| = 1$, the function $g'(t^Z, t^X)$ is the product of non-negative functions, therefore, it must be non-negative itself. When $|R(t)| = 2$ it is either the product of non-negative functions, or zero. Therefore, it must be non-negative. From here, it follows that $g(B_{\mathcal{T}})$ is non-negative for every measurable $B_{\mathcal{T}} \in \mathcal{B}_{\mathcal{T}}$.

Moreover, note that:

$$\int_{\mathcal{Y}} \frac{p(t^Z)}{\Psi_{x_l, \mathcal{Z}}} \psi_{x_l, \mathcal{Z}}(y) d\mu_Y = p(t^Z) \quad (72)$$

$$\int_{\mathcal{Y}} \left[\frac{p(t^Z)}{P_{x_l|z_k} - \tilde{p}_{x_l}} (f_{Y,X|Z=z_k}(y, x_l) - \frac{\tilde{p}_{x_l}}{\Psi_{x_l, \mathcal{Z}}} \psi_{x_l, \mathcal{Z}}(y)) \right] = p(t^Z) \quad (73)$$

From here, which follows that:

$$g(\mathcal{T}) = \sum_{t^Z \in \mathcal{T}_Z} \left[p(t^Z) \int_{\mathcal{T}_X} g'(t^Z, t^X) d\mu_{T^X} \right] \quad (74)$$

$$= \sum_{t^Z \in \mathcal{T}_Z} \left[p(t^Z) \right] \quad (75)$$

$$= 1 \quad (76)$$

Countable additivity follows mechanically. Let $\{B_{\mathcal{T}}^m\}_{m \in M}$ be a countable collection of disjoint subsets of $\mathcal{T}$ indexed by $m \in M$. Applying lemma A.1, each sets $B_{\mathcal{T}}^m$ can be sliced as follows:

$$B_{\mathcal{T}}^m = \bigcup_{t^Z \in \mathcal{T}_Z} \{t^Z\} \times B_{T_X}^{t^Z, m} \quad (77)$$

Since $B^m \cap B^{m'} = \emptyset$ for all $m, m' \in M$ such that $m \neq m'$ then its must be that $B_{T_X}^{t^Z, m} \cap B_{T_X}^{t^Z, m'} = \emptyset$.

Then:

$$g\left(\bigcup_{m \in M} B_{\mathcal{T}}^m\right) = g\left(\bigcup_{m \in M} \left[\bigcup_{t^Z \in \mathcal{T}_Z} \{t^Z\} \times B_{T_X}^{t^Z, m}\right]\right) \quad (78)$$

$$= g\left(\bigcup_{t^Z \in \mathcal{T}_Z} \left[\bigcup_{m \in M} \{t^Z\} \times B_{T_X}^{t^Z, m}\right]\right) \quad (79)$$

$$= g\left(\bigcup_{t^Z \in \mathcal{T}_Z} \left[\{t^Z\} \times \left\{\bigcup_{m \in M} B_{T_X}^{t^Z, m}\right\}\right]\right) \quad (80)$$

$$= \sum_{t^Z \in \mathcal{T}_Z} p(t^Z) \int_{\bigcup_{m \in M} B_{T_X}^{t^Z, m}} g'(t^Z, t^X) d\mu_{T_X} \quad (81)$$

$$= \sum_{t^Z \in \mathcal{T}_Z} \sum_{m \in M} p(t^Z) \int_{B_{T_X}^{t^Z, m}} g'(t^Z, t^X) d\mu_{T_X} \quad (82)$$

$$= \sum_{m \in M} \sum_{t^Z \in \mathcal{T}_Z} p(t^Z) \int_{B_{T_X}^{t^Z, m}} g'(t^Z, t^X) d\mu_{T_X} \quad (83)$$

$$= \sum_{m \in M} g(B_{\mathcal{T}}^m) \quad (84)$$

The first line is comes from application of lemma A.1 to every set $B_{\mathcal{T}}^m$, the second simply inverts the order of the unions, the third is a property of cartesian products, the fourth simply applies the definition of g and takes advantage of the fact that the set is already sliced, the fifth line uses countable additivity of the integral, the sixth changes the order of the sums, and the last one uses the definition of g but now applied to each set B_Y^m separately. This proves countable additivity.

It only remains to verify consistency with observed probabilities, let $B_{T_X}^{t^Z} = \mathcal{Y} \times \dots \times B_Y \times \dots \times \mathcal{Y}$ be the set obtained by taking the product of the outcome's support $L - 1$ times and letting the $l' - th$ element of the product coincide with B_Y instead of $\mathcal{Y}$. Then:

$$g\left(\{(t^Z, t^X) \in \mathcal{T} : (t^Z)_k = x_l, (t^X)_l \in B_Y\}\right) \quad (85)$$

$$= \sum_{t^Z : (t^Z)_k = x_l} \left[p(t^Z) \int_{B_{T^X}^{t^Z}} g'(t^Z, t^X) d\mu_{T^X} \right] \\ = \frac{\tilde{p}_{x_l}}{\Psi_{x_l, \mathcal{Z}}} \int_{B_Y} \psi_{x_l, \mathcal{Z}}(y) d\mu_Y \quad (86)$$

$$+ \frac{P_{x_l|z_k} - \tilde{p}_{x_l}}{P_{x_l|z_k} - \tilde{p}_{x_l}} \int_{B_Y} \left(f_{Y,X|Z=z_k}(y, x_l) - \frac{\tilde{p}_{x_l}}{\Psi_{x_l, \mathcal{Z}}} \psi_{x_l, \mathcal{Z}}(y) \right) d\mu_Y \\ = \int_{B_Y} f_{Y,X|Z=z_k}(y, x_l) d\mu_Y \quad (87)$$

The first equality is just the definition of g , the second uses the fact that at instrument value z_k there are only two instrument-response types that realize x_l (always- x_l takers, and $\{z_k\}$ -IFF- x_l -takers) combined with the definition of g' . The last line simply rearranges terms.

This completes the proof.

B Proofs for section 4

B.1 Proof for subsection 4.1

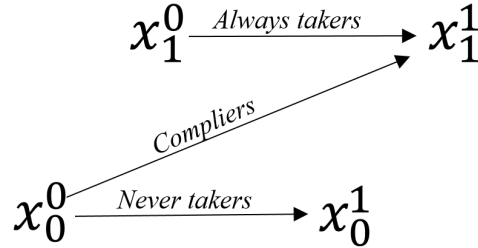
Before providing the proofs I formally define the transportation problem as well as the maximum flow and minimum cut problems.

First define the sets $\mathcal{X}_k$ for $k = 0, 1$ as copies of $\mathcal{X}$ but with elements $x_l^k \in \mathcal{X}_k$ labeled with the superscript k to make explicit that x_l^0 refers to treatment $x_l \in \mathcal{X}$ realized when $Z = 0$ and x_l^1 to treatment $x_l \in \mathcal{X}$ realized when $Z = 1$.

Next, define the directed graph with vertex set $\mathcal{X}_0 \cup \mathcal{X}_1$ and let the edges be $\mathcal{E}_R \subset \mathcal{X}_0 \times \mathcal{X}_1$, where the ordered pair $(x_l^0, x_{l'}^1) \in \mathcal{X}_0 \times \mathcal{X}_1$ belongs to $\mathcal{E}_R$ if and only if the instrument-response type $t^Z = (x_l, x_{l'})$ does not belong to R . In other words, an edge $(x_l^0, x_{l'}^1)$ belongs to the graph if the response type that realizes x_l at $Z = 0$ and $x_{l'}$ at $Z = 1$ is not ruled out by assumption 4.1.

For instance, in the case of the no defiers assumption, the set R would consist of the singleton $\{(x_1, x_0)\}$. Therefore, the edges (x_0^0, x_0^1) , (x_0^0, x_1^1) and (x_1^0, x_1^1) —corresponding to never takers, compliers, and always takers—would belong to $\mathcal{E}_R$, in contrast, the edge (x_1^0, x_0^1) would not. This particular case is illustrated in figure 5.

Figure 5: Graph representation of the no-defiers instrument-response type restriction.



Notes: This figure depicts the graph representation of the no-defiers/monotonicity assumption from [Imbens and Angrist \(1994\)](#). The edges with a super-index $k = 0, 1$ represent treatment values realized at instrument values $Z = z_k$. The edges with sub-index $l = 0, 1$ represent treatments x_0 and x_1 respectively. Since there is no edge from x_1^0 to x_0^1 , defiers are ruled out from the model.

The transportation problem is as follows: every node $x_l^0 \in \mathcal{X}_0$ starts with a mass $\mathbb{P}[X_i = x_l | Z_i = z_0]$ of units. Each unit can travel from its origin node x_l^0 through the admissible edges $(x_l^0, x_{l'}^1) \in \mathcal{E}_R$ in order to reach some destination node $x_{l'}^1 \in \mathcal{X}_1$ subject to the constraint that each destination node $x_{l'}$ cannot receive more than $\mathbb{P}[X_i = x_{l'} | Z_i = z_1]$ units. If it is possible to empty all origin nodes x_l^0 so that all

destination nodes end at full capacity, then, the transportation problem is solved.

In what follows, I prove that if the transportation problem is solved, a distribution over response types exists that is consistent with both the observed joint distribution of (X, Y) and the imposed response type restriction.

Through the definitions and proofs I alternate from edges of the graphs that define the transportation problem and response types. This is possible because there is a one-to-one correspondence between the response types t^Z that belong to $\mathcal{T}_Z \setminus R$ and edges $e \in \mathcal{E}_R$. This correspondence is given by $t^Z = (x_l, x_{l'}) \longrightarrow e = (x_l^0, x_{l'}^1)$. I denote the unique edge associated to $t^Z \in \mathcal{T}_Z$ as e_{t^Z} and the unique instrument-response type associated to $e \in \mathcal{E}_R$ as t_e^Z .

Definition B.1 (Circulation with demands problem associated to $\mathbf{R}$). *Here I introduce a version of the problem tailored to the instrument validity application. For a detailed discussion of discrete circulation and transportation problems see [Kleinberg and Tardos \(2006\)](#) chapter 7.5.*

Let $R \subset \mathcal{T}_Z$ be a collection of instrument-response types with associated graph $\mathcal{E}_R$. Define demands $d_{x_l^k}$ for $x_l^k \in \mathcal{X}_0 \cup \mathcal{X}_1$ as follows:

$$d_{(x,z)} = \begin{cases} -\mathbb{P}[X = x_l | Z = z_k] & \text{if } k = 0 \\ \mathbb{P}[X = x_l | Z = z_k] & \text{if } k = 1 \end{cases} \quad (88)$$

Furthermore, every edge $e = (x_l^0, x_{l'}^1) \in \mathcal{E}_R$ has a maximum capacity $c_e \in [0, 1]$ defined as $c_e = c_e((x_l^0, x_{l'}^1))$.

A candidate solution to the circulation with capacities problem is a function $\tilde{p} : E_R \longrightarrow \mathbb{R}_+$ such that, for each directed edge $e = (x_l^0, x_{l'}^1) \in \mathcal{E}_R$, the number $\tilde{p}(e)$, represents the flow from node x_l^0 to node $x_{l'}^1$. The flow $\tilde{p}$ solves the problem if:

1. *Capacity constraints are satisfied:*

$$0 \leq \tilde{p}(e) \leq c_e \text{ for all } e \in E_R \quad (89)$$

2. *Demand is satisfied at every node (note that supply is represented as negative*

demand):

$$d_{x_l^0} = - \sum_{e \in E^{out}(x_l^0)} \tilde{p}(e) \quad \text{for all } l = 1, 2, \dots, L \quad (90)$$

$$d_{x_{l'}^1} = \sum_{e \in E^{in}(x_{l'}^1)} \tilde{p}(e) \quad \text{for all } l = 1, 2, \dots, L \quad (91)$$

Where $E^{out}(x_l^0) = \{e \in \mathcal{E}_R : e = (x_l^0, x_{l'}^1) \text{ for some } x_{l'}^1 \in \mathcal{X}_1\}$ and $E^{in}(x_{l'}^1) = \{e \in \mathcal{E}_R : e = (x_l^0, x_{l'}^1) \text{ for some } x_l^0 \in \mathcal{X}_0\}$ represent the sets of edges that depart from x_l^0 and end in $x_{l'}^1$ respectively.

Definition B.2 (Maximum flow problem associated to R). Let $R \subset \mathcal{T}_Z$ be a collection of instrument-response types with associated graph $\mathcal{E}_R$. Let $(\mathcal{X}_0 \cup \mathcal{X}_1, \mathcal{E}_R)$ be the graph from definition B.1. From theorem 7.50 in Kleinberg and Tardos (2006) that any circulation with demands problem is equivalent to a maximum flow problem. The maximum flow problem associated to R is obtained by applying this principle to the associated circulation with demands problem.

The directed graph for the maximum flow problem is defined as follows:

Two vertices are added: a source s^* and a sink t^* to obtain the set of vertices $\mathcal{X}_0 \cup \mathcal{X}_1 \cup \{s^*, t^*\}$.

A directed edge points from the source to every node x_l^1 and from every node $x_{l'}^1$ to the sink:

$$\mathcal{E}_R^{MaxFlow} = \mathcal{E}_R \cup \{(x_l^0, t^*) : x_l^0 \in \mathcal{X}_0\} \cup \{(s^*, x_{l'}^1) : x_{l'}^1 \in \mathcal{X}_1\} \quad (92)$$

The capacity of all edges of the form $e = (x_l^0, t^*)$ is set to $c_e = \mathbb{P}[X = x_l^0 | Z = z_0]$, the capacity of all edges of the form $e = (s^*, x_{l'}^1)$ is set to $c_e = \mathbb{P}[X = x_{l'}^1 | Z = z_1]$. All remaining edges have the same capacities as in definition B.1.

A flow is a function $\tilde{p} : \mathcal{E}_R^{MaxFlow} \rightarrow \mathbb{R}_+$ such that every intermediate node (i.e. all nodes except s^* and t^* has exactly the same inflow and outflow). Formally:

1. For every $x_l^0 \in \mathcal{X}_0$:

$$\sum_{e' \in E^{out}(x_l^0)} \tilde{p}(e') = \tilde{p}((s^*, x_l^0)) \quad (93)$$

2. For every $x_\nu^1 \in \mathcal{X}_1$:

$$\sum_{e' \in E^{in}(x_\nu^1)} \tilde{p}(e') = \tilde{p}((x_\nu^1, t^*)) \quad (94)$$

Let $\tilde{P}$ denote the set of all flows. The problem is:

$$\max_{\tilde{p} \in \tilde{P}} \sum_{x_l^0 \in \mathcal{X}_0} \tilde{p}((s^*, x_l^0)) \quad (95)$$

$$\text{subject to: } \tilde{p}(e) \leq c_e. \quad (96)$$

Definition B.3 ($s^* - t^*$ cut of $(\mathcal{X}_0 \cup \mathcal{X}_1 \cup \{s^*, t^*\}, \mathcal{E}_R^{MaxFlow})$). Let $(\mathcal{X}_0 \cup \mathcal{X}_1 \cup \{s^*, t^*\}, \mathcal{E}_R^{MaxFlow})$ be the graph from definition B.2.

1. A cut $C = (S_C, T_C)$ is a partition (with elements S and T) of the set $\mathcal{X}_0 \cup \mathcal{X}_1 \cup \{s^*, t^*\}$ such that $s^* \in S_C$ and $t^* \in T_C$. I denote the set of all cuts of $(\mathcal{X}_0 \cup \mathcal{X}_1 \cup \{s^*, t^*\}, \mathcal{E}_R^{MaxFlow})$ as $\mathcal{C}(R, \mathcal{X})$.
2. The cut set of $C = (S_C, T_C)$ denoted Ξ_C , is defined as the set of edges that, if removed, would disconnect S_C from T_C , formally:

$$\Xi_C = \{e = (e_1, e_2) \in \mathcal{E}_R^{MaxFlow} \mid e_1 \in S_C \text{ and } e_2 \in T_C\} \quad (97)$$

3. The capacity of a cut C , denoted $Cap(C)$, is defined as the sum of the capacities of the edges in Ξ_C :

$$Cap(C) = \sum_{e \in \Xi_C} c_e \quad (98)$$

4. The minimum cut associated to R is $\max_{C \in \mathcal{C}(R, \mathcal{X})} Cap(C)$

Proposition B.1. The following statements are equivalent:

1. A distribution over instrument response types exists that satisfies assumptions (2.1), (4.1) and $p(x_l, x_\nu) \leq c_e = c_e((x_l^0, x_\nu^1))$ exists.
2. A solution to the circulation with demands problem from definition B.1 exists.

3. The maximum flow attainable in the problem from definition B.2 is equal to 1.
4. The minimum cut attainable in definition B.3 is equal to 1.

Proof of proposition B.1. I prove the equivalence between 1 and 2. The equivalence of 2, 3 and 4 follows from well known results in operations research (See Kleinberg and Tardos (2006) theorems 7.9 and 7.50).

for reference.

2 $\implies$ 1 :

Let $\tilde{p}$ be a flow that satisfies the circulation with demands problem associated to R .

For any instrument-response type $t^Z = (x_l, x_{l'}) \in \mathcal{T}_Z$ define the distribution over instrument-response types p as follows:

$$p(t^Z) = p((x_l, x_{l'})) := \begin{cases} \tilde{p}(e_{tz}) & \text{if } e_{tz} \in \mathcal{E}_R \\ 0 & \text{otherwise} \end{cases} \quad (99)$$

Consider any node x_l^0 :

$$-\mathbb{P}[X = x_l | Z = 0] = d_{x_l^0} \quad (100)$$

$$= - \sum_{e \in E^{out}(x_l^0)} \tilde{p}(e) \quad (101)$$

$$= - \sum_{x_{l'} \in \mathcal{X}} p((x_l, x_{l'})) \quad (102)$$

The first two equalities follow from the definition of the transportation problem, the last from $p(t^Z) = \tilde{p}(e_{tz})$.

An analogous argument shows that for $z = 1$:

$$\mathbb{P}[X = x_l | Z = 1z] = \sum_{x' \in \mathcal{X}} p((x_{l'}, x_l)) \quad (103)$$

Which shows that p is consistent with observed conditional treatment probabilities.

Capacity constraints are trivially satisfied since $\tilde{p}(e_{tz}) \leq c_e = c_e((x_l^0, x_{l'}^1))$.

Finally, from the definition of $\mathcal{E}_R$ it follows that $\tilde{p}$ only assigns non-negative flows to edges in $e \in \mathcal{E}_R$, therefore, p only assigns non-negative probabilities to response types $t^Z \in R$. This shows consistency with assumption 4.1.

1 $\implies$ 2 :

Let p be a distribution over response types that satisfies assumptions (2.1) and (4.1). For every edge $e = (x_l^0, x_{l'}^1)$ define the flow $\tilde{p}$ as follows:

$$\tilde{p}(e) = \tilde{p}((x_l^0, x_{l'}^1)) = p(x_l, x_{l'}) \quad (104)$$

Note that e is always well defined since every response type is assigned non-negative probability.

Now, consider an arbitrary node $x_l^0 \in \mathcal{X}_0$:

$$d_{x_l^0} = -\mathbb{P}[X = x_l | Z = z_0] \quad (105)$$

$$= -\sum_{x_{l'} \in \mathcal{X}} p((x_l, x_{l'})) \quad (106)$$

$$= -\sum_{e \in E^{out}(x_l^0)} \tilde{p}(e) \quad (107)$$

An analogous argument shows that for $x_{l'}^1$:

$$d_{(x,z)} = \sum_{e \in E^{in}(x_{l'}^1)} \tilde{p}(e) \quad (108)$$

Thus, demand is satisfied at every node. Again, capacity constraints are trivially satisfied since $\tilde{p}((x_l^0, x_{l'}^1)) \leq c_e = c_e((x_l^0, x_{l'}^1))$. This completes the proof. $\square$

Proof of theorem 4.1.

Part 1: First note that the restriction $c_{tz} = 1$ simply states that instrument-response probabilities are not bounded from above other than by the requirement that they are less than 1. This is, response types in R can occur with arbitrary frequency in the population.

$\Rightarrow$

$$\mathbb{P}[X \in S | Z = z_0] = p(\{t^Z \in \mathcal{T}^Z : t^Z = (x_l, x_{l'}) \in S \times \mathcal{X}\}) \quad (109)$$

$$= \sum_{x_{l'} \in \mathcal{X}} p(\{t^Z \in \mathcal{T}^Z : t^Z = (x_l, x_{l'}) \in S \times \{x_{l'}\}\}) \quad (110)$$

$$= \sum_{x_{l'} \in \{x_{l'} \in \mathcal{X} : (s, x_{l'}) \in R \text{ and } s \in S\}} (\{t^Z \in \mathcal{T}^Z : t^Z = (x_l, x_{l'}) \in S \times \{x_{l'}\}\}) \quad (111)$$

$$= \sum_{x_{l'} \in L_{\geq R}(S)^C} p(\{t^Z \in \mathcal{T}^Z : t^Z = (x_l, x_{l'}) \in S \times \{x_{l'}\}\}) \quad (112)$$

$$\leq \sum_{x_{l'} \in L_{\geq R}(S)^C} \mathbb{P}[X = x_{l'} | Z = 1] \quad (113)$$

$$= \mathbb{P}[X \in L_{\geq R}(S)^C | Z = z_1] \quad (114)$$

The first equality comes from the consistency of p with observed probabilities, the second from decomposing the probability in terms of all the different response types that realize x_l when $Z = 0$, the third equality comes from the fact that p assigns zero probability to response types outside of R , and the fourth noting that if $(x_l, x_{l'}) \in R$ then $x_l \geq^R x_{l'}$ which implies $x_{l'} \notin L_{\geq R}(S)$. The inequality comes from noting that p is consistent with observed probabilities and $\{x_{l'}\}$ could also be realized when $Z = 1$ by types whose potential outcome when $Z = 0$ lies outside of S . The last equality comes from adding disjoint probabilities across the elements of $L_{\geq R}(S)^C$.

$\Leftarrow$

I prove the counterpositive, this is, if no p that satisfies the hypotheses exists, then, a violation of equation 21 must exist. To this end I use the minimum cut characterization.

First, note that by setting $C = \{e \in \mathcal{E}_R : e = (s^*, x_l^0) \text{ for some } x_l \in \mathcal{X}_0\}$ a cut of capacity one can always be attained. Therefore, edges of the form $(x_l^0, x_{l'}^1)$ will never be part of a minimum cut since their capacity is 1 and removing a single edge will not induce a cut. Thus, minimum cut sets will only contain edges of the form (s^*, x_l^0) or $(x_{l'}^1, t^*)$.

Suppose that C is a minimum cut such that $\text{Cap}(C) \leq 1$. Let S be the set of all $x_l \in \mathcal{X}$ such that the edge (s^*, x_l^0) belongs to Ξ_C . Then, by construction, any node of the form $(s^*, x_{l'}^0)$ for $x_{l'} \in S^C = \mathcal{X} \setminus S'$ does not belong to the cut.

Define $S' := \mathcal{X} \setminus S$ and note that by construction for any $x_{l'} \in S'$ and any $x_{l''} \in \mathcal{X}$ such that $(x_{l'}, x_{l''}) \in R$, the edge $(x_{l''}^1, t^*)$ belongs to C . Otherwise, the path $(s^*, x_{l'}), (x_{l'}, x_{l''}), (x_{l''}, t^*)$ would connect t^* and s^* .

But the set of elements $x_{l''} \in \mathcal{X}$ such that $(x_{l'}, x_{l''}) \in R$ for some $x_{l'} \in S'$ is actually the set $L_{\geq R}(S')^C$.

Therefore:

$$\sum_{c_e \in \Xi_C} c_e < 1 \quad (115)$$

$$\implies \sum_{e \in \mathcal{E}_R \cap \{s^*\} \times S} c_e + \sum_{e \in \mathcal{E}_R \cap L_{\geq R}(S') \times \{t^*\}} c_e < 1 \quad (116)$$

$$\implies \sum_{e \in \mathcal{E}_R \cap L_{\geq R}(S') \times \{t^*\}} c_e < 1 - \sum_{e \in \mathcal{E}_R \cap \{s^*\} \times S} c_e \quad (117)$$

$$\implies \sum_{e \in \mathcal{E}_R \cap L_{\geq R}(S') \times \{t^*\}} c_e < \sum_{e \in \mathcal{E}_R \cap \{s^*\} \times S'} c_e \quad (118)$$

$$\implies \mathbb{P}[X \in L_{\geq R}(S') | Z = z_1] < \mathbb{P}[X \in S' | Z = z_0] \quad (119)$$

The first implication follows from the previous discussion. The second from rearranging terms. The third from $S' = S^C$ and the fourth from the definition of the capacities. The last equation is a violation of equation 21.

Part 2: $\implies$: Suppose that a distribution over instrument-response types $p \in \Delta(\mathcal{T}_Z)$ exists that satisfies assumptions 1 and 3.

For $\emptyset \neq S \subset \mathcal{X}$, let (Λ_S, Λ'_S) be a partition of $L_{\geq R}(S)^C$ such that the following three conditions are satisfied:

1. $\emptyset \neq \Lambda'_S \subset S$.
2. For every $x_{l'} \in \Lambda'_S$ the set $L_{\geq R}(x_{l'})^C \setminus \{x_{l'}\}$ is contained in Λ_S .
3. $L_{\geq R}(S \setminus \Lambda'_S)^C \subset \Lambda_S$.

Take equation 112:

$$\mathbb{P}[X \in S|Z = z_0] = \sum_{x_{l'} \in L_{\geq R}(S)^C} p(\{t^Z \in \mathcal{T}^Z : t^Z = (x_l, x_{l'}) \in S \times \{x_{l'}\}\}) \quad (120)$$

$$= \sum_{x_{l'} \in \Lambda} p(\{t^Z \in \mathcal{T}^Z : t^Z = (x_l, x_{l'}) \in S \times \{x_{l'}\}\}) \quad (121)$$

$$+ \sum_{x_{l'} \in \Lambda'} p(\{t^Z \in \mathcal{T}^Z : t^Z = (x_l, x_{l'}) \in S \times \{x_{l'}\}\})$$

$$= \sum_{x_{l'} \in \Lambda} p(\{t^Z \in \mathcal{T}^Z : t^Z = (x_l, x_{l'}) \in S \times \{x_{l'}\}\}) \quad (122)$$

$$+ \sum_{x_{l'} \in \Lambda'} p(\{t^Z \in \mathcal{T}^Z : t^Z = (x_l, x_{l'}) \in (S \setminus \Lambda'_S) \times \{x_{l'}\}\})$$

$$+ \sum_{x_{l'} \in \Lambda'} p(\{t^Z \in \mathcal{T}^Z : t^Z = (x_l, x_{l'}) \in \Lambda'_S \times \{x_{l'}\}\})$$

The second term from the last equation is equal to zero by condition 3. Condition 2 implies that the only possible value that x_l can take in the last term is $x_{l'}$. Therefore, the equation becomes:

$$\mathbb{P}[X \in S|Z = z_0] = \sum_{x_{l'} \in \Lambda} p(\{t^Z \in \mathcal{T}^Z : t^Z = (x_l, x_{l'}) \in S \times \{x_{l'}\}\}) \quad (123)$$

$$+ \sum_{x_{l'} \in \Lambda'} p(\{t^Z \in \mathcal{T}^Z : t^Z = (x_l, x_l) \text{ such that } x_l \in \Lambda'\})$$

$$\leq \mathbb{P}[X \in \Lambda|Z = z_1] + \sum_{x_l \in \Lambda'} p((x_l, x_l)) \quad (124)$$

$$\leq \mathbb{P}[X \in \Lambda|Z = z_1] + \sum_{x_l \in \Lambda'} \Psi_{x_l} \quad (125)$$

The first inequality follows from consistency of p with observed probabilities and the fact that response types that realize treatments outside of S when $Z = 0$ might also realize treatments in Λ when $Z = 1$ along with the fact that $L_{\geq R}(x_l)^C \setminus \{x_l\} \subset \Lambda$ for $x_l \in \Lambda'$. The second inequality follows from theorem 3.1.

$\Leftarrow$

As for part 1, I prove the contrapositive statement. That is, I suppose that a cut with capacity less than 1 exists and show that it implies that inequality (22) from theorem (4.1) is not satisfied for some $S \subset \mathcal{X}$.

Now the capacities of edges of the form (x_l^0, x_l^1) are potentially binding. For cuts that only contain edges connected to s^* or t^* the argument from part 1 still applies. Edges of the form $(x_l^0, x_{l'}^1)$ for $x_l \neq x_{l'}$ still have capacity 1 and therefore will not belong to a minimum cut that is strictly less than one.

Suppose that C is a minimum cut with at least one element of the form (x_l^0, x_l^1) .

Let S' be the set of elements in $x_l \in \mathcal{X}$ such that $(s^*, x_l^0) \in C$. Define $S = \mathcal{X} \setminus S'$. Note that for any $x_l \in S$ and any $x_{l'} \in L_{\geq R}(x_l)^C$ one of the following sets of conditions must hold:

1. The following holds:

- $(x_{l'}^1, t^*) \in C$

2. The following three conditions hold

- $x_l = x_{l'} \in S \cap L_{\geq R}(S)^C$
- $(x_{l'}^0, x_{l'}^1) \in C$.
- $(x_{l''}^1, t^*) \in C$ for all $x_{l''} \in L_{\geq R}(x_{l'})^C \setminus \{x_{l'}\}$

The first condition prevents paths of the form $(s^*, x_l^0), (x_l^0, x_{l'}^1), (x_{l'}^1, t^*)$ as in part 1.

If condition 1 is not satisfied and $(x_{l'}^1, t^*) \notin C$, then, the path $(s^*, x_l^0), (x_l^0, x_{l'}^1), (x_{l'}^1, t^*)$ must be cut at the middle at edge $(x_l^0, x_{l'}^1)$. Because edges associated to $x_l \neq x_{l'}$ cannot belong to the cut, it must be that $x_l = x_{l'} \in S' \cap L_{\geq R}(S')^C$. Furthermore, for every $x_{l''} \in L_{\geq R}(x_l)^C \setminus \{x_l\}$, the path $(s^*, x_l^0), (x_l^0, x_{l''}^1), (x_{l''}^1, t^*)$ must be blocked at some point. By construction, this can only happen at $(x_{l''}^1, t^*)$, thus, $(x_{l''}^1, t^*) \in C$.

In other words, a path $(s^*, x_l^0), (x_l^0, x_{l'}^1), (x_{l'}^1, t^*)$ might be blocked by condition 1, if not, then, conditions 2 must hold.

Let Λ'_S be the set $\Lambda'_S = \{x_l \in \mathcal{X} : (x_l^0, x_l^1) \in C\} \subset L_{\geq R}(S)^C \cap S$. Set $\Lambda_S = L_{\geq R}(S)^C \setminus \Lambda'_S$. By construction the sets Λ'_S and Λ_S partition $L_{\geq R}(S)^C$.

Note that $\{x_{l'} \in \mathcal{X} : (x_{l'}^1, t^*) \in C\}$ must be a subset of $L_{\geq R}(S)^C$, otherwise, $x_{l'}$ would not be reachable from $\{x_l^0 : x_l \in S\}$ and eliminating $(x_{l'}^1, t^*)$ from C would attain a cut with strictly lower capacity. Thus, $\{x_{l'} \in \mathcal{X} : (x_{l'}^1, t^*) \in C\} \subset \Lambda_S$.

Similarly, note that if $x_{l'} \in \Lambda_S$ it must be that $x_l \in \{x_{l'} \in \mathcal{X} : (x_{l'}^1, t^*) \in C\}$, otherwise, a path through some $x_l \in S$ such that $L_{\geq R}(x_l)^C \cap \Lambda_S \neq \emptyset$ to $x_{l'}$ would

connect s^* to t^* . Hence:

$$\{x_{l'} \in \mathcal{X} : (x_{l'}, t^*) \in C\} = \Lambda \quad (126)$$

Finally, suppose that for some $x_l \in S \setminus \Lambda'_S$ exists some $x_{l'} \in L_{\geq R}(x_l)^C \cap \Lambda'_S \neq \emptyset$. By definition it must be that $x_{l'} \neq x_l$. Then, the path $(s^*, x_l), (x_l, x_{l'}), (x_{l'}, t^*)$ would connect s^* to t^* . Hence, it must be that:

$$L_{\geq R}(S \setminus \Lambda'_S)^C \subset \Lambda_S \quad (127)$$

Then:

- $\emptyset \neq \Lambda' \subset S$ (By conditions 2).
- For all $x_{l'} \in \Lambda'$, $L_{\geq R}(x_{l'})^C \setminus \{x_{l'}\} \subset \Lambda$ (By conditions 2 and equation (126)).
- $L_{\geq R}(S \setminus \Lambda'_S)^C \subset \Lambda_S$ by expression (127).

Moreover, it must be that:

$$\sum_{c_e \in \Xi_C} c_e < 1 \quad (128)$$

$$\Rightarrow \sum_{c \in C \cap \{s^*\} \times S'} c_e + \sum_{\{c \in C : c=(x_l^0, x_l^1) \text{ s.t. } x_l \in \Lambda'\}} c_e + \sum_{\{c \in C : c=(x_l^1, t^*) \text{ s.t. } x_l \in \Lambda\}} c_e < 1 \quad (129)$$

$$\Rightarrow \sum_{\{c \in C : c=(x_l^0, x_l^1) \text{ s.t. } x_l \in \Lambda'\}} c_e + \sum_{\{c \in C : c=(x_l^1, t^*) \text{ s.t. } x_l \in \Lambda\}} c_e < 1 - \sum_{c \in C \cap \{s^*\} \times S'} c_e \quad (130)$$

$$\Rightarrow \sum_{\{c \in C : c=(x_l^0, x_l^1) \text{ s.t. } x_l \in \Lambda'\}} c_e + \mathbb{P}[X \in \Lambda | Z = 1] < \mathbb{P}[X \in S | Z = z_0] \quad (131)$$

Where the first implication follows from the previous discussion, the second from rearranging terms and the last from the definition of the capacities.

Setting $c_e((x_l^0, x_{l'}^1)) = \Psi_{x_l}$ completes the proof. □

B.2 Proofs for subsection 4.2

To prove theorem 4.2 I begin by formally defining latent preference types. To this end, let $\mathcal{L}_s(\mathcal{X})$ denote the set of all strict preference relations over $\mathcal{X}$ and let $\mathcal{N}(x_l) =$

$\{\succ \in \mathcal{L}_s(\mathcal{X}) : x_l \succ x_{l'} \text{ for all } x_{l'} \in \mathcal{X}\}$ denote the subset of strict preference relations that has x_l as a maximizer.

Definition B.4 (Latent preference types). *A latent preference type $t^\succ \in \mathcal{T}_\succ$ is a function that maps instrument realizations to strict preference profiles: $t^\succ : \mathcal{Z} \rightarrow \mathcal{L}_s(\mathcal{X})$.*

Note that $t^\succ(z)$ is a preference relation. Therefore, the equation $x_l t^\succ(z) x_{l'}$ reads as “latent preference types $t^\succ$ strictly prefer x_l over $x_{l'}$ when $Z = z$ ”. To improve readability, I denote $t^\succ(z)$ as $\succ^{t(z)}$ so that the previous equation becomes $x_l \succ^{t(z)} x_{l'}$.

Definition B.5 (Distribution over latent preference types). *A distribution over latent preference types is a function $h : \mathcal{H} \rightarrow [0, 1]$ that satisfies:*

$$h(t^\succ) \geq 0 \text{ for all } t^\succ \in \mathcal{T}_\succ \quad (132)$$

$$\sum_{t^\succ \in \mathcal{T}_\succ} h(t^\succ) = 1 \quad (133)$$

Remark B.1. *Let $h : \mathcal{T}_\succ \rightarrow [0, 1]$ be a distribution over latent preference types. The distribution over response types implied by h is the function $p_h : \Omega \rightarrow [0, 1]$ defined as:*

$$p_h(x_l, x_{l'}) = \sum_{t^\succ \in N(x_l, x_{l'})} h(t^\succ) \quad (134)$$

Where $N(x_l, x_{l'}) = \{t^\succ \in \mathcal{T}_\succ | t^\succ(0) \in \mathcal{N}(x_l) \text{ and } t^\succ(1) \in \mathcal{N}(x_{l'})\}$.

Definition B.6. *A distribution over latent preference types h is $\geq^*$ -Submonotone if $h(t^\succ) = 0$ for every $t^\succ$ that violates point 3 of assumption 4.2. This is, types $t^\succ$ such that $x_l \geq^* x_{l'}$ and $x_l \succ^{t(z_0)} x_{l'}$ but $x_{l'} \succ^{t(z_1)} x_l$ are not allowed.*

Assumption B.1 (No S always takers). *Let $S \subset \mathcal{X}$ and $R \subset \mathcal{T}_Z$ a collection of instrument-response types. R has no S always takers if $(x_l, x_l) \notin R$ for all $x_l \in S$.*

Lemma B.1. *Let h be a distribution over latent preference types. Let R be a collection of instrument-response types. Let $\geq^*$ be a binary relation (not necessarily transitive or complete) over $\mathcal{X}$. Let $\text{Ref}_{\geq^*} = \{x_l \in \mathcal{X} | x_l \geq^* x_l\}$ denote the elements that belong to the reflexive part of $\geq^*$. Let $\geq_{\text{Ref}}^*$ and $\geq_{\text{Irr}}^*$ denote the reflexive and irreflexive part of $\geq^*$ respectively. Then:*

1. h satisfies $\geq_{Irr}^*$ -Submonotonicity if and only if h satisfies $\geq^*$ -Submonotonicity.
2. If $\geq_{Irr}^* \setminus \geq^R \neq \emptyset$, then, a distribution over latent preference types h exists such that h satisfies $\geq^R$ -Submonotonicity and $p_h(t^Z) = 0$ for all $t^Z \in R$ but h is not $\geq^*$ -Submonotone.
3. If $\geq^R \setminus \geq_{Irr}^* \neq \emptyset$, then, a distribution over latent preference types h exists that is $\geq^*$ -Submonotone but $p_h(t^Z) > 0$ for some $t^Z \in R$.
4. If $\geq_{Ref}^R \neq \emptyset$ a distribution over latent response types exists that is $\geq^R$ -Submonotone but $p_h(t^Z) > 0$ for some $t^Z \in R$.
5. If $\geq^R$ -Submonotonicity holds and no $R_{\geq M}$ always takers holds, then, $p(t^Z) = 0$ for all $t^Z \notin R$.
6. If a distribution over response types p that satisfies equation ?? and $p(t^Z) = 0$ for all $t^Z \in R$ exists, then, a distribution h over latent preference types exists such that h is $\geq^R$ -Submonotone and $p = p_h$. This last equality implies that h is consistent with the data and that if p satisfies no S always takers, so does p_h .

Proof of lemma B.1.

Part 1:

Suppose $x_l \geq^* x_{l'}$ and $x_l \succ_1^{(0)} x_{l'}$. If $x_l = x_{l'}$, then $x_l \succ_1^{(1)} x_{l'}$ holds trivially, otherwise, $\geq_{Irr}^* -QM$ implies that $x_l \succ_1^{(1)} x_{l'}$. The converse is trivial since $\geq_{Irr}^* \subset \geq^*$.

Part 2:

Take the ordering induced by the indexing of the elements of $\mathcal{X}$. Let $\succ^{x_l, x_{l'}, *}$ be the preference relation that ranks x_l as the best alternative, $x_{l'}$ as the second best, and all remaining alternatives according to their index: $x_l \succ^{x_l, x_{l'}, *} x_{l'} \succ^{x_l, x_{l'}, *} x_1 \succ^{x_l, x_{l'}, *} \dots \succ^{x_l, x_{l'}, *} x_{L-1}$.

Suppose that $x_l \neq x_{l'}$ and $(x_l, x_{l'}) \in \geq_{Irr}^* \setminus \geq^R$. Let $t^\succ$ be the unique latent preference type that satisfies $t^\succ(0) = \succ^{x_l, x_{l'}, *}$ and $t^\succ(1) = \succ^{x_{l'}, x_l, *}$. Define h as:

$$h(t^\succ) = 1 \tag{135}$$

$$h(t^{\succ'}) = 0 \quad \text{for all } t^{\succ'} \neq t^\succ \tag{136}$$

Note that $(x_l, x_{l'}) \notin \geq^R$ implies that h is $\geq^R$ -Submonotone, but h is not $\geq^*$ -Submonotone (since preference reversals between x_l and $x_{l'}$ occur with positive

probability). Moreover, note that p_h assigns zero probability to response types outside of R .

Part 3:

If $Ref_{\geq R}$ is non-empty, a reasoning analogous to part 1 proves the claim. Otherwise, pick some $x_l \in Ref_{\geq^*}$ and define h as follows:

$$h((\succ^{x_l,*}, \succ^{x_l,*})) = 1 \quad (137)$$

$$h(t^Z) = 0 \quad \text{for all } t^Z \neq (\succ^{x_l,*}, \succ^{x_l,*}) \quad (138)$$

h is $\geq^*$ -QM but p_h assigns positive probability to the response type $(x_l, x_l) \notin R$.

Part 4

This part follows immediately from part 3 by setting $\geq_{Irr}^* = \emptyset$.

Part 5:

Let R' be the set of response types allowed by $\geq^R$ -QM and no stayers at $Ref_{\geq R}$. The goal is to show that $R \subset R'$.

Suppose that $(x_l, x_{l'}) \notin R$. This implies that $x_l \geq^R x_{l'}$.

If $x_l \neq x_{l'}$, consider a decision maker i such that $X(0) = x_{l'}$, then, rationality implies that $x_{l'} = X(0) \succ^{(0)} x_l$. Therefore, by $\geq^R$ -Submonotonicity, it must be that $x_{l'} \succ^{(1)} x_l$. This implies that $x_{l'}$ is not chosen by decision maker i when $Z = 1$ a.s.. Therefore $(x_l, x_{l'}) \notin R'$.

If $x_{l'} = x_l$, then, no stayers at $Ref_{\geq R}$ implies that $(x_l, x_{l'}) \notin R'$.

Therefore, $t^Z \notin R \implies t^Z \notin R'$ which is equivalent to $t^Z \in R' \implies t^Z \in R$.

Part 6:

Let $t_{x_l, x_{l'},*}^\succ$ be the latent preference type such that $t_{x_l, x_{l'},*}^\succ(0) = \succ^{x_l, x_{l'},*}$ and $t_{x_l, x_{l'},*}^\succ(1) = \succ^{x_{l'}, x_l,*}$. Define h as follows:

$$h(t^\succ) = \begin{cases} p(x_l, x_{l'}) & \text{if } t^\succ = t_{x_l, x_{l'},*}^\succ \\ 0 & \text{otherwise} \end{cases} \quad (139)$$

Note that h is $\geq^R$ -QM and that for all $(x_l, x'_l) \in R_Z$:

$$p_h((x_l, x'_l)) = \sum_{t^\succ \in N(x_l, x'_l)} h(t^\succ) = h(t^\succ_{x_l, x'_l, *}) = p(x_l, x'_l) \quad (140)$$

□

Now I comment Lemma 1. The first part shows that only the irreflexive part of binary relations is relevant for submonotonicity. Part 2 shows that when a binary relation $\geq^*$ is stronger than the one induced by R , restricting response types to R does not guarantee $\geq^*$ -submonotonicity. Parts 3 and 4 show the converse, that imposing weaker than $\geq^R$ does not guarantee that response types outside of R are ruled out. Part 5 shows that R can be guaranteed by imposing $\geq^R$ -submonotonicity and ruling out a specific set of always takers. 6 shows that $\geq^R$ -submonotonicity is only falsified if the response type restriction associated to R is falsified. Thus, in the binary instrument case $\geq^R$ -submonotonicity is just an interpretation for response type restrictions but yields no additional identifying power.

Theorem 4.2 follows immediately from lemma 6 by noting there is a unique correspondence between R and $\geq^R$ and therefore their cardinalities.

C Proofs for section

C.1 Estimation algorithm 1

1. Partition the sample in two splits: S_1 and S_2 of size n_1 and n_2 respectively.
2. Using the first sample, train a non-parametric classifier $\eta_{n_1}^*(y, x)$ as follows:

(a) Define:

$$\phi_{z_k, n}(y, x_l) := P_{k, n_1}^{-1} \left[\frac{1}{n_1 h_n^d} \sum_{i=1}^{n_1} \mathbf{1}\{X_i = x_l, Z_i = z_k\} K_n(y - Y_i) \right] \quad (141)$$

Where $n_{k,1} = |\{i \in S_1 : Z_i = z_k\}|$ and $P_{k, n_1} = n_{k,1}/n_1$.

(b) Then, define:

$$\Delta_{n_1}(y, x_l) := \phi_{z_1, n}(y, x_l) - \phi_{z_0, n}(y, x_l) \quad (142)$$

(c) And:

$$\eta_{n_1}^*(y, x_l) := \mathbf{1}\{\Delta_{n_1}(y, x_l) \geq 0\}. \quad (143)$$

3. In the second sample, estimate the probabilities $\mathbb{P}[Z = z_k]$ for $k = 0, 1$. Denote the estimators as P_{k, n_2} .
4. In the second sample, evaluate the weighted empirical loss:

$$L_{n_2}(\eta_{n_1}^*, x_l) := \frac{1}{n_2} \sum_{i=1}^{n_2} \left(\frac{\mathbf{1}\{\eta_{n_1}^*(Y_i, X_i) = 1, Z_i = z_0, X_i = x_l\}}{P_{0, n_2}} + \frac{\mathbf{1}\{\eta_{n_1}^*(Y_i, X_i) = 0, Z_i = z_1, X_i = x_l\}}{P_{1, n_2}} \right) \quad (144)$$

C.2 Proofs for subsection 5.1

Decompose:

$$\sqrt{n_2}(L_{n_2}(\eta_{n_1}^*, x_l) - \Psi_{x_l}) = \underbrace{\sqrt{n_2}(L_{n_2}(\eta_{n_1}^*, x_l) - L(\eta_{n_1}^*, x_l))}_{:= A_{1, n}(x_l)} + \underbrace{\sqrt{n_2}(L(\eta_{n_1}^*, x_l) - L(\eta^*, x_l))}_{:= A_{2, n}(x_l)} \quad (145)$$

And:

$$n_2 \text{Var}\left(L_{n_2}(\eta_{n_1}^*, x_l) - \Psi_{x_l}\right) = \underbrace{n_2 \text{Var}\left(L_{n_2}(\eta_{n_1}^*, x_l) - L(\eta_{n_1}^*, x_l)\right)}_{:=V_1} + \underbrace{n_2 \text{Var}\left(L(\eta_{n_1}^*, x_l) - L(\eta^*, x_l)\right)}_{:=V_2} \quad (146)$$

$$+ \underbrace{2n_2 \text{Cov}\left(L_{n_2}(\eta_{n_1}^*, x_l) - L(\eta_{n_1}^*, x_l), L(\eta_{n_1}^*, x_l) - L(\eta^*, x_l)\right)}_{:=V_3} \quad (147)$$

The first term corresponds to how well the empirical loss evaluated in the sub-sample S_2 approximates the expected loss of the classifier $\eta_{n_1}^*$, the second term corresponds to the loss difference between the estimated classifier $\eta_{n_1}^*$ and the optimal unfeasible classifier η^* .

Propositions C.1 shows that, under assumptions 5.1 and 5.2, the classifier trained in sub-sample S_1 is sufficiently good so that its expected missclassification loss (and its variance) are small relative to the expected loss of the optimal classifier. Thus, the asymptotic behavior of the estimator is governed by the term $A_{1,n}(x_l)$.

Proposition C.1.

1. $A_{2,n}(x_l) = o_p(1)$.
2. $\text{Cov}(A_{1,n}(x_l), A_{2,n}(x_l)) = 0$.
3. $\text{Var}(A_{2,n}(x_l)) = o(1)$.

Proof of proposition 5.1.

$$L(\eta^*, x_l) = \frac{\mathbb{P}[\eta^*(Y, X) = 0, Z = z_1, X = x_l]}{\mathbb{P}[Z = z_1]} + \frac{\mathbb{P}[\eta^*(Y, X) = 1, Z = z_0, X = x_l]}{\mathbb{P}[Z = z_0]} \quad (148)$$

$$= \mathbb{P}[X = x_l] \left(\frac{\mathbb{P}[\eta^*(Y, X) = 0, Z = z_1 | X = x_l]}{\mathbb{P}[Z = z_1]} + \frac{\mathbb{P}[\eta^*(Y, X) = 1, Z = z_0 | X = x_l]}{\mathbb{P}[Z = z_0]} \right) \quad (149)$$

$$= \mathbb{P}[X = x_l] \int_{\mathcal{Y}} \left(\frac{\mathbb{P}[\eta^*(y, x_l) = 0, Z = z_1 | Y = y, X = x_l]}{\mathbb{P}[Z = z_1]} \right. \quad (150)$$

$$\left. + \frac{\mathbb{P}[\eta^*(y, x_l) = 1, Z = z_0 | Y = y, X = x_l]}{\mathbb{P}[Z = z_0]} \right) f_{Y|X=x_l}(y) dy$$

$$= \mathbb{P}[X = x_l] \int_{\mathcal{Y}} \left(\mathbf{1}\{\eta^*(y, x_l) = 0\} \frac{\mathbb{P}[Z = z_1 | Y = y, X = x_l]}{\mathbb{P}[Z = z_1]} \right. \quad (151)$$

$$\left. + \mathbf{1}\{\eta^*(y, x_l) = 1\} \frac{\mathbb{P}[Z = z_0 | Y = y, X = x_l]}{\mathbb{P}[Z = z_0]} \right) f_{Y|X=x_l}(y) dy \quad (152)$$

$$= \mathbb{P}[X = x_l] \int_{\mathcal{Y}} \left(\mathbf{1}\{\eta^*(y, x_l) = 0\} \frac{f(y, x_l, z_1)}{\mathbb{P}[Z = z_1] f(y, x_l)} \right. \quad (153)$$

$$\left. + \mathbf{1}\{\eta^*(y, x_l) = 1\} \frac{f(y, x_l, z_0)}{\mathbb{P}[Z = z_0] f(y, x_l)} \right) f(y, x_l) dy \quad (154)$$

$$= \int_{\mathcal{Y}} \left(\mathbf{1}\{\eta^*(y, x_l) = 0\} f_{Y,X|Z=z_1}(y, x_l) + \mathbf{1}\{\eta^*(y, x_l) = 1\} f_{Y,X|Z=z_0}(y, x_l) \right) dy \quad (155)$$

$$= \Psi_{x_l} \quad (156)$$

□

Proof of proposition C.1.

Proof of part 1: Consider the difference between the loss of the estimated and target classifier:

$$A_{2,n}(x_l) = L(\eta_{n_1}^*, x_l) - L(\eta^*, x_l) \quad (157)$$

$$\begin{aligned} &= \int_{\mathcal{Y}} \left(\mathbf{1}\{\eta_{n_1}^*(y, x_l) = 0\} f_{Y,X|Z=z_1}(y, x_l) + \mathbf{1}\{\eta_{n_1}^*(y, x_l) = 1\} f_{Y,X|Z=z_0}(y, x_l) \right) \\ &\quad - \left(\mathbf{1}\{\eta^*(y, x_l) = 0\} f_{Y,X|Z=z_1}(y, x_l) + \mathbf{1}\{\eta^*(y, x_l) = 1\} f_{Y,X|Z=z_0}(y, x_l) \right) dy \end{aligned} \quad (158)$$

$$= \int_{\mathcal{Y}} \mathbf{1}\{\eta_{n_1}^*(y, x_l) \neq \eta^*(y, x_l)\} |\Delta(y, x_l)| dy \quad (159)$$

Where $\Delta(y, x_l) := f_{Y,X|Z=z_1}(y, x_l) - f_{Y,X|Z=z_0}(y, x_l)$.

Partition the support of the outcome as follows:

$$\mathcal{Y} = \mathcal{S}_{x_l, n_1} \cup \mathcal{M}_{x_l, n_1} \cup \mathcal{C}_{x_l} \quad (160)$$

Where:

$$\mathcal{S}_{x_l, n_1} = \left\{ y \in \mathcal{Y} : |\Delta(y, x_l)| > C_0 \sqrt{\frac{\log(h_n^{-1})}{n_1 h_n^d}} \right\} \quad (161)$$

$$\mathcal{M}_{x_l, n_1} = \left\{ y \in \mathcal{Y} : 0 < |\Delta(y, x_l)| \leq C_0 \sqrt{\frac{\log(h_n^{-1})}{n_1 h_n^d}} \right\} \quad (162)$$

$$\mathcal{C}_{x_l} = \left\{ y \in \mathcal{Y} : |\Delta(y, x_l)| = 0 \right\} \quad (163)$$

Then, equation 159 becomes:

$$\begin{aligned} A_{2,n}(x_l) &= \int_{\mathcal{S}_{x_l, n_1}} \left(\mathbf{1}\{\eta^*(y, x_l) \neq \eta_{n_1}^*(y, x_l)\} |\Delta(y, x_l)| \right) dy \\ &\quad + \int_{\mathcal{M}_{x_l, n_1}} \left(\mathbf{1}\{\eta^*(y, x_l) \neq \eta_{n_1}^*(y, x_l)\} |\Delta(y, x_l)| \right) dy \\ &\quad + \int_{\mathcal{C}_{x_l}} \left(\mathbf{1}\{\eta^*(y, x_l) \neq \eta_{n_1}^*(y, x_l)\} |\Delta(y, x_l)| \right) dy \end{aligned} \quad (164)$$

Now I show that $A_{2,n}(x_l) = o_p(n^{-\frac{1}{2}})$. For the first term I use the classic results on rates of uniform convergence for kernel density estimators from Audibert and

Tsybakov and Talagrand to prove that, when the sub-densities are well separated, missclassification occurs with low probability. For the second term I invoke the margin condition from assumption 5.1.3 combined with the fact that miss-classification loss near the boundary of the contact set is small. The third term is exactly zero since $|\Delta(y, x_l)| = 0$ for every $y \in \mathcal{C}_{x_l}$.

Consider the term associated with $\mathcal{S}_{x_l, n_1}$:

$$\int_{\mathcal{S}_{x_l, n_1}} \left(\mathbf{1}\{\eta^*(y, x_l) \neq \eta_{n_1}^*(y, x_l)\} |\Delta(y, x_l)| \right) dy \quad (165)$$

$$\begin{aligned} &\leq \int_{\mathcal{S}_{x_l, n_1}} \mathbf{1}\{\eta^*(y, x_l) \neq \eta_{n_1}^*(y, x_l)\} \left(f_{Y, X|Z=z_1}(y, x_l) + f_{Y, X|Z=z_0}(y, x_l) \right) dy \\ &\leq B^{-1} \int_{\mathcal{S}_{x_l, n_1}} \mathbf{1}\{\eta^*(y, x_l) \neq \eta_{n_1}^*(y, x_l)\} f_Y(y) dy \end{aligned} \quad (166)$$

$$= B^{-1} \mathbb{P}[\eta^*(Y, x_l) \neq \eta_{n_1}^*(Y, x_l), Y \in \mathcal{S}_{x_l, n_1} | S_1] \quad (167)$$

Where the first line follows from the triangle inequality, the second from the definition of $f_{Y, X|Z=z_k}$ combined with assumption 5.1.1, and the third is a standard property of indicator expectations.

Note that for any $y \in \mathcal{S}_{x_l, n_1}$, $|\Delta(y, x_l)| > C_0 \sqrt{\frac{\log(h_n^{-1})}{n_1 h_n^d}}$. Hence, if $\|\Delta_{n_1}(\cdot, x_l) - \Delta(\cdot, x_l)\|_\infty \leq r_{n_1}$, then $\Delta_{n_1}(y, x_l)$ has the same sign as $\Delta(y, x_l)$ for every $y \in \mathcal{S}_{x_l, n_1}$. Therefore:

$$\mathbb{P}[\eta^*(Y, x_l) \neq \eta_{n_1}^*(Y, x_l), Y \in \mathcal{S}_{x_l, n_1} | S_1] \leq \mathbf{1}\left\{ \|\Delta_{n_1}(\cdot, x_l) - \Delta(\cdot, x_l)\|_\infty > C_0 \sqrt{\frac{\log(h_n^{-1})}{n_1 h_n^d}} \right\}. \quad (168)$$

By Lemma C.3, the right-hand side is $o_p(n^{-1})$. Hence

$$\mathbb{P}[\eta^*(Y, x_l) \neq \eta_{n_1}^*(Y, x_l), Y \in \mathcal{S}_{x_l, n_1} | S_1] = o_p(n^{-1}). \quad (169)$$

Combining 167 and 169 yields:

$$\int_{\mathcal{S}_{x_l, n_1}} \left(\mathbf{1}\{\eta^*(y, x_l) \neq \eta_{n_1}^*(y, x_l)\} |\Delta(y, x_l)| \right) dy = o_p(n^{-1}) \quad (170)$$

Now consider the term associated with $\mathcal{M}_{x_l, n_1}$. Assumption ?? implies that the conditions from the first part of lemma C.4 is satisfied for $r_n = \sqrt{\frac{\log(h_n^{-1})}{n_1 h_n^d}}$ and $\mathcal{M}_{x_l}(r_n) = \mathcal{M}_{x_l, n_1}$. Therefore:

$$\int_{\mathcal{M}_{x_l, n_1}} \left(\mathbf{1}\{\eta^*(y, x_l) \neq \eta_{n_1}^*(y, x_l)\} |\Delta(y, x_l)| \right) dy = o_p(n^{-1/2}) \quad (171)$$

Thus, $A_{2,n}(x_l) = o_p(n^{-1/2})$. This completes the proof of part 1.

Proof of part 2:

Lemma C.5 implies that:

$$\text{Cov}\left(L_{n_2}(\eta_{n_1}^*, x_l), L(\eta_{n_1}^*, x_l)\right) = \mathbb{E}\left[L_{n_2}(\eta_{n_1}^*, x_l)L(\eta_{n_1}^*, x_l)\right] - \mathbb{E}\left[L_{n_2}(\eta_{n_1}^*, x_l)\right]\mathbb{E}\left[L(\eta_{n_1}^*, x_l)\right] \quad (172)$$

$$= \text{Var}\left(L(\eta_{n_1}^*, x_l)\right) \quad (173)$$

Then:

$$\text{Cov}\left(L_{n_2}(\eta_{n_1}^*, x_l) - L(\eta_{n_1}^*, x_l), L(\eta_{n_1}^*, x_l) - L(\eta^*, x_l)\right) \quad (174)$$

$$= \text{Cov}\left(L_{n_2}(\eta_{n_1}^*, x_l), L(\eta_{n_1}^*, x_l)\right) - \text{Var}\left(L_{n_2}(\eta_{n_1}^*, x_l)\right) \quad (175)$$

$$= 0$$

Proof of part 3:

The argument replicates the idea from part 1, but now applied to the variance. In the contact set, any classification decision yields the exact same loss, therefore, even if the classifier is random inside the contact set, the loss associated to such randomness is deterministic. On a shrinking region near the boundary of the contact set, classification mistakes occur with non-negligible probability. Nonetheless, the difference in loss between the optimal classifier and a random classifier is small and so is the variance of such loss, furthermore, the margin condition implies that such region shrinks at a sufficiently fast rate. Finally, far from the closure of the contact set, missclassification events are unlikely.

First note that:

$$Var(A_{2,n}(x_l)) = Var\left(\int_{\mathcal{Y}} \mathbf{1}\{\eta_{n_1}^*(y, x_l) \neq \eta^*(y, x_l)\} |\Delta(y, x_l)| dy\right) \quad (176)$$

$$= Cov\left(\int_{\mathcal{Y}} \mathbf{1}\{\eta_{n_1}^*(y, x_l) \neq \eta^*(y, x_l)\} |\Delta(y, x_l)| dy, \quad (177)$$

$$\int_{\mathcal{Y}} \mathbf{1}\{\eta_{n_1}^*(y', x_l) \neq \eta^*(y', x_l)\} |\Delta(y', x_l)| dy'\right) \\ = \int_{\mathcal{Y}} \int_{\mathcal{Y}} Cov(\mathbf{1}\{\eta_{n_1}^*(y, x_l) \neq \eta^*(y, x_l)\} |\Delta(y, x_l)|, \quad (178)$$

$$\mathbf{1}\{\eta_{n_1}^*(y', x_l) \neq \eta^*(y', x_l)\} |\Delta(y', x_l)|) dy dy' \\ = \int_{\mathcal{Y}} \int_{\mathcal{Y}} \left(\mathbb{E}[\mathbf{1}\{\eta_{n_1}^*(y, x_l) \neq \eta^*(y, x_l)\} \mathbf{1}\{\eta_{n_1}^*(y', x_l) \neq \eta^*(y', x_l)\}] \quad (179) \right. \\ \left. - \mathbb{E}[\mathbf{1}\{\eta_{n_1}^*(y, x_l) \neq \eta^*(y, x_l)\}] \mathbb{E}[\mathbf{1}\{\eta_{n_1}^*(y', x_l) \neq \eta^*(y', x_l)\}] \right) \\ \times |\Delta(y, x_l)| |\Delta(y', x_l)| dy dy'$$

$$= \int_{\mathcal{Y}} \int_{\mathcal{Y}} \left(\mathbb{P}[\eta_{n_1}^*(y, x_l) \neq \eta^*(y, x_l), \eta_{n_1}^*(y', x_l) \neq \eta^*(y', x_l)] \quad (180) \right.$$

$$\left. - \mathbb{P}[\eta_{n_1}^*(y, x_l) \neq \eta^*(y, x_l)] \mathbb{P}[\eta_{n_1}^*(y', x_l) \neq \eta^*(y', x_l)] \right) \\ \times |\Delta(y, x_l)| |\Delta(y', x_l)| dy dy' \quad (181)$$

$$= \int_{\mathcal{Y}} \int_{\mathcal{Y}} \mathbf{C}_P(y, x_l) dy dy' \quad (182)$$

$$\leq \int_{\mathcal{Y}} \int_{\mathcal{Y}} |\mathbf{C}_P(y, x_l)| dy dy' \quad (183)$$

Where:

$$C_P(y, y', x_l) := \mathbb{P}[\eta_{n_1}^*(y, x_l) \neq \eta^*(y, x_l), \eta_{n_1}^*(y', x_l) \neq \eta^*(y', x_l)] \quad (184) \\ - \mathbb{P}[\eta_{n_1}^*(y, x_l) \neq \eta^*(y, x_l)] \mathbb{P}[\eta_{n_1}^*(y', x_l) \neq \eta^*(y', x_l)] |\Delta(y, x_l)| |\Delta(y', x_l)|$$

Now decompose 183 as follows:

$$\begin{aligned}
Var(A_{2,n}(x_l)) &\leq \int_{\mathcal{S}_{x_l,n_1}} \int_{\mathcal{Y}_{x_l,n_1}} |C_P(y, y', x_l)| dy dy' + \int_{\mathcal{M}_{x_l,n_1}} \int_{\mathcal{M}_{x_l,n_1}} |C_P(y, y', x_l)| dy dy' \\
&\quad + \int_{\mathcal{M}_{x_l,n_1}} \int_{\mathcal{C}_{x_l}} |C_P(y, y', x_l)| dy dy' + \int_{\mathcal{C}_{x_l}} \int_{\mathcal{C}_{x_l}} |C_P(y, y', x_l)| dy dy'
\end{aligned} \tag{185}$$

Where the sets $\mathcal{S}_{x_l,n_1}$, $\mathcal{M}_{x_l,n_1}$ and $\mathcal{C}_{x_l,n_1}$ are as defined in part 1 of the proof.

All the terms that involve integrals over $\mathcal{C}_{x_l,n_1}$ are exactly zero, thus:

$$\begin{aligned}
Var(A_{2,n}(x_l)) &\leq \int_{\mathcal{S}_{x_l,n_1}} \int_{\mathcal{Y}_{x_l,n_1}} |C_P(y, y', x_l)| dy dy' + \int_{\mathcal{M}_{x_l,n_1}} \int_{\mathcal{M}_{x_l,n_1}} |C_P(y, y', x_l)| dy dy'
\end{aligned} \tag{186}$$

Let $M(y, x_l)$ denote the event $\{\eta_{n_1}^*(y, x_l) \neq \eta^*(y, x_l)\}$. Observe that the term $\mathbb{P}[\eta_{n_1}^*(y, x_l) \neq \eta^*(y, x_l), \eta_{n_1}^*(y', x_l) \neq \eta^*(y', x_l)] - \mathbb{P}[\eta_{n_1}^*(y, x_l) \neq \eta^*(y, x_l)] \mathbb{P}[\eta_{n_1}^*(y', x_l) \neq \eta^*(y', x_l)]$ is the covariance of the random variables $v_{y,x_l} := M(y, x_l)$ and $v_{y',x_l} := M(y', x_l)$ that indicate missclassification events at fixed values of Y and X with respect to the distribution of sub-sample S_1 .

Thus, Cauchy-Schwarz's inequality implies:

$$|Cov(v_{y,x_l}, v_{y',x_l})| \leq Var(v_{y,x_l}) Var(v_{y',x_l}) \tag{187}$$

$$\begin{aligned}
&= (\mathbb{P}[M(y, x_l)] - \mathbb{P}[M(y, x_l)]^2) (\mathbb{P}[M(y', x_l)] - \mathbb{P}[M(y', x_l)]^2) \\
&\tag{188}
\end{aligned}$$

$$\leq \mathbb{P}[M(y, x_l)] \mathbb{P}[M(y', x_l)] \tag{189}$$

Therefore:

$$\int_{\mathcal{S}_{x_l, n_1}} \int_{\mathcal{Y}_{x_l, n_1}} |C_P(y, y', x_l)| dy dy' \leq \int_{\mathcal{S}_{x_l, n_1}} \int_{\mathcal{Y}_{x_l, n_1}} \mathbb{P}[M(y, x_l)] \mathbb{P}[M(y', x_l)] |\Delta(y, x_l)| |\Delta(y', x_l)| dy dy' \quad (190)$$

$$\leq B \int_{\mathcal{S}_{x_l, n_1}} \mathbb{P}[M(y, x_l)] dy \quad (191)$$

$$\leq B \mathbb{P} \left[\|\Delta_{n_1}(\cdot, x_l) - \Delta(\cdot, x_l)\|_\infty > C_0 \sqrt{\frac{\log(h_n^{-1})}{n_1 h_n^d}} \right] \quad (192)$$

$$= o_p(n^{-1}) \quad (193)$$

Where the first inequality follows from 189, the second from the fact that $\mathbb{P}[M(y', x_l)] |\Delta(y, x_l)| |\Delta(y', x_l)|$ is uniformly bounded, and the last from the fact that for any $y \in \mathcal{S}_{x_l, n_1}$, $|\Delta(y, x_l)| > C_0 \sqrt{\frac{\log(h_n^{-1})}{n_1 h_n^d}}$. Hence, if $\|\Delta_{n_1}(\cdot, x_l) - \Delta(\cdot, x_l)\|_\infty \leq r_{n_1}$, then $\Delta_{n_1}(y, x_l)$ has the same sign as $\Delta(y, x_l)$ for every $y \in \mathcal{S}_{x_l, n_1}$. The final identity follows from lemma C.3.

Now consider the term associated with $\mathcal{M}_{x_l, n_1}$:

$$\int_{\mathcal{M}_{x_l, n_1}} \int_{\mathcal{M}_{x_l, n_1}} |C_P(y, y', x_l)| dy dy' \leq \int_{\mathcal{M}_{x_l, n_1}} \int_{\mathcal{M}_{x_l, n_1}} \mathbb{P}[M(y, x_l)] \mathbb{P}[M(y', x_l)] |\Delta(y, x_l)| |\Delta(y', x_l)| dy dy' \quad (194)$$

But, part 2 of lemma C.4 implies that the right hand side is $o(n^{-1})$. This completes the proof. $\square$

Proof of proposition 5.1.

Proposition C.1 implies that:

$$\sqrt{n_2}(\boldsymbol{\theta}_n - \boldsymbol{\mu}_\theta) = \sqrt{n_2}(\tilde{\boldsymbol{\theta}}_n - \boldsymbol{\mu}_\theta) + o(1) \quad (195)$$

Where $\tilde{\boldsymbol{\theta}}_n = (\mathbf{P}_{\boldsymbol{x}_0, n}^T, \mathbf{P}_{\boldsymbol{x}_1, n}^T, n_2^{-1/2} \mathbf{A}_{1, n}^T)$ and $\mathbf{A}_{1, n}^T = (n_2^{-1/2} A_{1, n}(x_0), \dots, A_{1, n}(x_{L'-1}))$. Furthermore, lemma C.5 and proposition C.1 imply that $\mathbb{E}[\tilde{\boldsymbol{\theta}}_n] - \boldsymbol{\mu}_\theta = o(n^{-1})$.

Define the random variable $\xi(Y, X, Z; z_k, x_l) = \mathbf{1}\{\eta_{n_1}^*(Y, X, Z; z_k, x_l) \neq k, X =$

$x_l, Z = z_k\}$ and note that:

$$A_{1,n}(x_l) = \frac{\sqrt{n_2}}{n_2} \sum_{i=1}^{n_2} \left(\frac{n_2}{n_{2,0}} \xi(Y_i, X_i, Z_i; z_0, x_l) + \frac{n_2}{n_{2,1}} \xi(Y_i, X_i, Z_i; z_1, x_l) - L(\eta_{n_1}^*, x_l) \right) \quad (196)$$

Let the vector $\mathbf{P}_{n_2}$ and the function $\mathbf{g}$ be as defined in lemmas C.6 and C.7 respectively. Equation 196 implies that $\tilde{\boldsymbol{\theta}}_n$ is a subvector of $\mathbf{g}(\mathbf{P}_{n_2})$. Formally, if J denotes the set of indices associated with all the entries of $\mathbf{g}(\mathbf{P}_{n_2})$ that comprise $\tilde{\boldsymbol{\theta}}_n$, and e_j is the j -th canonical vector in $\mathbb{R}^{3L}$. Then, the selector matrix $S_{\boldsymbol{\theta}}$ is the matrix that stacks the vectors e_j for every $j \in J$ so that $\tilde{\boldsymbol{\theta}}_n = S_{\boldsymbol{\theta}} \mathbf{g}(\mathbf{P}_{n_2})$.

By construction, every component of $(\mathbf{P}_{\mathbf{x}_0,n}^T, \mathbf{P}_{\mathbf{x}_1,n}^T)$ is linearly independent from the rest asymptotically, and all the components of $\mathbf{A}_{1,n}$ are linearly independent from each other with probability approaching one. Moreover, assumption ?? guarantees that every component of $A_{1,n}$ of $\mathbf{A}_{1,n}$ is linearly independent from $P_{x_l|z_0,n_2}$ and $P_{x_l|z_0,n_2}$ with probability approaching one. Thus, all the elements of $\tilde{\boldsymbol{\theta}}_n$ are asymptotically linearly independent with probability approaching one.

Hence, with probability approaching one, the matrix $S_{\boldsymbol{\theta}} \mathbf{V}_{S_1,n} S_{\boldsymbol{\theta}}^T$ is symmetric positive-definite and, for every non-zero $\mathbf{c} \in \mathbb{R}^m$, a fixed constant $\epsilon_{\mathbf{c}}$ that depends on $\mathbf{c}$ exists such that the probability $\mathbb{P}[\mathbf{c}^T S_{\boldsymbol{\theta}} \mathbf{V}_{S_1,n} S_{\boldsymbol{\theta}}^T \mathbf{c} > \epsilon_{\mathbf{c}}]$ tends to one.

Then, lemma C.7 implies that:

$$\sup_{t \in \mathbb{R}} \left| \mathbb{P} \left[\frac{\mathbf{c}^T \sqrt{n_2} (\boldsymbol{\theta}_n - \boldsymbol{\mu}_{\boldsymbol{\theta}})}{\sqrt{\mathbf{c}^T S_{\boldsymbol{\theta}} \mathbf{V}_{S_1,n} S_{\boldsymbol{\theta}}^T \mathbf{c}}} \leq t \middle| S_1 \right] - \Phi(t) \right| \xrightarrow{p} 0, \quad \forall \mathbf{c} \in \mathbb{R}^m, \mathbf{c} \neq \mathbf{0}, \quad (197)$$

and the Cramér-Wold device yields the desired result. □

Lemma C.1 (KDE bias). *Let $\tilde{\Delta}_{n_1}(Y, x_l) = \frac{1}{n_1 h_n^d} \sum_{i=1}^{n_1} H_n(Y_i, X_i, Z_i; x_l, y)$, then:*

$$\sup_{y \in \mathcal{Y}} \left| \mathbb{E}[\tilde{\Delta}_{n_1}(Y, x_l)] - \Delta(y, x_l) \right| = O(h_n^2) \quad (198)$$

Lemma C.2. *For any $r_n > 0$ such that $\frac{n_1 r_n^2}{\log(n_1)} \rightarrow \infty$ and $r_n \rightarrow 0$:*

$$\mathbb{P} \left(\left\| R_n(y, x_l) \right\|_{\infty} \geq r_n \right) = o(n_1^{-1}) \quad (199)$$

Lemma C.3. *Let K be a kernel density function satisfying assumption 5.2.1, h_n a bandwidth sequence satisfying assumption 5.2.2 and suppose that $f_{Y|X=x_l, Z=z_k}$ satisfies assumption 5.1.2. Then, a constant $C_0 > 0$ exists such that:*

$$\mathbb{P}\left[\left\|\Delta_{n_1}(y, x_l) - \Delta(y, x_l)\right\|_{\infty} > C_0 \sqrt{\frac{\log(h_n^{-1})}{n_1 h_n^d}}\right] = o(n_1^{-1}) \quad (200)$$

Lemma C.4 (Margin missclassification loss).

Let r_n be a sequence such that $r_n \rightarrow 0$ and $r_n^{1+\alpha} \sqrt{n_1} \rightarrow 0$, and let $\mathcal{M}_{x_l}(r_n)$ be the set:

$$\mathcal{M}_{x_l}(r_n) = \{y \in \mathcal{Y} : 0 < |\Delta(y, x_l)| \leq r_n\} \quad (201)$$

Then:

$$\int_{\mathcal{M}_{x_l}(r_n)} \mathbf{1}\{M(y, x_l)\} |\Delta(y, x_l)| dy = o(n^{-1/2}) \quad (202)$$

And:

$$\int_{\mathcal{M}_{x_l}(r_n)} \int_{\mathcal{M}_{x_l}(r_n)} \mathbb{P}[M(y, x_l)] \mathbb{P}[M(y', x_l)] |\Delta(y, x_l)| |\Delta(y', x_l)| dy dy' = o(n^{-1}) \quad (203)$$

Where $M(y, x_l)$ denotes the event $\eta^(y, x_l) \neq \eta_{n_1}^*(y, x_l)$*

Lemma C.5 (The expected empirical loss of $\eta_{n_1}^*$ is its theoretical loss).

$$\mathbb{E}\left[L_{n_2}(\eta_{n_1}^*, x_l) | S_1\right] = L(\eta_{n_1}^*, x_l) \quad (204)$$

$$(205)$$

Lemma C.6.

For $i \in S_2$, let $\boldsymbol{\pi}_{i,n}$ denote the random vector:

$$\begin{aligned} \boldsymbol{\pi}_{i,n} = & (\mathbf{1}\{Z_i = z_0\}, \mathbf{1}\{Z_i = z_1\}, \\ & \mathbf{1}\{Z_i = z_0, X_i = x_0\}, \dots, \mathbf{1}\{Z_i = z_0, X_i = x_{L-1}\}, \\ & \mathbf{1}\{Z_i = z_1, X_i = x_1\}, \dots, \mathbf{1}\{Z_i = z_1, X_i = x_{L-1}\}, \\ & \xi(Y_i, X_i, Z_i; z_0, x_0), \dots, \xi(Y_i, X_i, Z_i; z_0, x_{L-1}) \\ & \xi(Y_i, X_i, Z_i; z_1, x_0), \dots, \xi(Y_i, X_i, Z_i; z_1, x_{L-1})) \end{aligned} \quad (206)$$

With mean $\boldsymbol{\mu}_{S_1,n} := \mathbb{E}[\boldsymbol{\pi}_{i,n}|S_1]$ and variance $\boldsymbol{\Sigma}_{S_1,n} := \text{Var}(\boldsymbol{\pi}_{i,n}|S_1)$

Let $\mathbf{P}_{n_2}$ denote the sample average:

$$\mathbf{P}_{n_2} := \frac{1}{n_2} \sum_{i=1}^{n_2} \boldsymbol{\pi}_{i,n} \quad (207)$$

For any measurable vector-valued function $\mathbf{b}_{S_1,n} = \mathbf{b}_{S_1,n}(S_1, n)$ mapping realizations of S_1 to $\mathbb{R}^{2+3L}$ for which constants M_b and c_b exist such that $\mathbb{P}[\|\mathbf{b}_{S_1,n}\|_1 \leq M_b] \rightarrow 1$ and $\mathbb{P}[\mathbf{b}_{S_1,n}^T \boldsymbol{\Sigma}_{S_1,n} \mathbf{b}_{S_1,n} > c_b] \rightarrow 1$:

$$\sup_{t \in \mathbb{R}} \left| \mathbb{P} \left[\sqrt{n_2} \frac{\mathbf{b}_{S_1,n}^T (\mathbf{P}_{n_2} - \boldsymbol{\mu}_{S_1,n})}{\sqrt{\mathbf{b}_{S_1,n}^T \boldsymbol{\Sigma}_{S_1,n} \mathbf{b}_{S_1,n}}} \leq t \middle| S_1 \right] - \Phi(t) \right| \xrightarrow{p} 0 \quad (208)$$

Where $\Phi(\cdot)$ is the cumulative distribution of a standard normal random variable.

Lemma C.7. Consider the sample averages that comprise the vector $\mathbf{P}_{n_2}$ defined in lemma C.6:

$$P_{z_k, n_2} = \frac{1}{n_2} \sum_{i=1}^{n_2} \mathbf{1}\{Z_i = z_k\} \quad (209)$$

$$P_{z_k, x_l, n_2} = \frac{1}{n_2} \sum_{i=1}^{n_2} \mathbf{1}\{X_i = x_l, Z_i = z_k\} \quad (210)$$

$$P_{\xi, z_k, x_l, n_2} = \frac{1}{n_2} \sum_{i=1}^{n_2} \xi(Y_i, X_i, Z_i; z_0, x_l) \quad (211)$$

Let $\mathbf{P}_{n_2}$, $\boldsymbol{\mu}_{S_1,n}$ and $\boldsymbol{\Sigma}_{S_1,n}$ be defined as in Lemma C.6. Define the map $\mathbf{g} : \mathbb{R}^{2+3L} \rightarrow$

$\mathbb{R}^{3L-2}$ as:

$$\mathbf{g}(\mathbf{P}_{n_2}) = \begin{pmatrix} r_{0,0}(\mathbf{P}_{n_2}), \dots, r_{0,L-1}(\mathbf{P}_{n_2}), \\ r_{1,0}(\mathbf{P}_{n_2}), \dots, r_{1,L-1}(\mathbf{P}_{n_2}), \\ g_0(\mathbf{P}_{n_2}), \dots, g_{L-1}(\mathbf{P}_{n_2}), \end{pmatrix}^T,$$

where, for each $l = 0, \dots, L-1$,

$$g_l(\mathbf{P}_{n_2}) := \frac{P_{\xi, z_0, x_l, n_2}}{P_{z_0, n_2}} + \frac{P_{\xi, z_1, x_l, n_2}}{1 - P_{z_0, n_2}}, \quad (212)$$

and, for $k \in \{0, 1\}$ and $l = 0, \dots, L-1$,

$$r_{k,l}(\mathbf{P}_{n_2}) := \begin{cases} \frac{P_{z_0, x_l, n_2}}{P_{z_0, n_2}} & \text{if } k = 0 \\ \frac{P_{z_1, x_l, n_2}}{1 - P_{z_0, n_2}} & \text{if } k = 1 \end{cases}. \quad (213)$$

Let $\mathbf{J}_{S_1, n}$ denote the Jacobian of $\mathbf{g}$ evaluated at $\boldsymbol{\mu}_{S_1, n}$, and define $\mathbf{V}_{S_1, n} := \mathbf{J}_{S_1, n} \boldsymbol{\Sigma}_{S_1, n} \mathbf{J}_{S_1, n}^T$.

Suppose that $P[Z = z_k] > c_z > 0$ for $k = 0, 1$. Then, for any fixed $\mathbf{a} \in \mathbb{R}^{3L}$ such that there exists $c_a > 0$ that satisfies $\mathbb{P}[\mathbf{a}^T \mathbf{V}_{S_1, n} \mathbf{a} > c_a] \rightarrow 1$:

$$\sup_{t \in \mathbb{R}} \left| \mathbb{P} \left[\frac{\mathbf{a}^T \sqrt{n_2} (\mathbf{g}(\mathbf{P}_{n_2}) - \mathbf{g}(\boldsymbol{\mu}_{S_1, n}))}{\sqrt{\mathbf{a}^T \mathbf{V}_{S_1, n} \mathbf{a}}} \leq t \right] - \Phi(t) \right| \xrightarrow{p} 0. \quad (214)$$

Lemma C.8.

For every $x_l \in \mathcal{X}$ and $z_k \in \mathcal{Z}$:

$$n_2 \text{Var}(\Psi_{x_l, n} - \Psi_{x_l} | S_1) = \sum_{k=0,1} \frac{P_{\xi, x_l | z_k, n_2} (1 - P_{\xi, x_l | z_k, n_2})}{\mathbb{P}[Z = z_k]} + o(1) \quad (215)$$

$$n_2 \text{Cov}(\Psi_{x_l, n} - \Psi_{x_l}, \Psi_{x_l, n} - \Psi_{x_l'} | S_1) = - \sum_{k=0,1} \frac{P_{\xi, x_l | z_k, n_2} P_{\xi, x_l' | z_k, n_2}}{\mathbb{P}[Z = z_k]} + o(1) \quad (216)$$

$$n_2 \text{Cov}(\Psi_{x_l, n} - \Psi_{x_l}, P_{x_{l'} | z_k, n_2} - P_{x_{l'} | z_{k'}} | S_1) = \frac{P_{\xi, x_l | z_k, n_2} (\mathbf{1}\{l = l'\} - P_{x_l | z_k, n_2})}{\mathbb{P}[Z = z_k]} + o(1) \quad (217)$$

$$n_2 \text{Cov}(P_{x_l | z_k, n_2} - P_{x_l | z_k}, P_{x_{l'} | z_{k'}, n_2} - P_{x_{l'} | z_{k'}} | S_1) = \mathbf{1}\{k = k'\} \frac{P_{x_l | z_k} (\mathbf{1}\{l = l'\} - P_{x_l | z_{k'}})}{\mathbb{P}[Z = z_k]} + o(1) \quad (218)$$

Where $P_{\xi, x_l | z_k, n_2} := \mathbb{P}[\eta_{n_1}^*(X, Y) \neq z_k, X = x_l | Z = z_k, S_1]$.

Proof of lemma C.1.

Note that:

$$\mathbb{E}[\tilde{\Delta}_{n_1}(y, x_l)] - \Delta(y, x_l) = \mathbb{P}[X = x_l | Z = z_1] \mathbb{E}\left[\frac{K_n(y - Y)}{h_n^d} | X = x_l, Z = z_1\right] \quad (219)$$

$$\begin{aligned} & - \mathbb{P}[X = x_l | Z = z_0] \mathbb{E}\left[\frac{K_n(y - Y)}{h_n^d} | X = x_l, Z = z_0\right] - \Delta(y, x_l) \\ & = \mathbb{P}[X = x_l | Z = z_1] \left(\mathbb{E}\left[\frac{K_n(y - Y)}{h_n^d} | X = x_l, Z = z_1\right] - f_{Y|X=x_l, Z=z_1}(y) \right) \\ & \quad - \mathbb{P}[X = x_l | Z = z_0] \left(\mathbb{E}\left[\frac{K_n(y - Y)}{h_n^d} | X = x_l, Z = z_0\right] - f_{Y|X=x_l, Z=z_0}(y) \right) \end{aligned} \quad (220)$$

Where the first equality follows from the law of total probability and the definition of H_n , and the second equality from re-writing $\Delta(y, x_l)$ in terms of conditional distributions.

Then, an exact Taylor expansion of $f_{Y|X=x_l, Z=z_k}$, assumptions 5.2.1, 5.2.2, 5.1.2, and standard properties of kernel density estimators imply that if $\sup_{||u||=1} |(\nabla^2 f_{Y|X=x_l, Z=z_k})u|$

is bounded, then:

$$\mathbb{E}\left[\frac{K_n(y - Y)}{h_n^d} | X = x_l, Z = z_k\right] - f_{Y|X=x_l, Z=z_k}(y) = O(h_n^2) \quad (221)$$

uniformly over y, x_l and z_k .

□

Proof of lemma C.2. First note that:

$$\mathbb{P}\left[\left\|R_n(y, x_l)\right\|_\infty \geq r_n\right] \leq \mathbb{P}\left[\left\|R_n^1(y, x_l)\right\|_\infty + \left\|R_n^0(y, x_l)\right\|_\infty \geq r_n\right] \quad (222)$$

$$\leq \mathbb{P}\left[\left\|R_n^1(y, x_l)\right\|_\infty \geq \frac{r_n}{2}\right] + \mathbb{P}\left[\left\|R_n^0(y, x_l)\right\|_\infty \geq \frac{r_n}{2}\right] \quad (223)$$

To bound the previous probabilities, define the class of functions:

$$\mathcal{K}_n = \left\{ \kappa_n(y, x, z; t, x_l, z_k) = \mathbf{1}\{X = x_l, Z = z_k\} K_n(t - y) : x_l \in \mathcal{X}, z_k \in \mathcal{Z}, t \in \mathcal{Y} \right\} \quad (224)$$

And the associated sub-density estimators:

$$f_{z_k, n}(y, x_l) := \left[\frac{1}{n_1 h_n^d} \sum_{i=1}^{n_1} \mathbf{1}\{X_i = x_l, Z_i = z_k\} K_n(y - Y_i) \right] \quad (225)$$

Note that if $K(\cdot)$ is bounded, for any constant $B' > 0$:

$$\begin{aligned} & \mathbb{P}\left[\left\|f_{z_k, n}(y, x_l) - \mathbb{E}f_{z_k, n}(y, x_l)\right\|_\infty \geq B'\right] \\ &= \mathbb{P}\left[\left\|\frac{1}{n_1 h_n^d} \sum_{i=1}^{n_1} (\kappa_n(Y_i, X_i, Z_i; y, x_l, z_k) - \mathbb{E}[\kappa_n(Y, X, Z; y, x_l, z_k)])\right\|_\infty \geq B'\right] \end{aligned} \quad (226)$$

$$= \mathbb{P}\left[\left\|\sum_{i=1}^{n_1} (\kappa_n(Y_i, X_i, Z_i; y, x_l, z_k) - \mathbb{E}[\kappa_n(Y, X, Z; y, x_l, z_k)])\right\|_\infty \geq B' n_1 h_n^d\right] \quad (227)$$

Let $\sigma_{\kappa_n}^2 = \sup_{\kappa \in \mathcal{K}_n} \text{Var}(\kappa(Y, X, Z)) = O(h_n^d)$ and $U = \sup_{\kappa \in \mathcal{K}_n} \|\kappa\|_\infty$. Note that for any $(t, x_l, z_k) \in \mathcal{Y} \times \mathcal{X} \times \mathcal{Z}$ the function $\kappa_n(\cdot, \cdot, \cdot; t, x_l, z_k)$ satisfies the conditions from assumption 5.2.1. Moreover, the map $(y, x, z) \mapsto \kappa_n(y, x, z; t, x_l, z_k)$ is jointly

measurable and, for a sufficiently large n , the condition $0 < \sigma_{\kappa_n}^2 < U/2$ is satisfied. Thus, corollary 2.2. from Gine and Gillou 2002 implies that for any sufficiently large B' and n , positive constants C and L depending only on the VC class of the family $\mathcal{K}_n$ exist, such that for every λ such that for $B'h_n^d \leq \frac{\lambda}{U}\sigma_{\kappa_n}^2$:

$$\mathbb{P}\left[\left\|f_{z_k,n}(y, x_l) - \mathbb{E}f_{z_k,n}(y, x_l)\right\|_{\infty} \geq B'\right] \leq L \exp\left(\frac{-\log(1 + \lambda/(4L))}{\lambda L} \frac{n_1(B'h_n^d)^2}{\sigma_{\kappa_n}^2}\right) = o(n_1^{-1}) \quad (228)$$

Moreover, since $K(\cdot)$ and f are bounded, the sequence $\left\|\mathbb{E}f_{z_k,n}(y, x_l)\right\|_{\infty}$ is also bounded. Thus, for some $B \geq B'$:

$$\mathbb{P}\left[\left\|f_{z_k,n}(y, x_l)\right\|_{\infty} \geq B\right] = \mathbb{P}\left[\left\|f_{z_k,n}(y, x_l) - \mathbb{E}f_{z_k,n}(y, x_l) + \mathbb{E}f_{z_k,n}(y, x_l)\right\|_{\infty} \geq B\right] \quad (229)$$

$$\leq \mathbb{P}\left[\left\|f_{z_k,n}(y, x_l) - \mathbb{E}f_{z_k,n}(y, x_l)\right\|_{\infty} + \left\|\mathbb{E}f_{z_k,n}(y, x_l)\right\|_{\infty} \geq B\right] \quad (230)$$

$$\leq \mathbb{P}\left[\left\|f_{z_k,n}(y, x_l) - \mathbb{E}f_{z_k,n}(y, x_l)\right\|_{\infty} \geq B'\right] \quad (231)$$

Now let P_k denote $\mathbb{P}[Z = z_k]$ and note that for any $B > 0$:

$$\mathbb{P}\left[\left|\frac{1}{P_{k,n_1}} - \frac{1}{P_k}\right| B \geq \frac{r_n}{2}\right] = \mathbb{P}\left[\left|\frac{P_k - P_{k,n_1}}{P_k P_{k,n_1}}\right| B \geq \frac{r_n}{2}\right] \quad (232)$$

$$\begin{aligned} &= \mathbb{P}\left[\left|\frac{P_k - P_{k,n_1}}{P_k P_{k,n_1}}\right| B \geq \frac{r_n}{2}, P_{k,n_1} \geq \frac{P_k}{2}\right] \\ &\quad + \mathbb{P}\left[\left|\frac{P_k - P_{k,n_1}}{P_k P_{k,n_1}}\right| B \geq \frac{r_n}{2}, P_{k,n_1} < \frac{P_k}{2}\right] \end{aligned} \quad (233)$$

$$\leq \mathbb{P}\left[|P_k - P_{k,n_1}| \geq \frac{P_k^2 r_n}{2B}\right] + \mathbb{P}\left[|P_{k,n_1} - P_k| \geq \frac{P_k}{2}\right] \quad (234)$$

Where the second equality follows from the law of total probability and the inequality from the fact that $P_{k,n_1} < \frac{P_k}{2} \Rightarrow |P_{k,n_1} - P_k| \geq \frac{P_k}{2}$.

Thus, for a sufficiently large n_1 , Hoeffding's inequality implies:

$$\mathbb{P}\left[\left|\frac{1}{P_{k,n_1}} - \frac{1}{P_k}\right| B \geq \frac{r_n}{2}\right] \leq 2\mathbb{P}\left[|P_k - P_{k,n_1}| \geq \frac{P_k^2 r_n}{2B}\right] \quad (235)$$

$$\leq 4 \exp\left(-\frac{P_k^4}{2B^2} n_1 r_n^2\right) = o(n_1^{-1}). \quad (236)$$

Then, constants B, B' and λ must exist such that $B'h_n^d \leq \frac{\lambda}{U} \sigma_{\kappa_n}^2$, $B \geq B'$ and:

$$\mathbb{P}\left[\left\|\left|R_n^k(y, x_l)\right\|_{\infty} \geq \frac{r_n}{2}\right] = \mathbb{P}\left[\left\|\left(\frac{1}{P_{k,n_1}} - \frac{1}{P_k}\right)f_{z_k,n}(y, x_l)\right\|_{\infty} \geq \frac{r_n}{2}\right] \quad (237)$$

$$= \mathbb{P}\left[\left|\frac{1}{P_{k,n_1}} - \frac{1}{P_k}\right| \|f_{z_k,n}(y, x_l)\|_{\infty} \geq \frac{r_n}{2}\right] \quad (238)$$

$$\leq \mathbb{P}\left[\left|\frac{1}{P_{k,n_1}} - \frac{1}{P_k}\right| B \geq \frac{r_n}{2}\right] + \mathbb{P}\left[\|f_{z_k,n}(y, x_l)\|_{\infty} \geq B\right] \quad (239)$$

$$\leq 4 \exp\left(-\frac{P_k^4}{2B^2} n_1 r_n^2\right) + L \exp\left(\frac{-\log(1 + \lambda/(4L))}{\lambda L} \frac{n_1 (B'h_n^d)^2}{\sigma_{\kappa_n}^2}\right) \quad (240)$$

$$= o(n_1^{-1}) \quad (241)$$

Where the last inequality follows from 231 and 236. $\square$

Proof of lemma C.3.

This lemma shows that the asymptotic behavior of the difference $\Delta_{n_1}(y, x_l)$ is determined by the kernel density estimator component, and that the difference $|\Delta_{n_1}(y, x_l) - \Delta_n(y, x_l)|$ exceeds the rate $\sqrt{\frac{\log(h_n^{-1})}{n_1 h_n^d}}$ with probability $o(n^{-1})$.

The result uses lemmas C.1 and C.2 which prove that the bias from kernel density estimation and the randomness from estimating conditional treatment and instrument probabilities are negligible at the rate $\sqrt{\frac{\log(h_n^{-1})}{n_1 h_n^d}}$.

For a given value of $x_l \in \mathcal{X}$, define:

$$H_n(y', x, z; x_l, y) = \left[\frac{\mathbf{1}\{x = x_l, z = z_1\}}{\mathbb{P}[Z = z_1]} - \frac{\mathbf{1}\{x = x_l, z = z_0\}}{\mathbb{P}[Z = z_0]} \right] K_n(y - y') \quad (242)$$

Then, $\Delta_{n_1}(y, x_l)$ can be decomposed into a main component depending on $H_n(y', x, z; x_l, y)$ plus a bias term and a residual that depends on the difference $\left(\frac{1}{P_{1,n_1}} - \frac{1}{\mathbb{P}[Z=z_1]}\right)$:

$$\Delta_{n_1}(y, x_l) - \Delta(y, x_l) = P_{1,n_1}^{-1} \left[\frac{1}{n_1 h_n^d} \sum_{i=1}^{n_1} \mathbf{1}\{X_i = x_l, Z_i = z_1\} K_n(y - Y_i) \right] \quad (243)$$

$$\begin{aligned} & - P_{0,n_1}^{-1} \left[\frac{1}{n_1 h_n^d} \sum_{i=1}^{n_1} \mathbf{1}\{X_i = x_l, Z_i = z_0\} K_n(y - Y_i) \right] - \Delta(y, x_l) \\ & = \frac{1}{n_1 h_n^d} \sum_{i=1}^{n_1} H_n(Y_i, X_i, Z_i; x_l, y) - \Delta(y, x_l) \end{aligned} \quad (244)$$

$$\begin{aligned} & + \underbrace{\left(\frac{1}{P_{1,n_1}} - \frac{1}{\mathbb{P}[Z = z_1]} \right) \left[\frac{1}{n_1 h_n^d} \sum_{i=1}^{n_1} \mathbf{1}\{X_i = x_l, Z_i = z_1\} K_n(y - Y_i) \right]}_{:= R_n^1(y, x_l)} \\ & - \underbrace{\left(\frac{1}{P_{0,n_1}} - \frac{1}{\mathbb{P}[Z = z_0]} \right) \left[\frac{1}{n_1 h_n^d} \sum_{i=1}^{n_1} \mathbf{1}\{X_i = x_l, Z_i = z_0\} K_n(y - Y_i) \right]}_{:= R_n^0(y, x_l)} \\ & = \underbrace{\frac{1}{n_1 h_n^d} \sum_{i=1}^{n_1} H_n(Y_i, X_i, Z_i; x_l, y) - \Delta(y, x_l)}_{:= \tilde{\Delta}_{n_1}(y, x_l)} + R_n(y, x_l) \end{aligned} \quad (245)$$

$$\begin{aligned} & = \left(\tilde{\Delta}_{n_1}(y, x_l) - \mathbb{E}[\tilde{\Delta}_{n_1}(Y, x_l)] \right) + \underbrace{\left(\mathbb{E}[\tilde{\Delta}_{n_1}(Y, x_l)] - \Delta(y, x_l) \right)}_{Bias} + R_n(y, x_l) \end{aligned} \quad (246)$$

Where $R_n(y, x_l) = R_n^0(y, x_l) - R_n^1(y, x_l)$.

Define the class of functions:

$$\mathcal{H}_n = \left\{ H_n(y, x, z; x_l, t) : x_l \in \mathcal{X}, t \in \mathcal{Y} \right\} \quad (247)$$

Let $\sigma_{H_n}^2 = \sup_{H \in \mathcal{H}_n} \text{Var}(H(Y, X, Z)) = O(h_n^d)$ and $U = \sup_{H \in \mathcal{H}_n} \|H\|_\infty$. Note that for any $(t, x_l) \in \mathcal{Y} \times \mathcal{X}$ the function $H_n(\cdot, \cdot, \cdot; t, x_l)$ satisfies the conditions from assumption 5.2.1. Moreover, the map $(y, x, z) \mapsto H_n(y, x, z; x_l, t)$ is jointly measurable and, for a sufficiently large n_1 , the condition $0 < \sigma_{H_n}^2 < U/2$ is satisfied. Thus, part 2 of corollary 2.2. from Gine and Gillou 2002 implies that positive constants C and L

depending only on the VC class of the family $\mathcal{H}_n$ exist, such that for every $C' > C$:

$$\mathbb{P}\left[\left\|\sum_{i=1}^{n_1} (H_n(Y_i, X_i, Z_i; x_l, y) - \mathbb{E}[H_n(Y_i, X_i, Z_i; x_l, y)])\right\|_{\infty} > C' \sqrt{n_1 \sigma_{H_n}^2} \sqrt{\log(U \sigma_{H_n}^{-1})}\right] \quad (248)$$

$$\leq L \exp\left\{-\frac{C' \log(1 + C'/(4L))}{L} \log(U \sigma_{H_n}^{-1})\right\} \quad (249)$$

$$= L \exp\left\{-K(C') \log(U \sigma_{H_n}^{-1})\right\} \quad (250)$$

$$= L [U \sigma_{H_n}^{-1}]^{-K(C')} \quad (251)$$

$$= M [\sigma_{H_n}^{K(C')}] \quad (252)$$

Where M is positive and depends on L and C' , and $K'(C')$ denotes the term $\frac{C' \log(1+C'/(4L))}{L}$.

Moreover, by standard properties of kernel density estimators:

$$\mathbb{E}[H_n(Y, X, Z; x_l, y)^2] = O(h_n^d) \quad (253)$$

Thus, $\sigma_{H_n} = O(h_n^{d/2})$ and $\sigma_{H_n}^{K(C')} = O(n_1^{-\gamma d K(C')/2})$.

Observe that the function $K(C')$ is strictly increasing in C' , and thus, for a sufficiently large C' , the upper bound from 252 is $o(n_1^{-1})$.

For any fixed C' such that $\sigma_{H_n}^{K(C')} = o(n_1^{-1})$, elementary algebra shows that a positive constant $C_0 \geq C'$ and a fixed integer N exist such that $\frac{C_0}{2} \sqrt{n_1 h_n^d \log(h_n^{-1})} \geq C' \sqrt{n_1 \sigma_{H_n}^2 \log(U \sigma_{H_n}^{-1})}$ for any $n_1 \geq N$.

Fix such C_0 and consider the probability $\mathbb{P}\left[\left\|\Delta_{n_1}(y, x_l) - \Delta(y, x_l)\right\|_{\infty} > C_0 \sqrt{\frac{\log(h_n^{-1})}{n_1 h_n^d}}\right]$. Then, lemma C.1 applied to the bias term in equation 246 yields:

$$\begin{aligned} & \mathbb{P}\left[\left\|\Delta_{n_1}(y, x_l) - \Delta(y, x_l)\right\|_{\infty} > C_0 \sqrt{\frac{\log(h_n^{-1})}{n_1 h_n^d}}\right] \\ &= \mathbb{P}\left[\left\|\left(\tilde{\Delta}_{n_1}(y, x_l) - \mathbb{E}[\tilde{\Delta}_{n_1}(y, x_l)]\right) + O(h_n^2) + R_n(y, x_l)\right\|_{\infty} > C_0 \sqrt{\frac{\log(h_n^{-1})}{n_1 h_n^d}}\right] \end{aligned} \quad (254)$$

Lemma C.2 applied with $r_n = \sqrt{\frac{\log(h_n^{-1})}{n_1 h_n^d}}$ implies that:

$$\mathbb{P}\left[\left\|\Delta_{n_1}(y, x_l) - \Delta(y, x_l)\right\|_\infty > C_0 \sqrt{\frac{\log(h_n^{-1})}{n_1 h_n^d}}\right] \quad (255)$$

$$\begin{aligned} &= \mathbb{P}\left[\left\|\left(\tilde{\Delta}_{n_1}(y, x_l) - \mathbb{E}[\tilde{\Delta}_{n_1}(y, x_l)]\right) + O(h_n^2) + o\left(\sqrt{\frac{\log(h_n^{-1})}{n_1 h_n^d}}\right)\right\|_\infty > C_0 \sqrt{\frac{\log(h_n^{-1})}{n_1 h_n^d}}\right] + o(n_1^{-1}) \\ &\leq \mathbb{P}\left[\left\|\left(\tilde{\Delta}_{n_1}(y, x_l) - \mathbb{E}[\tilde{\Delta}_{n_1}(y, x_l)]\right)\right\|_\infty + \left\|o\left(\sqrt{\frac{\log(h_n^{-1})}{n_1 h_n^d}}\right)\right\|_\infty > C_0 \sqrt{\frac{\log(h_n^{-1})}{n_1 h_n^d}}\right] + o(n_1^{-1}) \end{aligned} \quad (256)$$

Where the last line follows from assumption 5.2.2 (which implies $h_n^2 = o\left(\sqrt{\frac{\log(h_n^{-1})}{n_1 h_n^d}}\right)$) and the triangle inequality.

Then, for some integer N' , if $n_1 \geq \max\{N, N'\}$, the bound from 252 yields:

$$\mathbb{P}\left[\left\|\Delta_{n_1}(y, x_l) - \Delta(y, x_l)\right\|_\infty > C_0 \sqrt{\frac{\log(h_n^{-1})}{n_1 h_n^d}}\right] \quad (257)$$

$$\begin{aligned} &\leq \mathbb{P}\left[\left\|\sum_{i=1}^{n_1} \left(H_n(Y_i, X_i, Z_i; x_l, y) - \mathbb{E}[H_n(Y_i, X_i, Z_i; x_l, y)]\right)\right\|_\infty > \frac{C_0}{2} \sqrt{n_1 h_n^d \log(h_n^{-1})}\right] + o(n_1^{-1}) \\ &= o(n_1^{-1}) \end{aligned} \quad (258)$$

□

Proof of lemma C.4.

For a sufficiently large n :

$$\int_{\mathcal{M}_{x_l}(r_n)} \left(\mathbf{1}\{\eta^*(y, x_l) \neq \eta_{n_1}^*(y, x_l)\} |\Delta(y, x_l)|\right) dy \leq \int_{\mathcal{M}_{x_l}(r_n)} \left(\mathbf{1}\{\eta^*(y, x_l) \neq \eta_{n_1}^*(y, x_l)\} r_n\right) dy \quad (259)$$

$$= (\mu\{\mathcal{M}_{x_l}(r_n)\}) r_n \quad (260)$$

$$\leq B r_n^{1+\alpha} = O(r_n^{1+\alpha}) \quad (261)$$

Where the first inequality follows from the definition of the set $\mathcal{M}_{x_l}(r_n)$, and the second inequality from direct application of assumption 5.1.3.

For part 2, observe that:

$$\int_{\mathcal{M}_{x_l}(r_n)} \int_{\mathcal{M}_{x_l}(r_n)} \mathbb{P}[M(y, x_l)] \mathbb{P}[M(y, x_l)] |\Delta(y, x_l)| |\Delta(y', x_l)| dy dy' \leq (\mu\{\mathcal{M}_{x_l}(r_n)\})^2 r_n^2 dy dy' \quad (262)$$

$$= O(r_n^{2\alpha+2}) \quad (263)$$

□

Proof of lemma C.5.

$$\mathbb{E}[L_{n_2}(\eta_{n_1}^*, x_l) | S_1] \quad (264)$$

$$= \mathbb{E}\left[\frac{1}{n_2} \sum_{i=1}^{n_2} \left(\frac{\mathbf{1}\{\eta_{n_1}^*(Y_i, X_i) = 0, Z_i = z_1, X_i = x_l\}}{P_{1,n_2}} + \frac{\mathbf{1}\{\eta_{n_1}^*(Y_i, X_i) = 1, Z_i = z_0, X_i = x_l\}}{P_{0,n_2}} \right) \middle| S_1 \right] \quad (265)$$

$$= \mathbb{E}\left[\mathbb{E}\left\{\frac{1}{n_2} \sum_{i=1}^{n_2} \left(\frac{\mathbf{1}\{\eta_{n_1}^*(Y_i, X_i) = 0, Z_i = z_1, X_i = x_l\}}{P_{1,n_2}} + \frac{\mathbf{1}\{\eta_{n_1}^*(Y_i, X_i) = 1, Z_i = z_0, X_i = x_l\}}{P_{0,n_2}} \right) \middle| \{Z_i\}_{i=1}^{n_2} \right\} \middle| S_1 \right] \quad (266)$$

$$= \mathbb{E}\left[\mathbb{E}\left\{\frac{1}{n_{2,z_1}} \sum_{i \in S_2, Z_i = z_1}^{n_2} \mathbf{1}\{\eta_{n_1}^*(Y_i, X_i) = 0, Z_i = z_1, X_i = x_l\} + \frac{1}{n_{2,z_0}} \sum_{i \in S_2, Z_i = z_0}^{n_2} \mathbf{1}\{\eta_{n_1}^*(Y_i, X_i) = 1, Z_i = z_0, X_i = x_l\} \middle| \{Z_i\}_{i=1}^{n_2}, S_1 \right\} \middle| S_1 \right] \quad (267)$$

$$= \mathbb{E}\left[\mathbb{E}\left\{\mathbb{P}[\eta_{n_1}^*(Y, X) = 1, X = x_l | Z = z_0, S_1] + \mathbb{P}[\eta_{n_1}^*(Y, X) = 0, X = x_l | Z = z_1, S_1] \middle| \{Z\}_{i=1}^{n_2}, S_1 \right\} \middle| S_1 \right] \quad (268)$$

$$= \mathbb{P}[\eta_{n_1}^*(Y, X) = 1, X = x_l | Z = z_0, S_1] + \mathbb{P}[\eta_{n_1}^*(Y, X) = 0, X = x_l | Z = z_1, S_1] \quad (269)$$

$$= \frac{\mathbb{P}[\eta_{n_1}^*(Y, X) = 1, Z = z_0, X = x_l | S_1]}{\mathbb{P}[Z = z_0]} + \frac{\mathbb{P}[\eta_{n_1}^*(Y, X) = 0, Z = z_1, X = x_l | S_1]}{\mathbb{P}[Z = z_1]} \quad (270)$$

$$= L(\eta_{n_1}^*, x_l) \quad (271)$$

□

Proof of lemma C.6.

Note that all the coordinates of $\mathbf{P}_i$ are indicator variables, thus, its support is bounded, and all its conditional moments exist.

Given the measurable function $\mathbf{b}_{S_1,n}$, define:

$$U_{i,n} = \mathbf{b}_{S_1,n}^T (\boldsymbol{\pi}_{i,n} - \boldsymbol{\mu}_{S_1,n}) \quad (272)$$

By definition, $\mathbb{E}[U_{i,n}|S_1] = 0$ and $\sigma_{b,S_1,n} := \text{Var}(U_i|S_1) = \mathbf{b}_{S_1,n}^T \boldsymbol{\Sigma}_{S_1,n} \mathbf{b}_{S_1,n}$.

Now consider the standardized sum:

$$\sqrt{n_2} \frac{\mathbf{b}_{S_1,n}^T (\boldsymbol{\pi}_{n_2} - \boldsymbol{\mu}_{S_1,n})}{\sqrt{\mathbf{b}_{S_1,n}^T \boldsymbol{\Sigma}_{S_1,n} \mathbf{b}_{S_1,n}}} = \frac{1}{\sqrt{n_2}} \sum_{i=1}^{n_2} \frac{U_{i,n}}{\sigma_{b,S_1,n}} \quad (273)$$

Let $s_n^2 = \sum_{i=1}^{n_2} \sigma_{b,S_1,n}^2 = n_2 \sigma_{b,S_1,n}^2$. Then, on the events $\sigma_{b,S_1,n}^2 > c_b$ and $\|\mathbf{b}_{S_1,n}\|_1 \leq M_b$, for any fixed $\epsilon > 0$ and a sufficiently large n :

$$\frac{1}{s_n^2} \sum_{i=1}^{n_2} \mathbb{E}[U_{i,n}^2 \mathbf{1}\{|U_i| \geq \epsilon s_n\} | S_1] \leq \frac{4\|\mathbf{b}_{S_1,n}\|_1^2}{s_n^2} \sum_{i=1}^{n_2} \mathbb{E}[\mathbf{1}\{|U_i| \geq \epsilon s_n\} | S_1] \quad (274)$$

$$= \frac{4M_b^2}{s_n^2} \sum_{i=1}^{n_2} \mathbb{P}[|U_i| \geq \epsilon s_n | S_1] \quad (275)$$

$$= 0 \quad (276)$$

Where the inequality follows from the fact that $|U_2| \leq 2\|\mathbf{b}_{S_1,n}\|_1 \leq 2M_b$ which in turn implies that $\mathbb{P}[|U_i| \geq \epsilon s_n | S_1] = 0$.

Thus, the Lindeberg condition holds which implies that, conditional on S_1 , $\sigma_{b,S_1,n}^2 > c_b$ and $\|\mathbf{b}_{S_1,n}\|_1 \leq M_b$, the CDF of the sum $\frac{1}{\sqrt{n_2}} \sum_{i=1}^{n_2} \frac{U_{i,n}}{\sigma_{b,S_1,n}}$ converges uniformly to $\Phi(\cdot)$.

Hence:

$$\sup_{t \in \mathbb{R}} \left| \mathbb{P} \left[\sqrt{n_2} \frac{\mathbf{b}_{S_1,n}^T (\mathbf{P}_{n_2} - \boldsymbol{\mu}_{S_1,n})}{\sqrt{\mathbf{b}_{S_1,n}^T \boldsymbol{\Sigma}_{S_1,n} \mathbf{b}_{S_1,n}}} \leq t \middle| S_1 \right] - \Phi(t) \right| \xrightarrow{p} 0 \quad (277)$$

□

Proof of Lemma C.7. The proof follows from the Delta method and Lemma C.6.

First note that, by construction:

$$g_l(\boldsymbol{\pi}_{n_2}) = \frac{\pi_{\xi, z_0, x_l, n_2}}{\pi_{z_0, n_2}} + \frac{\pi_{\xi, z_1, x_l, n_2}}{\pi_{z_1, n_2}} = L_{n_2}(\eta_{n_1}^*, x_l), \quad (278)$$

and

$$g_l(\boldsymbol{\mu}_{S_1, n}) = \frac{\mu_{\xi, z_0, x_l, S_1, n}}{P[Z = z_0]} + \frac{\mu_{\xi, z_1, x_l, S_1, n}}{P[Z = z_1]} = L(\eta_{n_1}^*, x_l). \quad (279)$$

Similarly:

$$r_{k, l}(\boldsymbol{\pi}_{n_2}) = \frac{\pi_{z_k, x_l, n_2}}{\pi_{z_k, n_2}} \quad (280)$$

is the sample analogue of $P[X = x_l \mid Z = z_k]$, and:

$$r_{k, l}(\boldsymbol{\mu}_{S_1, n}) = \frac{P[X = x_l, Z = z_k]}{P[Z = z_k]} = P[X = x_l \mid Z = z_k]. \quad (281)$$

Since $P[Z = z_k] > c_z > 0$ for $k = 0, 1$, the map $\mathbf{g}$ is continuously differentiable in a neighborhood of $\boldsymbol{\mu}_{S_1, n}$ with probability approaching one. Moreover, by the law of large numbers conditional on S_1 ,

$$\boldsymbol{\Pi}_{n_2} - \boldsymbol{\mu}_{S_1, n} = o_p(1). \quad (282)$$

Therefore, a first-order Taylor expansion yields

$$\sqrt{n_2}(\mathbf{g}(\boldsymbol{\Pi}_{n_2}) - \mathbf{g}(\boldsymbol{\mu}_{S_1, n})) = \mathbf{J}_{S_1, n} \sqrt{n_2}(\boldsymbol{\Pi}_{n_2} - \boldsymbol{\mu}_{S_1, n}) + \mathbf{R}_n, \quad (283)$$

where, for every fixed $\mathbf{a} \in \mathbb{R}^{3L}$, $\mathbf{a}^T \mathbf{R}_n = o_p(1)$.

Fix $\mathbf{a} \in \mathbb{R}^{3L}$ and define

$$\mathbf{b}_{S_1, n} := \mathbf{J}_{S_1, n}^T \mathbf{a}. \quad (284)$$

Because $\mathbf{J}_{S_1, n}$ is a function of $\boldsymbol{\mu}_{S_1, n}$, and $\boldsymbol{\mu}_{S_1, n}$ is measurable with respect to S_1 , the vector $\mathbf{b}_{S_1, n}$ is S_1 -measurable. Furthermore, since all coordinates of $\boldsymbol{\mu}_{S_1, n}$ are probabilities and $P[Z = z_k]$ is bounded away from zero, there exists a finite constant

M_b such that

$$\mathbb{P} [\|\mathbf{b}_{S_1,n}\|_1 \leq M_b] \longrightarrow 1. \quad (285)$$

Also,

$$\mathbf{b}_{S_1,n}^T \Sigma_{S_1,n} \mathbf{b}_{S_1,n} = \mathbf{a}^T \mathbf{J}_{S_1,n} \Sigma_{S_1,n} \mathbf{J}_{S_1,n}^T \mathbf{a} = \mathbf{a}^T \mathbf{V}_{S_1,n} \mathbf{a}. \quad (286)$$

By assumption,

$$\mathbb{P} [\mathbf{b}_{S_1,n}^T \Sigma_{S_1,n} \mathbf{b}_{S_1,n} > c_a] \longrightarrow 1. \quad (287)$$

Thus, Lemma C.6 applies to the random direction $\mathbf{b}_{S_1,n} = \mathbf{J}_{S_1,n}^T \mathbf{a}$ and implies

$$\sup_{t \in \mathbb{R}} \left| \mathbb{P} \left[\frac{\mathbf{a}^T \mathbf{J}_{S_1,n} \sqrt{n_2} (\mathbf{\Pi}_{n_2} - \boldsymbol{\mu}_{S_1,n})}{\sqrt{\mathbf{a}^T \mathbf{V}_{S_1,n} \mathbf{a}}} \leq t \middle| S_1 \right] - \Phi(t) \right| \xrightarrow{p} 0. \quad (288)$$

Finally, since:

$$\mathbf{a}^T \sqrt{n_2} (g(\mathbf{\Pi}_{n_2}) - g(\boldsymbol{\mu}_{S_1,n})) = \mathbf{a}^T \mathbf{J}_{S_1,n} \sqrt{n_2} (\mathbf{\Pi}_{n_2} - \boldsymbol{\mu}_{S_1,n}) + o_p(1), \quad (289)$$

and since: $\mathbf{a}^T \mathbf{V}_{S_1,n} \mathbf{a}$ is bounded away from zero with probability approaching one, Slutsky's theorem gives:

$$\sup_{t \in \mathbb{R}} \left| \mathbb{P} \left[\frac{\mathbf{a}^T \sqrt{n_2} (g(\mathbf{\Pi}_{n_2}) - g(\boldsymbol{\mu}_{S_1,n}))}{\sqrt{\mathbf{a}^T \mathbf{V}_{S_1,n} \mathbf{a}}} \leq t \middle| S_1 \right] - \Phi(t) \right| \xrightarrow{p} 0. \quad (290)$$

Now set $T_n := \frac{\mathbf{a}^T \sqrt{n_2} (g(\mathbf{\Pi}_{n_2}) - g(\boldsymbol{\mu}_{S_1,n}))}{\sqrt{\mathbf{a}^T \mathbf{V}_{S_1,n} \mathbf{a}}}$ and note that:

$$\sup_t |\mathbb{P}[T_n = t] - \Phi(t)| = \sup_t |\mathbb{E}[\mathbb{P}[T_n \leq t | S_1]] - \Phi(t)| \quad (291)$$

$$\leq \mathbb{E} \left[\sup_t |\mathbb{P}[T_n \leq t | S_1] - \Phi(t)| \right] \quad (292)$$

Thus, 290 becomes:

$$\sup_{t \in \mathbb{R}} \left| \mathbb{P} \left[\frac{\mathbf{a}^T \sqrt{n_2} (\mathbf{g}(\mathbf{\Pi}_{n_2}) - \mathbf{g}(\boldsymbol{\mu}_{S_{1,n}}))}{\sqrt{\mathbf{a}^T \mathbf{V}_{S_{1,n}} \mathbf{a}}} \leq t \right] - \Phi(t) \right| \xrightarrow{p} 0. \quad (293)$$

This proves the claim. $\square$

Proof of lemma C.8.

First consider $Var(A_{1,n}(x_l))$:

$$Var(A_{1,n}(x_l)) = Var(\mathbb{E}[A_{1,n}(x_l)|S_1]) + \mathbb{E}(Var[A_{1,n}(x_l)|S_1]) \quad (294)$$

$$= \mathbb{E}(Var[A_{1,n}(x_l)|S_1]) \quad (295)$$

Where the first line follows from the law of total variance and the second from lemma 5.1.

Note that:

$$E[L_{n_2}(\eta_{n_1}^*, x_l)|S_1] = \mathbb{E} \left[\sum_{i=1}^{n_2} \left(\sum_{k=0,1} \frac{1}{n_{2,k}} \xi(Y_i, X_i, Z_i; z_k, x_l) \right) | S_1 \right] \quad (296)$$

$$= \mathbb{E} \left\{ \mathbb{E} \left[\sum_{i=1}^{n_2} \left(\sum_{k=0,1} \frac{1}{n_{2,k}} \xi(Y_i, X_i, Z_i; z_k, x_l) \right) \middle| \{Z_i\}_{i \in S_2}, S_1 \right] \middle| S_1 \right\} \quad (297)$$

$$= \mathbb{E} \left\{ \sum_{k=0,1} \mathbb{E}[\xi(Y_i, X_i, Z_i; z_k, x_l) | Z_i = z_k, S_1] \middle| S_1 \right\} \quad (298)$$

$$= \mathbb{P}[\eta_{n_1}^*(Y, X, Z; z_0, x_l) \neq k, X = x_k | Z = z_0, S_1] \quad (299)$$

$$+ \mathbb{P}[\eta_{n_1}^*(Y, X, Z; z_1, x_l) \neq k, X = x_k | Z = z_1, S_1]$$

Where the second line comes from conditioning on realized instrument values, the third from the fact that observations are i.i.d. conditional on instrument values, and the last from the definition of $\xi(\cdot)$.

And:

$$\mathbb{E}\left[L_{n_2}(\eta_{n_1}^*, x_l)^2 | S_1\right] = \mathbb{E}\left[\sum_{i=1}^{n_2} \sum_{j=1}^{n_2} \left(\sum_{k=0,1} \frac{1}{n_{2,k}} \xi(Y_i, X_i, Z_i; z_k, x_l)\right) \left(\sum_{k=0,1} \frac{1}{n_{2,k}} \xi(Y_j, X_j, Z_j; z_k, x_l)\right) \middle| S_1\right] \quad (300)$$

$$= \mathbb{E}\left[\sum_{i=1}^{n_2} \left(\sum_{k=0,1} \frac{1}{n_{2,k}^2} \xi(Y_i, X_i, Z_i; z_k, x_l)\right) \right. \quad (301)$$

$$\left. + \sum_{j \neq i, j \in S_2} \sum_{k'=0,1} \frac{1}{n_{2,k} n_{2,k'}} \xi(Y_i, X_i, Z_i; z_k, x_l) \xi(Y_j, X_j, Z_j; z_k, x_l)\right] \middle| S_1$$

Now expand the second term and condition on instrument realizations:

$$\mathbb{E}\left[\sum_{i=1}^{n_2} \sum_{k=0,1} \sum_{j \neq i, j \in S_2} \sum_{k'=0,1} \frac{1}{n_{2,k} n_{2,k'}} \xi(Y_i, X_i, Z_i; z_k, x_l) \xi(Y_j, X_j, Z_j; z_k, x_l)\right] \middle| S_1 \quad (302)$$

$$= \mathbb{E}\left\{\mathbb{E}\left[\sum_{i=1}^{n_2} \sum_{k=0,1} \sum_{j \neq i, j \in S_2} \sum_{k'=0,1} \frac{1}{n_{2,k} n_{2,k'}} \xi(Y_i, X_i, Z_i; z_k, x_l) \right. \right. \quad (303)$$

$$\left. \left. \times \xi(Y_j, X_j, Z_j; z_k, x_l)\right) \middle| \{Z_i\}_{i \in S_2}, S_1\right\}$$

$$= \mathbb{E}\left[\frac{n_{2,0}(n_{2,0}-1)}{n_{2,0}^2} \mathbb{E}[\xi(Y, X, Z; z_0, x_l) | Z = z_0, S_1]^2 + \frac{n_{2,1}(n_{2,1}-1)}{n_{2,1}^2} \mathbb{E}[\xi(Y, X, Z; z_1, x_l) | Z = z_1, S_1]^2 \right. \quad (304)$$

$$\left. + 2(\mathbb{E}[\xi(Y, X, Z; z_0, x_l) | Z = z_0, S_1] \mathbb{E}[\xi(Y, X, Z; z_1, x_l) | Z = z_1, S_1]) \right] \middle| S_1$$

Combining equations 300, 302 and 305 yields:

$$\text{Var}(A_{1,n}(x_l) | S_1) = \mathbb{E}\left[\frac{n_2}{n_{2,k}} \sum_{k=0,1} \mathbb{P}[\eta_{n_1}^*(X, Y) \neq z_k, X = x_l | Z = z_k, S_1] \right. \quad (305)$$

$$\left. \times (1 - \mathbb{P}[\eta_{n_1}^*(X, Y) \neq z_k, X = x_l | Z = z_k, S_1]) \right] \middle| S_1$$

Moreover, the terms $\mathbb{P}[\eta_{n_1}^*(X, Y) \neq z_k, X = x_l | Z = z_k] \times (1 - \mathbb{P}[\eta_{n_1}^*(X, Y) \neq z_k, X = x_l | Z = z_k])$ for $k = 0, 1$ are bounded and the difference $\mathbb{E}\left[\frac{n_2}{n_{2,k}}\right] - \frac{1}{\mathbb{P}[Z=z_k]}$ is $o(1)$ and

independent from S_1 , thus:

$$Var(A_{1,n}(x_l)|S_1) = \mathbb{E} \left[\sum_{k=0,1} \frac{\mathbb{P}[\eta_{n_1}^*(X, Y) \neq z_k, X = x_l | Z = z_k, S_1]}{\mathbb{P}[Z = z_k]} \right] \quad (306)$$

$$\begin{aligned} & \times (1 - \mathbb{P}[\eta_{n_1}^*(X, Y) \neq z_k, X = x_l | Z = z_k, S_1]) \Big| S_1 \Big] + o(1) \\ & = \sum_{k=0,1} \frac{\mathbb{P}[\eta_{n_1}^*(X, Y) \neq z_k, X = x_l | Z = z_k, S_1]}{\mathbb{P}[Z = z_k]} \end{aligned} \quad (307)$$

$$\times (1 - \mathbb{P}[\eta_{n_1}^*(X, Y) \neq z_k, X = x_l | Z = z_k, S_1]) + o(1) \quad (308)$$

$$= \frac{P_{\xi, x_l | z_0, n_2}(1 - P_{\xi, x_l | z_0, n_2})}{\mathbb{P}[Z = z_0]} + \frac{P_{\xi, x_l | z_1, n_2}(1 - P_{\xi, x_l | z_1, n_2})}{\mathbb{P}[Z = z_1]} + o(1) \quad (309)$$

Where $P_{\xi, x_l | z_k, n_2} := \mathbb{P}[\eta_{n_1}^*(X, Y) \neq z_k, X = x_l | Z = z_k, S_1]$ is the missclassification probability for the pre-trained classifier associated with $X = x_l$, conditional on $Z = z_k$

To derive $Cov(A_{1,n}(x_l), A_{1,n}(x_{l'}))$ consider the variance of the sum $A_{1,n}(x_l) + A_{1,n}(x_{l'})$, define $\zeta(Y, X, Z; z_l, x_l, x_{l'}) := \xi(Y, X, Z; z_l, x_l) + \xi(Y, X, Z; z_l, x_{l'})$ and note that:

$$\mathbb{E}[n_2^{-1/2}(L_{n_2}(\eta_{n_1}^*, x_l) + L_{n_2}(\eta_{n_1}^*, x_{l'})) | S_1] = \mathbb{E} \left[\sum_{i=1}^{n_2} \left(\sum_{k=0,1} \frac{1}{n_{2,k}} \zeta(Y_i, X_i, Z_i; z_k, x_l) \right) \Big| S_1 \right] \quad (310)$$

Then, a derivation analogous to the variance of $A_{1,n}(x_l)$ yields:

$$\begin{aligned} Var(A_{1,n}(x_l) + A_{1,n}(x_{l'}) | S_1) &= \frac{(P_{\xi, x_l | z_0, n_2} + P_{\xi, x_{l'} | z_0, n_2})(1 - P_{\xi, x_l | z_0, n_2} - P_{\xi, x_{l'} | z_0, n_2})}{\mathbb{P}[Z = z_0]} \\ &+ \frac{(P_{\xi, x_l | z_1, n_2} + P_{\xi, x_{l'} | z_1, n_2})(1 - P_{\xi, x_l | z_1, n_2} - P_{\xi, x_{l'} | z_1, n_2})}{\mathbb{P}[Z = z_1]} + o(1) \end{aligned} \quad (311)$$

Thus:

$$Cov(A_{1,n}(x_l), A_{1,n}(x_{l'}) | S_1) = -\frac{P_{\xi, x_l | z_0, n_2} P_{\xi, x_{l'} | z_0, n_2}}{\mathbb{P}[Z = z_0]} - \frac{P_{\xi, x_l | z_1, n_2} P_{\xi, x_{l'} | z_1, n_2}}{\mathbb{P}[Z = z_1]} + o(1) \quad (312)$$

Now consider the conditional treatment probability estimators $P_{x_l|z_k, n_2} := \frac{P_{z_k, n_2}}{P_{z_k, x_l, n_2}}$. Note that:

$$\mathbb{E}\left[L_{n_2}(\eta_{n_1}^*, x_l)P_{x_{l'}|z_k, n_2}\right] \quad (313)$$

$$\begin{aligned} &= \mathbb{E}\left[\sum_{i=1}^{n_2} \sum_{j=1}^{n_2} \left(\sum_{k'=0,1} \frac{1}{n_{2,k}} \xi(Y_i, X_i, Z_i; z_{k'}, x_l)\right) \left(\frac{1}{n_{2,k}} \mathbf{1}\{X_j = x_{l'}, Z_j = z_k\}\right) \middle| S_1\right] \\ &= \mathbb{E}\left[\sum_{i=1}^{n_2} \left(\frac{\mathbf{1}\{l = l'\}}{n_{2,k}^2} \xi(Y_i, X_i, Z_i; z_k, x_l)\right. \right. \\ &\quad \left. \left. + \sum_{j \neq i, j \in S_2} \sum_{k'=0,1} \frac{1}{n_{2,k} n_{2,k'}} \xi(Y_i, X_i, Z_i; z_{k'}, x_l) \mathbf{1}\{X_j = x_{l'}, Z_j = z_k\}\right) \middle| S_1\right] \end{aligned} \quad (314)$$

Again, expand the second term and condition on instrument realizations to obtain:

$$\begin{aligned} &\mathbb{E}\left[\sum_{i=1}^{n_2} \sum_{j \neq i, j \in S_2} \sum_{k'=0,1} \frac{1}{n_{2,k} n_{2,k'}} \xi(Y_i, X_i, Z_i; z_{k'}, x_l) \mathbf{1}\{X_j = x_{l'}, Z_j = z_k\} \middle| S_1\right] \\ &= \mathbb{E}\left[\sum_{k'=0,1} \frac{(n_{2,k} - \mathbf{1}\{k = k'\})n_{2,k'}}{n_{2,k} n_{2,k'}} P_{\xi, x_l|z_k, n_2} P_{x_{l'}|z_0, n_2} \middle| S_1\right] \end{aligned} \quad (315)$$

Therefore:

$$Cov\left(A_{1,n}(x_l), \sqrt{n_2}(P_{x_{l'}|z_{k'}, n_2} - P_{x_{l'}|z_k, n_2})\right) = \frac{P_{\xi, x_l|z_k, n_2}(\mathbf{1}\{l = l'\} - P_{x_l|z_k, n_2})}{\mathbb{P}[Z = z_k]} + o(1) \quad (316)$$

Finally, the delta method yields:

$$\begin{aligned} &Cov\left(\sqrt{n_2}(P_{x_l|z_k, n_2} - P_{x_l|z_k}), \sqrt{n_2}(P_{x_{l'}|z_{k'}, n_2} - P_{x_{l'}|z_k})\right) \\ &= \mathbf{1}\{k = k'\} \frac{\mathbf{1}\{l = l'\} P_{x_l|z_k} - P_{x_l|z_k} P_{x_l|z_{k'}}}{\mathbb{P}[Z = z_k]} + o(1) \end{aligned} \quad (317)$$

Equations 309, 312, 316 and 317 characterize the entire variance-covariance matrix of the vector $\mathbf{\Pi}_{n_2}$ conditional on S_1 .

□

C.3 Estimation algorithm 2

1. Choose a measurable set $D \subset \mathbb{R}^d$.
2. Partition the sample in two $\tilde{K}$ folds $\kappa = 1, \dots, \tilde{K}$ of size $\tilde{n} = \lfloor n/\tilde{K} \rfloor$.
3. For every $\kappa = 1, \dots, \tilde{K}$:
 - (a) In J_κ^C , train a non-parametric classifier $\eta_{\kappa,n}^d(y, x)$ as follows:

i. Define:

$$\phi_{z_k, \kappa, n}(y, x_l) := P_{k, \kappa, n}^{-1} \left[\frac{1}{(\tilde{K} - 1)\tilde{n}h_n^d} \sum_{i \in J_\kappa^C} \mathbf{1}\{X_i = x_l, Z_i = z_k\} K_n(y - Y_i) \right] \quad (318)$$

Where $n_{\kappa,1} = |\{i \in J_\kappa^C : Z_i = z_k\}|$ and $P_{k, \kappa, n} = n_{k, \kappa}/\tilde{n}$.

ii. Then, define:

$$\Delta_{\kappa, n}(y, x_l) := \phi_{z_1, \kappa, n}(y, x_l) - \phi_{z_0, \kappa, n}(y, x_l) \quad (319)$$

iii. And:

$$\eta_{\kappa, n}^d(y, x_l) := \begin{cases} \mathbf{1}\{\Delta_{\kappa, n}(y, x_l) \geq 0\} & \text{if } |\Delta_{\kappa, n}(y, x_l)| \geq d_n \\ \mathbf{1}\{y \in D\} & \text{otherwise} \end{cases} \quad (320)$$

- (b) In J_κ , estimate the probabilities $\mathbb{P}[Z = z_k]$ for $k = 0, 1$. Denote the estimators as $P_{k, \kappa, n}$.
- (c) In J_κ , evaluate the weighted empirical loss:

$$L_\kappa(\eta_{\kappa, n}^d, x_l) := \frac{1}{\tilde{n}} \sum_{i \in J_\kappa} \left(\frac{\mathbf{1}\{\eta_{\kappa, n}^d(Y_i, X_i) = 1, Z_i = z_0, X_i = x_l\}}{P_{0, \kappa, n}} + \frac{\mathbf{1}\{\eta_{\kappa, n}^d(Y_i, X_i) = 0, Z_i = z_1, X_i = x_l\}}{P_{1, \kappa, n}} \right) \quad (321)$$

4. Calculate:

$$\Psi_{x_l}^d = \frac{1}{\tilde{K}} \sum_{\kappa=1}^{\tilde{K}} L_\kappa(\eta_{\kappa, n}^d, x_l) \quad (322)$$

C.4 Proofs for subsection 5.4

Proposition C.2.

Decompose:

$$\sqrt{\tilde{n}}(L_{\kappa}(\eta_{\kappa,n}^d, x_l) - \Psi_{x_l}) = \underbrace{\sqrt{\tilde{n}}(L_{\kappa}(\eta_{\kappa,n}^d, x_l) - L(\eta_{\kappa,n}^d, x_l))}_{:=A_{1,n}^d(x_l)} + \underbrace{\sqrt{\tilde{n}}(L(\eta_{\kappa,n}^d, x_l) - L(\eta^d, x_l))}_{:=A_{2,n}^d(x_l)} \quad (323)$$

1. $A_{2,n}^d = o_p(1)$.
2. $Cov(A_{1,n}^d, A_{2,n}^d) = 0$.
3. $Var(A_{2,n}^d) = o(1)$.

Proof of proposition C.2.

The proof follows by the same arguments from C.1 but now considering a region around the contact set that shrinks at a strictly slower rate than $r_n = \sqrt{\frac{\log(n_1)}{n_1 h_n}}$. In this region, the classifier is intentionally defined so that it incurs in classification mistakes. The only additional requirement is a stronger margin condition necessary to bound the potential additional loss from worsening the classifier.

As in the proof of C.1: for any possible realization of the classifier $\eta_{n_1}^d$ i) the classification loss from the contact set $\mathcal{C}_{x_l}$ is invariant, and ii) the margin condition implies that the classification loss near the boundary of the contact set is $\{y \in \mathcal{Y} : |\Delta(y, x_l)| \leq d_n\}$ is of order $o(n^{-1})$. Thus, asymptotic properties of the classifier $\eta_{n_1}^d$ might only differ from those of $\eta_{n_1}^*$ if its missclassification loss far from the boundary of the contact set is non-negligible.

For any $y \in \mathcal{Y}$ such that $|\Delta(y, x_l)| \geq d_n$, missclassification might occur if $|\Delta(y, x_l) - \Delta_n(y, x_l)| \geq d_n \geq r_n$ or if $|\Delta_n(y, x_l)| < d_n$. If $|\Delta_n(y, x_l)| < d_n$, there are two cases: i) when $|\Delta(y, x_l)| - r_n \geq d_n$, and ii) when $d_n \geq |\Delta(y, x_l)| - r_n$.

All the cases listed but the last imply $|\Delta(y, x_l) - \Delta_n(y, x_l)| \geq r_n$, and thus, are covered by the same argument used in the proof of proposition C.1 which uses lemma C.3 to bound the probability that $\|\Delta(y, x_l) - \Delta_n(y, x_l)\|_{\infty} \geq r_n$.

In the last case:

$$d_n \geq |\Delta(y, x_l)| - r_n \quad (324)$$

$$\implies d_n + r_n \geq |\Delta(y, x_l)| \quad (325)$$

$$\implies 2d_n \geq |\Delta(y, x_l)| \quad (326)$$

This implies that additional classification mistakes can only occur in the set $\{y \in \mathcal{Y} : |\Delta(y, x_l)| \leq 2d_n\}$. Thus, the arguments from the proof of proposition C.1 can be replicated replacing the rate $\sqrt{\frac{\log(n_1)}{n_1 h_n}}$ with the rate $d'_n = 2d_n$.

Partition the support of the outcome as follows:

$$\mathcal{Y} = \mathcal{S}_{x_l, d'_n} \cup \mathcal{M}_{x_l, d'_n} \cup \mathcal{C}_{x_l} \quad (327)$$

Where:

$$\mathcal{S}_{x_l, d'_n} = \left\{ y \in \mathcal{Y} : |\Delta(y, x_l)| > d'_n \right\} \quad (328)$$

$$\mathcal{M}_{x_l, d'_n} = \left\{ y \in \mathcal{Y} : 0 < |\Delta(y, x_l)| \leq d'_n \right\} \quad (329)$$

$$\mathcal{C}_{x_l} = \left\{ y \in \mathcal{Y} : |\Delta(y, x_l)| = 0 \right\} \quad (330)$$

Then, lemmas C.3 and C.4 and an argument analogous to the proof of proposition C.1 yield the desired results. □

Proposition C.3. *Set $\Psi_{x_l, \kappa, n}^d := L_\kappa(\eta_{\kappa, n}, x_l)$ and define:*

$$P_{x_l | z_k, \kappa, n} = \frac{\sum_{i \in J_\kappa} \mathbf{1}\{X_i = x_l, Z_i = z_k\}}{\sum_{i=1}^n \mathbf{1}\{Z_i = z_k\}}. \quad (331)$$

Define the random vector:

$$\boldsymbol{\theta}_{\kappa, n}^T = (P_{x_0 | z_0, \kappa, n}, \dots, P_{x_1 | z_0, \kappa, n}, P_{x_{L-1} | z_1, \kappa, n}, \dots, P_{x_{L-1} | z_1, \kappa, n}, \Psi_{x_0, \kappa, n}^d, \dots, \Psi_{x_{L-1}, \kappa, n}^d) \quad (332)$$

With expectation $\boldsymbol{\mu}_\theta^T = (P_{x_0 | z_0}, \dots, P_{x_1 | z_0}, P_{x_{L-1} | z_1}, \dots, P_{x_{L-1} | z_1}, \Psi_{x_0}, \dots, \Psi_{x_{L-1}})$ and and variance-covariance matrix $\text{Var}(\boldsymbol{\theta}_{\kappa, n}^T | S_1)$.

Then, a symmetric positive semi-definite matrix $\mathbf{\Omega}$ exist such that:

$$n\text{Var}(\boldsymbol{\theta}_{\kappa,n}^T | S_1) \xrightarrow{p} \mathbf{\Omega} \quad (333)$$

And:

$$\sqrt{n}(\boldsymbol{\theta}_{\kappa,n} - \boldsymbol{\mu}_\theta) \xrightarrow{d} \mathcal{N}(0, \mathbf{\Omega}) \quad (334)$$

Proof of proposition C.3.

An analogous argument to proposition 5.1 replacing $\mathbf{1}\{\eta_{n_1}^*(Y, X, Z; z_k, x_l) \neq k, X = x_l, Z = z_k\}$ with $\xi(Y, X, Z; z_k, x_l) = \mathbf{1}\{\eta_{n_1}^d(Y, X, Z; z_k, x_l) \neq k, X = x_l, Z = z_k\}$ and proposition C.1 with proposition C.2 yields asymptotic normality for non-degenerate directions, more specifically:

For any fixed $\mathbf{a} \in \mathbb{R}^{3L}$ such that there exists $c_a > 0$ that satisfies $\mathbb{P}[\mathbf{a}^T \mathbf{V}_{\kappa,n} \mathbf{a} > c_a] \rightarrow 1$:

$$\sup_{t \in \mathbb{R}} \left| \mathbb{P} \left[\frac{\mathbf{a}^T \sqrt{\tilde{n}} (\boldsymbol{\theta}_{\kappa,n} - \boldsymbol{\mu}_\theta)}{\sqrt{\mathbf{a}^T \mathbf{V}_{\kappa,n} \mathbf{a}}} \leq t \right] - \Phi(t) \right| \xrightarrow{p} 0. \quad (335)$$

Where $n_2 \mathbf{V}_{\kappa,n}$ is the variance of the vector $\boldsymbol{\theta}_{\kappa,n}$ conditional on J_κ^C (i.e. the data in which the classifier is trained).

By lemma C.8, the entries of $\mathbf{V}_{\kappa,n}$ depend on conditional treatment probabilities and missclassification probabilities $\mathbb{P}[\eta_{\kappa,n}^d(Y, X) \neq k, X = x_l, Z = z_k]$. Then, lemma C.9 implies that the sequence $\mathbf{V}_{S_1,n}$ converges to a deterministic limit $\mathbf{V}$. Thus:

$$\mathbf{a}^T \sqrt{\tilde{n}} (\boldsymbol{\theta}_{\kappa,n} - \boldsymbol{\mu}_\theta) \xrightarrow{d} \mathcal{N}(0, \mathbf{a}^T \mathbf{V} \mathbf{a}) \quad (336)$$

More over, for any degenerate direction $\mathbf{a}$ such that $\mathbf{a}^T \mathbf{V} \mathbf{a} = 0$, the sequence converges to a degenerate normal with variance zero.

Thus, the Cramér-Wold device implies that:

$$\sqrt{\tilde{n}}(\boldsymbol{\theta}_{\kappa,n} - \boldsymbol{\mu}_\theta) \xrightarrow{d} \mathcal{N}(0, \mathbf{\Omega}) \quad (337)$$

□

Proof of theorem 5.2. Proposition C.3 implies that each of the terms $\sqrt{\tilde{n}}(\boldsymbol{\theta}_{\kappa,n} - \boldsymbol{\mu}_\theta)$ is asymptotically normal with variance $\mathbf{\Omega}$ for $\kappa = 1, \dots, \tilde{K}$. Lemma C.10 implies

that each of the terms is asymptotically independent. Thus, the vector the vector $\sqrt{n}(\boldsymbol{\theta}_n - \boldsymbol{\theta})$ is asymptotically normal and its asymptotic variance is equal to $\boldsymbol{\Omega}$. $\square$

Lemma C.9.

$$\lim_{n \rightarrow \infty} \mathbb{P}[\eta_{\kappa,n}^d(y, x_l) \neq k, X = x_l, Z = z_k | S_1] = \mathbb{P}[\eta^*(y, x_l) \neq k, X = x_l, Z = z_k] \quad (338)$$

Lemma C.10. (*Influence function for K-folds cross-fitting*)

For any fixed classifier η , define the random variable $\xi_\eta(Y, X, Z; z_k, x_l) = \mathbf{1}\{\eta(Y, X) \neq k, X = x_l, Z = z_k\}$ and let $\mu_\eta(z_k, x_l)$ denote its mean.

Then:

$$\begin{aligned} L_\kappa(\eta_{\kappa,n}, x_l) - \Psi_{x_l}^d &= \frac{1}{\tilde{n}} \sum_{i \in J_\kappa} \left[\sum_{k=0,1} \frac{P_{z_k} - \mathbf{1}\{Z_i = z_k\}}{P_{z_k}^2} \mu_{\eta^d}(z_k, x_l) \right. \\ &\quad \left. + \frac{1}{P_{z_k}} \xi_{\eta^d}(Y, X, Z; z_k, x_l) \right] + o_p(n^{-1/2}) \end{aligned} \quad (339)$$

Proof of lemma C.9.

$$\mathbb{P}[\eta_{\kappa,n}^d(Y, X) \neq k, X = x_l, Z = z_k | S_1] - \mathbb{P}[\eta^*(Y, X) \neq k, X = x_l, Z = z_k] \quad (340)$$

$$\begin{aligned} &\leq \int_{\mathcal{Y}} \mathbf{1}\{\eta_{\kappa,n}^d(y, x_l) \neq \eta_{\kappa,n}^*(y, x_l)\} f_Y(y) dy \\ &= \int_{\mathcal{S}_{x_l, d'_n}} \mathbf{1}\{\eta_{\kappa,n}^d(y, x_l) \neq \eta_{\kappa,n}^*(y, x_l)\} f_Y(y) dy \end{aligned} \quad (341)$$

$$\begin{aligned} &\quad + \int_{\mathcal{M}_{x_l, d'_n}} \mathbf{1}\{\eta_{\kappa,n}^d(y, x_l) \neq \eta_{\kappa,n}^*(y, x_l)\} f_Y(y) dy \\ &\quad + \int_{\mathcal{C}_{x_l}} \mathbf{1}\{\eta_{\kappa,n}^d(y, x_l) \neq \eta_{\kappa,n}^*(y, x_l)\} f_Y(y) dy \\ &\leq \int_{\mathcal{Y}} \mathbf{1}\{|\Delta_{\kappa,n}^d(Y, x_l) - \Delta(Y, x_l)| > d'_n\} f_Y(y) dy + \int_{\mathcal{M}_{x_l, d'_n}} f_Y(y) dy \end{aligned} \quad (342)$$

$$\leq \mathbb{P}[|\Delta_{\kappa,n}^d(Y, x_l) - \Delta(Y, x_l)|_\infty > d'_n] + K d_n'^\alpha \quad (343)$$

$$= o(d_n'^\alpha) \quad (344)$$

Where the first inequality follows from elemental properties. The second line par-

titions $\mathcal{Y}$ into the sets defined in the proof of proposition C.2. The third line notes that in the sets $\mathcal{S}_{x_l, d'_n}$ and $\mathcal{C}_{x_l}$ the classifier $\eta_{\kappa, n}^d$ can only disagree with $\eta_{\kappa, n}^d$ if $|\Delta_n^d(Y, x_l) - \Delta(Y, x_l)| > d'_n$, and the last uses the margin condition from assumption 5.1.3. $\square$

Proof of lemma C.10.

Observe that $\mathbb{E}[\xi_\eta(Y, X, Z; z_k, x_l)] = L(\eta, x_l)$.

The estimator based on the semi-deterministic classifier trained on J_κ^C and evaluated in J_κ can be recast as:

$$L_\kappa(\eta_{\kappa, n}^d, x_l) = \frac{1}{\tilde{n}} \sum_{i \in J_\kappa} \sum_{k=0,1} \frac{1}{P_{z_k, \kappa, n}} \xi_{\eta_{\kappa, n}^d}(Y, X, Z; z_k, x_l) \quad (345)$$

$$= \frac{1}{\tilde{n}} \sum_{i \in J_\kappa} \sum_{k=0,1} \left[\frac{1}{P_{z_k}} \xi_{\eta^d}(Y, X, Z; z_k, x_l) + \left(\frac{1}{P_{z_k, \kappa, n}} - \frac{1}{P_{z_k}} \right) \xi_{\eta^d}(Y, X, Z; z_k, x_l) \right. \quad (346)$$

$$\left. + \frac{1}{P_{z_k}} \left(\xi_{\eta_{\kappa, n}^d}(Y, X, Z; z_k, x_l) - \xi_{\eta^d}(Y, X, Z; z_k, x_l) \right) + \left(\frac{1}{P_{z_k, \kappa, n}} - \frac{1}{P_{z_k}} \right) \left(\xi_{\eta_{\kappa, n}^d}(Y, X, Z; z_k, x_l) - \xi_{\eta^d}(Y, X, Z; z_k, x_l) \right) \right]$$

Now consider the random sum:

$$\frac{1}{\tilde{n}} \sum_{i \in J_\kappa} \sum_{k=0,1} \frac{1}{P_{z_k}} \left(\xi_{\eta_{\kappa, n}^d}(Y, X, Z; z_k, x_l) - \xi_{\eta^d}(Y, X, Z; z_k, x_l) \right) \quad (347)$$

Note that both $\xi_{\eta_{\kappa, n}^d}(Y, X, Z; z_k, x_l)$ and $\xi_{\eta^d}(Y, X, Z; z_k, x_l)$ are indicators, thus, their moments exist and are bounded. Moreover, all the terms in the sum are independent and identically distributed, and the probabilities P_{z_k} for $k = 0, 1$ bounded away from zero, furthermore, part 1 of proposition C.2 implies that:

$$\mathbb{E} \left[\sum_{k=0,1} \frac{1}{P_{z_k}} \left(\xi_{\eta_{\kappa, n}^d}(Y, X, Z; z_k, x_l) - \xi_{\eta^d}(Y, X, Z; z_k, x_l) \right) \middle| S_1 \right] = o_p(n^{-1/2}) \quad (348)$$

Moreover:

$$\mathbb{E}\left\{\left[\sum_{k=0,1} \frac{1}{P_{z_k}} \left(\xi_{\eta_{\kappa,n}^d}(Y, X, Z; z_k, x_l) - \xi_{\eta^d}(Y, X, Z; z_k, x_l)\right)\right]^2 \middle| S_1\right\} \quad (349)$$

$$= \int_{\mathcal{Y}} \left[\sum_{k=0,1} \frac{1}{P_{z_k}} \left(\xi_{\eta_{\kappa,n}^d}(y, x_l, z_k; z_k, x_l) - \xi_{\eta^d}(y, x_l, z_k; z_k, x_l)\right)\right]^2 f_{Y,X,Z}(y, x_l, z_k) dy \quad (350)$$

$$\leq K \int_{\mathcal{Y}} \sum_{k=0,1} \mathbf{1}\{\eta_{\kappa,n}^d(y, x_l) \neq \eta^d(y, x_l)\} f_{Y,X,Z}(y, x_l, z_k) dy \quad (351)$$

$$= o_p(1) \quad (352)$$

Where the last equality follows from a similar reasoning to the proof of lemma C.9.

Then, 352 and 348 imply that the sum in 347 is $o_p(n^{-1/2})$.

On the other hand, assumption 5.1.1, Slutsky's theorem and the delta method imply that:

$$\begin{aligned} & \frac{1}{\tilde{n}} \sum_{i \in J_{\kappa}} \sum_{k=0,1} \left(\frac{1}{P_{z_k, \kappa, n}} - \frac{1}{P_{z_k}} \right) \xi_{\eta^d}(Y, X, Z; z_k, x_l) \\ &= \frac{1}{\tilde{n}} \sum_{i \in J_{\kappa}} \sum_{k=0,1} \frac{P_{z_k} - \mathbf{1}\{Z_i = z_k\}}{P_{z_k}^2} \mu_{\eta^d}(z_k, x_l) + o_p(n^{-1/2}) \end{aligned} \quad (353)$$

□

C.5 Proofs for subsection 5.5

Proof of theorem 5.3: Let n' be either $n/2$ or n depending on the estimator. Let $\mathbf{\Omega}_n$ denote $n\text{Var}(\boldsymbol{\theta}_n)$ for the estimator based on a semi-deterministic classifier, or $\frac{n'}{2}\text{Var}(\boldsymbol{\theta}_n|S_1)$ for the main estimator.

Theorems 5.1 and 5.2 along with lemma C.8 imply that the sequence $n'\boldsymbol{\Sigma}_n^{-1/2}\text{Var}(\boldsymbol{\theta}_n)\boldsymbol{\Sigma}_n^{-1/2}$ converges in probability to $\mathbf{I}_m$ and the random sequence $\sqrt{n_2}\mathbf{\Omega}_n^{-1/2}(\boldsymbol{\theta}_n - \boldsymbol{\mu}_\theta)$ converges in distribution to a multivariate, non-degenerate normal with variance-covariance matrix $\mathbf{I}_m$.

Thus, assumption 3(a) from Cox and Shi (2023) is satisfied. $\square$

D Examples of shape restrictions encompassed by the framework

Multivalued ordered monotonicity: Ordered monotonicity captures the idea that treatment values are strictly ordered (e.g, by intensity) and that as the instrument increases, units may only switch to weakly higher treatment levels. That is, for any pair of indices $k' > k > 0$, response types such that $(t^Z)_k > (t^Z)_{k'}$ are ruled out. This restriction can be easily accommodated in the framework of assumption 2.2 by imposing that response types as the one described before occur with zero probability.

In their seminal paper about IV with heterogeneous treatment effects, [Imbens and Angrist \(1994\)](#) showed that if both the instrument and treatment are binary, and treatment is weakly monotonic in the instrument, the classical Wald estimand identifies a LATE for units that strictly increase their realized treatment when the instrument changes. This result has motivated the use of binary instruments and treatments in a wide range of empirical applications.

In a subsequent paper, [Angrist and Imbens \(1995\)](#) extended their analysis to multivalued ordered treatments. In this setting, no LATE is point identified, but the Wald estimand continues to identify a convex combination of LATEs among units that are induced to take strictly higher treatments.

Despite the interpretation provided by [Angrist and Imbens \(1995\)](#), it is common practice in applied work to binarize monotone multivalued treatments. That is, the researcher selects a threshold $\bar{x} \in \mathcal{X}$ and defines a binary treatment indicator $X^{Bin} = \mathbf{1}_{\{X \geq \bar{x}\}}$, thereby reducing the setting to the binary treatment case. A justification for this practice is that the resulting Wald estimand has a simpler interpretation.

However, this practice has two limitations, first, there are instances in which the binarized model is rejected by a specification test, even though the correctly specified multivalued model is not. Second, after binarization, the Wald estimand identifies a weighted sum of LATEs for certain response types, but there is no guarantee that the weights add up to one. As a result, the estimand cannot generally be interpreted as LATE or even as a convex combination of LATEs. An in-depth analysis of the second limitation is provided in [Rose and Shem-Tov \(2024\)](#).

To illustrate the first limitation, suppose that $\mathcal{X} = \{x_0, x_1, x_2\}$, $\mathcal{Y} = \{0, 1\}$ and $\mathcal{Z} = \{z_0, z_1\}$. Assume that half the population has the instrument-response type

(x_0, x_1) , one fourth (x_1, x_2) , and the last fourth (x_2, x_2) ¹¹. Suppose that potential outcomes are homogeneous so that $Y(x_0) = 0$, $Y(x_1) = 1$ and $Y(x_2) = 1$ for all $i \in \mathcal{I}$. Now suppose that a researcher sets $\bar{x} = x_2$. It is easy to see that units with instrument-response type (x_0, x_1) will not change their treatment status with respect to the artificially defined treatment variable X^{Bin} as Z increases, but, their potential outcomes will, violating the exclusion restriction in the binarized model.

Furthermore, [Kitagawa \(2015\)](#) shows that if instrument validity holds in the binary case, the following equation must be satisfied:

$$\mathbb{P}[Y = 1, X^{Bin} = 0 | Z = 0] \geq \mathbb{P}[Y = 1, X^{Bin} = 0 | Z = 1] \quad (354)$$

In the example, the left hand side is equal to $\frac{1}{4}$ while the right hand side is $\frac{1}{2}$, thus, instrument validity would be rejected for the binarized model, even if it holds in the true multivalued model with the right (multivalued ordered monotone) instrument-response restriction.

Furthermore, as studied by [Rose and Shem-Tov \(2024\)](#), even if equation 355 is not violated, the Wald estimand obtained using Z as an instrument for X^{Bin} may be difficult to interpret. To see this problem, suppose that the instrument-response types (x_0, x_1) , (x_1, x_2) , and (x_2, x_2) occur with equal probability $\frac{1}{3}$ in the population. In this case, the Wald estimand becomes:

$$W = \frac{\frac{1}{3}\mathbb{E}[Y(x_1) - Y(x_0) | i \in (x_0, x_1)] + \frac{1}{3}\mathbb{E}[Y(x_2) - Y(x_1) | i \in (x_1, x_2)]}{\frac{1}{3}} \quad (355)$$

While the numerator is a weighted sum of LATEs, the weights do not necessarily sum to one. Consequently, the estimand W cannot be interpreted as LATE or a convex combination of LATEs. Moreover, if instead of $\frac{1}{3}$, the denominator, i.e. the population probability associated with instrument-response type (x_1, x_2) , was small, the magnitude of the Wald estimand could be arbitrarily large. Thus, even if binarizing a monotone multivalued treatment may simplify the model, it can produce estimands that are both biased and difficult to interpret.

These concerns provide a strong motivation for retaining a correctly specified monotone model when the treatment is multivalued.

¹¹Recall that the instrument-response type $t = (x_l, x_{l'})$ represents units that realize treatment x_l when $Z = z_0$ and $x_{l'}$ when $Z = z_1$.

Generalizing monotonicity to settings with multiple instruments is nontrivial. Angrist and Imbens (1995) proposed one approach, which imposes binary monotonicity in one direction between every pair of treatment values. This definition was later criticized by Mogstad et al. (2021) who proposed an alternative extension. More recently, Goff (2020) introduced a tractable notion –vector monotonicity– which ensures identification of a series of LATEs.

Classical monotonicity and its three aforementioned generalizations can be expressed as restrictions that rule out specific response types, that is, they fall within assumption 2.2.

Unordered monotonicity and its refinements: Heckman and Pinto (2018) proposed an alternative notion of monotonicity for settings in which no natural ordering over treatment values exists. Intuitively, unordered monotonicity requires that, as the instrument changes from one value to another, a subset of treatments $X^I \subset \mathcal{X}$ are equally incentivized relative to the treatments in $\mathcal{X} \setminus X^I$. This restriction not only implies that take-up of the incentivized treatments must weakly increase but also rules out unit-level switches within the set of promoted treatments X^I , or its complement $\mathcal{X} \setminus X^I$. This is another restriction on heterogeneity obtained from ruling out particular response types and thus, it can be accommodated within the framework of this paper.

Heckman and Pinto (2018) argue that the identifying power of unordered monotonicity can be strengthened by further restricting the set of admissible response types. For example, suppose that $x_0 \in X^I$ represents a discounted car, and $x_1, x_2 \in \mathcal{X} \setminus X^I$ represent, respectively, a similar full-priced car and a pick-up truck. One could impose the restriction that only individuals who would have chosen x_1 in the absence of the discount may switch to x_0 as a result of the discount, ruling out switches from x_2 to x_0 . Unordered monotonicity augmented with this type of restriction remains within the framework of this paper.

Other response type restrictions: The identifying power of response type restrictions extends beyond ordered and unordered monotonicity. Goff (2024) shows that the ability of restrictions on treatment-response heterogeneity to point identify some LATE without restricting the outcome space is underpinned by the set of treatment-response types it allows. Leveraging this insight, Goff (2024) identifies a series of novel response-type restrictions that guarantee point identification of at least one LATE, and a procedure to find other such restrictions.

To advance the interpretability of IV models with multiple instruments and treatments, [Lee and Salanié \(2020\)](#) investigated identification when multiple instruments are available and each instrument targets (i.e. incentivizes) adoption of one particular treatment more than the rest. In particular, [Lee and Salanié \(2020\)](#) provide a general framework and then –to achieve point or partial identification– introduce the notions of strict targeting, and one-to-one targeting. Multiple instances of these two assumptions fit into the framework studied in this paper.

In a closely related paper, [Bai et al. \(2024c\)](#) study the case in which every treatment value is uniquely incentivized by a single instrument. For this particular targeting structure they show that multiple LATEs are point identified and derive the sharp testable implications of the model. However, the assumption that each instrument uniquely targets one –and only one– treatment (without affecting any other comparison) might be overly restrictive in multiple empirical settings. The framework developed in this paper can be used to assess the validity of the complete model proposed by [Bai et al. \(2024c\)](#), as well as more plausible relaxations.

Other linear restrictions on instrument-response types: Rather than ruling out specific response types, a less restrictive approach to IV identification is to place bounds on the prevalence or relative frequency of certain behaviors or instrument-response types. A notable example is the “more compliers than defiers” condition of [De Chaisemartin \(2017\)](#). Another example is [Richardson and Robins \(2010\)](#), who derive bounds on the outcome distribution under assumptions on the fraction of always takers in the population.

Similar assumptions are also used as intermediate steps in sensitivity analysis methods that assess the robustness of IV estimands to relaxations of monotonicity ([Huber \(2014\)](#), [Noack \(2021\)](#), [Yap \(2025\)](#)). The tests from this paper can be used to falsify these types of restrictions on their own or combined with response-type restrictions.

E Examples of submonotonicity

Ordered monotonicity: Ordered monotonicity captures the notion that an implicit ranking (a total order) of alternatives exists, and, as the instrument increases, adoption of higher ranked alternatives is weakly incentivized. This assumption is implicit in most random utility models in which a mean structural utility function is common to all members of the population and unit-specific unobservables are additively separable. Thus, holding unobserved heterogeneity fixed, as observables change all decision makers agree on which alternatives yield higher utility levels and thus are being promoted.

Because submonotonicity allows for general binary relations, ordered monotonicity is a particular case.

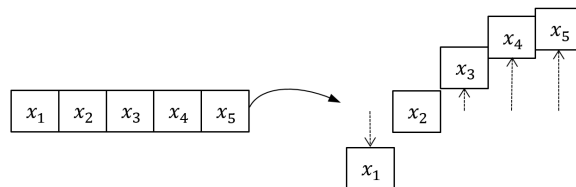


Figure 6: Ordered monotonicity

Notes: Visual representation of ordered monotonicity. The left of each figure represents alternatives before the intervention ($Z = z_0$). The right represents the effect of the intervention ($Z = z_1$). The higher an alternative is depicted on the left side, the more it is being incentivized. The black arrows represent the relative magnitude of the incentives.

Unordered monotonicity: Unordered monotonicity seeks to abstain from imposing an ex-ante ordering of alternatives and instead simply acknowledging that instruments incentivize adoption of some treatments relative to the rest. This idea is formalized by assuming that a subset of treatments $S \subset \mathcal{X}$ exists such that all alternatives in S are being promoted and no alternative in its complement $S' := \mathcal{X} \setminus S$ is.

While this assumption is general in spirit and in the multiple instruments setting unordered monotonicity is not implied nor implies ordered monotonicity, in the binary instrument case, it is a strictly more restrictive assumption. The reason is that unordered monotonicity not only requires that alternatives in S are being promoted, but also that they are being promoted with exactly the same intensity so that switches within the set S (and its complement) occur with zero probability.

The previous point is easy to see by noting that the binary relation associated to

unordered monotonicity is defined as follows:

$$x_l \geq^R x_{l'} \iff \begin{cases} x_l, x_{l'} \in S & or \\ x_l, x_{l'} \in S' & or \\ x_l \in S, x_{l'} \in S' \end{cases} \quad (356)$$

This restriction induces a partial order with two indifference classes S and S' , so that if $x_l, x_{l'} \in S$ then $x_l \geq^R x_{l'}$ and $x_{l'} \geq^R x_l$. The name partial order is misleading in this setting –it suggests weakening a total order–. The weakening is achieved by imposing that different treatments belong to the same equivalence class, that is, adding more ordered pairs to the binary relation which imposes additional restrictions on heterogeneity.

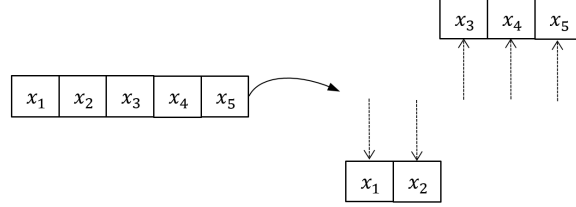


Figure 7: Unordered monotonicity

Notes: Visual representation of unordered monotonicity. The left of each figure represents alternatives before the intervention ($Z = z_0$). The right represents the effect of the intervention ($Z = z_1$). The higher an alternative is depicted, the more it is incentivized. The black arrows represent the relative magnitude of the incentives. The set $S = \{x_3, x_4, x_5\}$ represents the alternatives that are being incentivized relative to $S' = \{x_1, x_2\}$. Note that comparisons between alternatives within the same set are not altered.

Furthermore, note that if S has only one element s (i.e. $|S| = 1$), then, an instrument that encourages adoption of s (Lee and Salanié (2020)) or strictly targets s (Bai et al. (2024c)) satisfies a particular case of unordered monotonicity. Targeting or encouragement are not implied nor imply unordered monotonicity in the multiple instruments case.

The sets S and S' could be further partitioned into three sets instead of two. For example, let $S = S_1 \cap S_2$ and $S' = S'_1 \cap S'_2$ and let alternatives in S_1 and S'_1 represent compact cars and alternatives in S_2 and S'_2 SUVs. Suppose that S represents the set of all discounted vehicles (SUVs and compact) and S' the set of vehicles without a discount. In this setting, one plausible restriction is that the discount on compact cars

only affects decision makers who, absent the discount, would buy alternatives in S'_1 ; and the discount on SUVs only affects decision makers who would pick alternatives in S'_2 . Thus, switches from S'_1 to S_2 or S'_2 to S_1 are assumed to occur with zero probability. This restriction is no longer represented by a partial order, but it is derived from adding ordered pairs to the partial order described in equation 356, thus, it is a submonotone restriction associated to the extension of a partial order.

Relaxations of unordered monotonicity: The assumption that all alternatives are equally discounted might be too strong, instead, one could assume that there are N discount bins and that comparisons within the same bin are not affected but comparisons across different bins are –always in favor of alternatives with larger discounts. For example, if $N = 3$ there could be a set of heavily discounted cars, a set of cars with payment facilities, and a set of cars that have to be payed upfront. This situation also corresponds to submonotonicity with respect to a partial order but now with three indifference classes.

Additional restrictions, as the one considered in the compact cars and SUVs examples could also be added across all three discount bins. The resulting restriction would remain within the submontonicity framework.

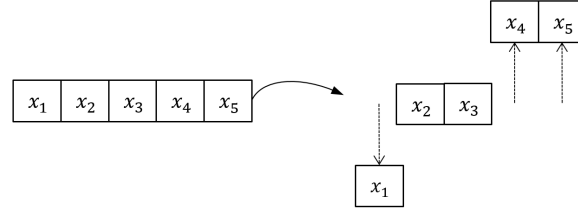


Figure 8: Submonotonicity with respect to a partial order with three indifference classes

Notes: Visual representation of submonotonicity with respect to a partial order with three indifference classes. The left of each figure represents alternatives before the intervention ($Z = z_0$). The right represents the effect of the intervention ($Z = z_1$). The higher an alternative is depicted, the more it is incentivized. The black arrows represent the relative magnitude of the incentives. The set $\{x_4, x_5\}$ represents the alternatives that are being incentivized the most (heavily discounted), the set $\{x_2, x_3\}$ the alternatives that for which payment facilities are offered (i.e. weakly incentivized). Alternative x_1 is not being incentivized in any way. Note that comparisons between alternatives within the same set are not altered and thus, their difference in height remains equal.

Relaxations of encouragement: Now consider an intervention in which an instrument encourages adoption of certain treatments in $S \subset \mathcal{X}$, but there is a concern that the instrument might also affect utility from other treatments. For example,

consider an an intervention aimed at increasing sports participation by providing information. It is clear that such an intervention could be used as an instrument for sports practice, but the intervention may also promote other activities that develop motor or social skills such as community service or playing board-games. Hence, the restriction that the instrument is not affecting comparisons between treatments other than practicing sports might not be plausible and a weaker assumption might be preferred.

One such weaker assumption could be that alternatives x_1 and x_2 which represent sports are certainly being promoted relative to all others (since they are the main target of the intervention), but pairwise comparisons between other treatments might also affected and the direction of this effect is heterogeneous across the population. Thus, restrictions on the direction of the flows between non-sports activities cannot be credibly imposed. This situation is encoded by submonotonicity with respect to a quasi-order, i.e. a binary relation that is transitive but not complete.

If x_5 and x_4 represent different sports and x_3 , x_2 and x_1 other activities, the specific submonotonicity assumption would be $x_l \geq^R x_{l'}$ for $l = 5, 4$ and $l' = 3, 2, 1$ but neither $x_l \geq^R x_{l'}$ nor $x_{l'} \geq^R x_l$ for $l, l' = 5, 4$ or $l, l' = 3, 2, 1$. A graphical representation is given in figure ??.

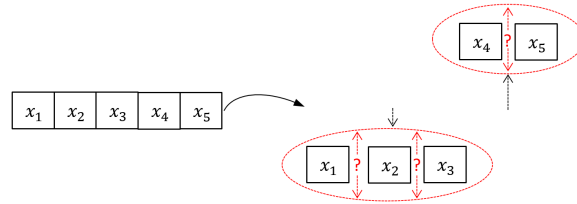


Figure 9: Submonotonicity with respect to a quasi-order - Sports example

Notes: Visual representation of submonotonicity with respect to a quasi-order. The left of each figure represents alternatives before the intervention ($Z = z_0$). The right represents the effect of the intervention ($Z = z_1$). The higher an alternative is depicted on the right side, the more it is incentivized. The set $\{x_4, x_5\}$ represents the alternatives that are being incentivized the most (sports), the set $\{x_1, x_2, x_3\}$ the alternatives that are not being incentivized. The black arrows represent the relative magnitude of the incentives but the red arrows represent relative uncertainty about this effect within the set of circled alternatives. That is, within the red ellipses no restriction on how pairwise comparisons are affected is imposed.

Note that relaxing the completeness requirement allows for more heterogeneity. In particular, it allows decision makers to change their relative ranking of the same pair of alternatives in different directions. This situation cannot be captured by a random utility model in which the effect of Z is common to all decision makers, instead, it is

allowing the effect of Z to be determined by unobserved unit characteristics in the spirit of a random coefficients model.

Now consider an intervention aimed at preventing students from dropping college to take part-time jobs. Suppose this goal is achieved by providing additional classes aimed at developing professional qualifications but also by providing information on the importance of college completion.

It is clear that such an intervention would promote staying in college relative to taking part-time jobs, however, the classes could help students find full-time jobs, even without the degree. On the other hand, the information could reduce the willingness of students to accept any type of job. If the latter effect is stronger for full-time jobs (because they are more demanding and thus increase the risk of not finishing college), some students might choose to leave a full-time job and take a part-time one. Thus, it might be desirable to only impose the restriction that college is being incentivized relative to part-time jobs, but nothing else.

This restriction can also be accommodated by submonotonicity by assuming that college degrees x_4 and x_5 are being incentivized relative to half-time jobs x_3 and x_2 but no restrictions involving full-time jobs x_1 are imposed. Formally, $x_l \geq^R x_{l'}$ for $l = 1, 2$ and $l' = 3, 4$.

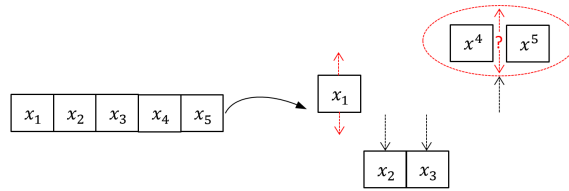


Figure 10: Submonotonicity with respect to a transitive quasi-order - College and full-time jobs example.

Notes: Visual representation of submonotonicity with respect to a quasi-order. The left of each figure represents alternatives before the intervention ($Z = 0$). The right represents the effect of the intervention ($Z = 1$). The higher an alternative is depicted, the more it is incentivized. The set $\{x_4, x_5\}$ represents the alternatives that are being incentivized the most (sports), the set $\{x_1, x_2, x_3\}$ the alternatives that are not being incentivized. The black arrows represent the relative magnitude of the incentives but the red arrows represent relative uncertainty about this effect relative to alternatives in the same class. Thus, within the red circle, no stance is taken on how pairwise comparisons are affected. Furthermore, no stance on whether x_1 is being promoted or not relative to x_4 , x_5 or even x_2 and x_3 is taken.

Relaxing transitivity/encouraging experimentation: The submonotonicity framework does not require transitivity of the binary relation $\geq^R$ –i.e. it allows for cycles in $\geq^R$. This observation raises the question of how can such a restriction be

interpreted –specially in light of [Goff \(2024\)](#) who shows that some models associated to non-transitive binary relations can point identify LATEs for some specific response groups.

To answer this question, consider a three alternatives example and suppose that $x_1 >^R x_2$, $x_2 >^R x_3$ and $x_3 >^R x_1$ (where $x_l >^R x_{l'}$ indicates $x_l \geq^R x_{l'}$ but not $x_{l'} \geq^R x_l$).

This restriction implies that:

- Any decision maker who choses x_1 when $Z = 0$ will now choose x_1 or x_3 .
- Any decision maker who choses x_2 when $Z = 0$ will now choose x_2 or x_1 .
- Any decision maker who choses x_3 when $Z = 0$ will now choose x_3 or x_2 .

Furthermore, if the assumption that there are no always takers at $\mathcal{X}$ is added (i.e. all types of always takers are ruled out), then:

- Any decision maker who choses x_1 when $Z = 0$ will now choose x_3 .
- Any decision maker who choses x_2 when $Z = 0$ will now choose x_1 .
- Any decision maker who choses x_3 when $Z = 0$ will now choose x_2 .

In the first case, choices when $Z = 0$ restrict potential choices when $Z = 1$. In the second, there is a one to one correspondence between one and the other. In both cases, if decision makers change their choice, they pick an alternative different than the one they would have chosen absent the instrument. The result is a situation where every alternative is promoted among subjects who do not choose it at the initial value of the instrument; and discouraged among those that already prefer it over the rest. The previous discussion suggests interpreting the effect of the instrument as promoting experimentation or variety.

Interpreting more general non-transitive restrictions is not straightforward, but, in all cases, submonotonicity induces restrictions on heterogeneity that can be expressed as a correspondence of choices when $Z = 0$, and thus, can be seen as restricting heterogeneity in terms of status-quo choices.

□

Table 3: Bias, overlap coverage, and joint overlaps and CTPs coverage for sample splitting estimators. Uniform kernel.

			$h_n = n^{-1/3}$			$h_n = n^{-1/4}$			$h_n = n^{-1/5}$			
$n = 1000$ $\mathbb{E}n_{z_0, x_l} = 165, \mathbb{E}n_{z_1, x_l} = 330$			$C_{\frac{2}{3}}$	$C_{\frac{1}{2}}$	$C_{\frac{1}{3}}$	$C_{\frac{2}{3}}$	$C_{\frac{1}{2}}$	$C_{\frac{1}{3}}$	$C_{\frac{2}{3}}$	$C_{\frac{1}{2}}$	$C_{\frac{1}{3}}$	
$w = 0$	$\mu_2 = 0$	$\Psi_{x_0, n} - \Psi_{x_0}$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
		$\Psi_{x_0} \in C_{0.95}$	0.94	0.97	0.94	0.95	0.96	0.95	0.96	0.96	0.94	
		$\theta_0 \in E_{0.95}$	0.94	0.94	0.96	0.93	0.95	0.92	0.95	0.94	0.93	
	$\mu_2 = 0.5$	$\Psi_{x_0, n} - \Psi_{x_0}$	0.03	0.04	0.04	0.02	0.03	0.03	0.02	0.02	0.03	
		$\Psi_{x_0} \in C_{0.95}$	0.90	0.84	0.74	0.93	0.90	0.80	0.94	0.92	0.87	
		$\theta_0 \in E_{0.95}$	0.85	0.75	0.58	0.89	0.82	0.67	0.89	0.85	0.80	
	$\mu_2 = \tau$	$\Psi_{x_0, n} - \Psi_{x_0}$	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	
		$\Psi_{x_0} \in C_{0.95}$	0.94	0.93	0.91	0.95	0.95	0.96	0.94	0.95	0.94	
		$\theta_0 \in E_{0.95}$	0.92	0.90	0.87	0.91	0.92	0.91	0.93	0.92	0.92	
$w = 1$	$\mu_2 = 0.5$	$\Psi_{x_0, n} - \Psi_{x_0}$	0.02	0.03	0.03	0.02	0.02	0.02	0.01	0.02	0.02	
		$\Psi_{x_0} \in C_{0.95}$	0.94	0.91	0.85	0.91	0.92	0.90	0.91	0.92	0.87	
		$\theta_0 \in E_{0.95}$	0.89	0.84	0.72	0.89	0.88	0.80	0.89	0.90	0.78	
	$\mu_2 = \tau$	$\Psi_{x_0, n} - \Psi_{x_0}$	0.01	0.01	0.01	0.01	0.01	0.01	0.00	0.00	0.00	
		$\Psi_{x_0} \in C_{0.95}$	0.91	0.94	0.93	0.95	0.94	0.95	0.92	0.94	0.95	
		$\theta_0 \in E_{0.95}$	0.93	0.92	0.90	0.93	0.93	0.91	0.93	0.94	0.92	
	$n = 2000$ $\mathbb{E}n_{z_0, x_l} = 330, \mathbb{E}n_{z_1, x_l} = 660$			$C_{\frac{2}{3}}$	$C_{\frac{1}{2}}$	$C_{\frac{1}{3}}$	$C_{\frac{2}{3}}$	$C_{\frac{1}{2}}$	$C_{\frac{1}{3}}$	$C_{\frac{2}{3}}$	$C_{\frac{1}{2}}$	$C_{\frac{1}{3}}$
	$w = 0$	$\mu_2 = 0$	$\Psi_{x_0, n} - \Psi_{x_0}$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
			$\Psi_{x_0} \in C_{0.95}$	0.95	0.95	0.95	0.96	0.95	0.93	0.95	0.97	0.95
$\theta_0 \in E_{0.95}$			0.91	0.95	0.95	0.94	0.94	0.94	0.92	0.96	0.95	
$\mu_2 = 0.5$		$\Psi_{x_0, n} - \Psi_{x_0}$	0.02	0.03	0.03	0.01	0.02	0.02	0.01	0.01	0.02	
		$\Psi_{x_0} \in C_{0.95}$	0.88	0.85	0.75	0.90	0.90	0.84	0.93	0.91	0.89	
		$\theta_0 \in E_{0.95}$	0.86	0.79	0.59	0.89	0.88	0.75	0.92	0.89	0.82	
$\mu_2 = \tau$		$\Psi_{x_0, n} - \Psi_{x_0}$	0.01	0.01	0.01	0.00	0.00	0.01	0.00	0.00	0.00	
		$\Psi_{x_0} \in C_{0.95}$	0.94	0.96	0.94	0.95	0.95	0.95	0.95	0.95	0.95	
		$\theta_0 \in E_{0.95}$	0.92	0.92	0.92	0.93	0.91	0.95	0.93	0.92	0.94	
$w = 1$	$\mu_2 = 0.5$	$\Psi_{x_0, n} - \Psi_{x_0}$	0.02	0.02	0.02	0.01	0.01	0.02	0.01	0.01	0.01	
		$\Psi_{x_0} \in C_{0.95}$	0.92	0.92	0.89	0.93	0.92	0.90	0.94	0.93	0.93	
		$\theta_0 \in E_{0.95}$	0.88	0.87	0.78	0.91	0.90	0.85	0.92	0.90	0.86	
	$\mu_2 = \tau$	$\Psi_{x_0, n} - \Psi_{x_0}$	0.00	0.01	0.01	0.00	0.00	0.00	0.00	0.00	0.00	
		$\Psi_{x_0} \in C_{0.95}$	0.97	0.95	0.96	0.95	0.95	0.94	0.96	0.97	0.95	
		$\theta_0 \in E_{0.95}$	0.94	0.93	0.96	0.94	0.95	0.96	0.92	0.95	0.95	
	$n = 4000$ $\mathbb{E}n_{z_0, x_l} = 660, \mathbb{E}n_{z_1, x_l} = 1320$			$C_{\frac{2}{3}}$	$C_{\frac{1}{2}}$	$C_{\frac{1}{3}}$	$C_{\frac{2}{3}}$	$C_{\frac{1}{2}}$	$C_{\frac{1}{3}}$	$C_{\frac{2}{3}}$	$C_{\frac{1}{2}}$	$C_{\frac{1}{3}}$
	$w = 0$	$\mu_2 = 0$	$\Psi_{x_0, n} - \Psi_{x_0}$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
			$\Psi_{x_0} \in C_{0.95}$	0.96	0.92	0.94	0.96	0.95	0.96	0.95	0.97	0.95
$\theta_0 \in E_{0.95}$			0.93	0.92	0.94	0.96	0.93	0.95	0.94	0.95	0.94	
$\mu_2 = 0.5$		$\Psi_{x_0, n} - \Psi_{x_0}$	0.01	0.01	0.02	0.01	0.01	0.01	0.00	0.00	0.01	
		$\Psi_{x_0} \in C_{0.95}$	0.89	0.86	0.82	0.93	0.94	0.89	0.95	0.95	0.94	
		$\theta_0 \in E_{0.95}$	0.87	0.78	0.67	0.92	0.90	0.81	0.94	0.92	0.85	
$\mu_2 = \tau$		$\Psi_{x_0, n} - \Psi_{x_0}$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
		$\Psi_{x_0} \in C_{0.95}$	0.95	0.96	0.95	0.95	0.95	0.96	0.94	0.95	0.96	
		$\theta_0 \in E_{0.95}$	0.94	0.95	0.93	0.93	0.94	0.94	0.93	0.94	0.95	
$w = 1$	$\mu_2 = 0.5$	$\Psi_{x_0, n} - \Psi_{x_0}$	0.01	0.01	0.01	0.00	0.01	0.01	0.00	0.00	0.01	
		$\Psi_{x_0} \in C_{0.95}$	0.95	0.91	0.88	0.95	0.93	0.90	0.94	0.94	0.91	
		$\theta_0 \in E_{0.95}$	0.93	0.82	0.77	0.94	0.91	0.85	0.94	0.93	0.90	
	$\mu_2 = \tau$	$\Psi_{x_0, n} - \Psi_{x_0}$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
		$\Psi_{x_0} \in C_{0.95}$	0.96	0.95	0.95	0.96	0.95	0.95	0.96	0.95	0.94	
		$\theta_0 \in E_{0.95}$	0.93	0.92	0.94	0.94	0.94	0.95	0.94	0.96	0.94	

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Table 4: Bias, overlap coverage, and joint overlaps and CTPs coverage for sample splitting estimators. Epanechnikov kernel.

			$h_n = n^{-1/3}$			$h_n = n^{-1/4}$			$h_n = n^{-1/5}$			
$n = 1000$ $\mathbb{E}n_{z_0, x_l} = 165, \mathbb{E}n_{z_1, x_l} = 330$			$C_{\frac{2}{3}}$	$C_{\frac{1}{2}}$	$C_{\frac{1}{3}}$	$C_{\frac{2}{3}}$	$C_{\frac{1}{2}}$	$C_{\frac{1}{3}}$	$C_{\frac{2}{3}}$	$C_{\frac{1}{2}}$	$C_{\frac{1}{3}}$	
$w = 0$	$\mu_2 = 0$	$\Psi_{x_0, n} - \Psi_{x_0}$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
		$\Psi_{x_0} \in C_{0.95}$	0.97	0.97	0.94	0.95	0.96	0.94	0.96	0.96	0.95	
		$\theta_0 \in E_{0.95}$	0.95	0.96	0.94	0.92	0.93	0.93	0.94	0.94	0.93	
	$\mu_2 = 0.5$	$\Psi_{x_0, n} - \Psi_{x_0}$	0.04	0.04	0.05	0.03	0.03	0.04	0.03	0.03	0.03	
		$\Psi_{x_0} \in C_{0.95}$	0.88	0.82	0.73	0.89	0.86	0.78	0.91	0.87	0.82	
		$\theta_0 \in E_{0.95}$	0.82	0.73	0.56	0.88	0.80	0.65	0.88	0.82	0.71	
	$\mu_2 = \tau$	$\Psi_{x_0, n} - \Psi_{x_0}$	0.01	0.01	0.02	0.01	0.01	0.01	0.01	0.01	0.01	
		$\Psi_{x_0} \in C_{0.95}$	0.94	0.91	0.90	0.94	0.94	0.95	0.94	0.95	0.95	
		$\theta_0 \in E_{0.95}$	0.91	0.89	0.83	0.91	0.90	0.90	0.92	0.92	0.91	
$w = 1$	$\mu_2 = 0.5$	$\Psi_{x_0, n} - \Psi_{x_0}$	0.03	0.03	0.03	0.02	0.02	0.03	0.02	0.02	0.02	
		$\Psi_{x_0} \in C_{0.95}$	0.92	0.89	0.82	0.93	0.92	0.87	0.93	0.93	0.88	
		$\theta_0 \in E_{0.95}$	0.88	0.85	0.72	0.88	0.88	0.78	0.88	0.89	0.77	
	$\mu_2 = \tau$	$\Psi_{x_0, n} - \Psi_{x_0}$	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	
		$\Psi_{x_0} \in C_{0.95}$	0.92	0.94	0.94	0.92	0.95	0.93	0.93	0.94	0.96	
		$\theta_0 \in E_{0.95}$	0.92	0.95	0.90	0.92	0.95	0.92	0.92	0.95	0.90	
	$n = 2000$ $\mathbb{E}n_{z_0, x_l} = 330, \mathbb{E}n_{z_1, x_l} = 660$			$C_{\frac{2}{3}}$	$C_{\frac{1}{2}}$	$C_{\frac{1}{3}}$	$C_{\frac{2}{3}}$	$C_{\frac{1}{2}}$	$C_{\frac{1}{3}}$	$C_{\frac{2}{3}}$	$C_{\frac{1}{2}}$	$C_{\frac{1}{3}}$
	$w = 0$	$\mu_2 = 0$	$\Psi_{x_0, n} - \Psi_{x_0}$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
			$\Psi_{x_0} \in C_{0.95}$	0.94	0.97	0.96	0.95	0.97	0.93	0.96	0.95	0.95
$\theta_0 \in E_{0.95}$			0.94	0.96	0.96	0.95	0.96	0.95	0.95	0.96	0.94	
$\mu_2 = 0.5$		$\Psi_{x_0, n} - \Psi_{x_0}$	0.03	0.03	0.04	0.01	0.02	0.03	0.01	0.02	0.02	
		$\Psi_{x_0} \in C_{0.95}$	0.87	0.83	0.68	0.92	0.89	0.79	0.90	0.91	0.85	
		$\theta_0 \in E_{0.95}$	0.85	0.74	0.54	0.89	0.83	0.69	0.89	0.87	0.77	
$\mu_2 = \tau$		$\Psi_{x_0, n} - \Psi_{x_0}$	0.01	0.01	0.01	0.00	0.01	0.01	0.00	0.00	0.00	
		$\Psi_{x_0} \in C_{0.95}$	0.95	0.96	0.93	0.96	0.94	0.95	0.95	0.94	0.95	
		$\theta_0 \in E_{0.95}$	0.92	0.92	0.92	0.93	0.90	0.94	0.93	0.92	0.95	
$w = 1$	$\mu_2 = 0.5$	$\Psi_{x_0, n} - \Psi_{x_0}$	0.02	0.02	0.02	0.01	0.01	0.02	0.01	0.01	0.02	
		$\Psi_{x_0} \in C_{0.95}$	0.92	0.90	0.87	0.92	0.92	0.90	0.95	0.92	0.89	
		$\theta_0 \in E_{0.95}$	0.85	0.81	0.75	0.90	0.88	0.83	0.94	0.89	0.85	
	$\mu_2 = \tau$	$\Psi_{x_0, n} - \Psi_{x_0}$	0.00	0.01	0.01	0.00	0.00	0.00	0.00	0.00	0.00	
		$\Psi_{x_0} \in C_{0.95}$	0.96	0.95	0.94	0.95	0.96	0.94	0.94	0.95	0.95	
		$\theta_0 \in E_{0.95}$	0.94	0.93	0.91	0.94	0.94	0.95	0.95	0.94	0.96	
	$n = 4000$ $\mathbb{E}n_{z_0, x_l} = 660, \mathbb{E}n_{z_1, x_l} = 1320$			$C_{\frac{2}{3}}$	$C_{\frac{1}{2}}$	$C_{\frac{1}{3}}$	$C_{\frac{2}{3}}$	$C_{\frac{1}{2}}$	$C_{\frac{1}{3}}$	$C_{\frac{2}{3}}$	$C_{\frac{1}{2}}$	$C_{\frac{1}{3}}$
	$w = 0$	$\mu_2 = 0$	$\Psi_{x_0, n} - \Psi_{x_0}$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
			$\Psi_{x_0} \in C_{0.95}$	0.97	0.95	0.95	0.95	0.93	0.95	0.97	0.95	0.95
$\theta_0 \in E_{0.95}$			0.96	0.94	0.93	0.94	0.94	0.94	0.93	0.94	0.94	
$\mu_2 = 0.5$		$\Psi_{x_0, n} - \Psi_{x_0}$	0.01	0.02	0.02	0.01	0.01	0.01	0.00	0.01	0.01	
		$\Psi_{x_0} \in C_{0.95}$	0.89	0.81	0.75	0.92	0.90	0.86	0.94	0.93	0.92	
		$\theta_0 \in E_{0.95}$	0.86	0.72	0.53	0.91	0.86	0.76	0.93	0.90	0.83	
$\mu_2 = \tau$		$\Psi_{x_0, n} - \Psi_{x_0}$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
		$\Psi_{x_0} \in C_{0.95}$	0.95	0.94	0.94	0.95	0.95	0.94	0.95	0.95	0.95	
		$\theta_0 \in E_{0.95}$	0.93	0.94	0.92	0.93	0.95	0.94	0.95	0.94	0.94	
$w = 1$	$\mu_2 = 0.5$	$\Psi_{x_0, n} - \Psi_{x_0}$	0.01	0.01	0.02	0.00	0.01	0.01	0.00	0.00	0.01	
		$\Psi_{x_0} \in C_{0.95}$	0.91	0.89	0.82	0.94	0.91	0.90	0.94	0.94	0.91	
		$\theta_0 \in E_{0.95}$	0.90	0.80	0.71	0.94	0.87	0.84	0.93	0.90	0.87	
	$\mu_2 = \tau$	$\Psi_{x_0, n} - \Psi_{x_0}$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
		$\Psi_{x_0} \in C_{0.95}$	0.97	0.96	0.94	0.95	0.96	0.95	0.97	0.96	0.94	
		$\theta_0 \in E_{0.95}$	0.94	0.93	0.91	0.93	0.94	0.94	0.95	0.95	0.94	

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Table 5: Bias, overlap coverage, and joint overlaps and CTPs coverage for semi-deterministic full sample and K-fold cross-fitting estimators. Gaussian kernel.

			$d_n = 0$				$d_n = (nh_n^{1.5})^{-1/2}$				$d_n = (nh_n^3)^{-1/2}$				
$n = 1000$ $\mathbb{E}n_{20,x_I} = 165, \mathbb{E}n_{21,x_I} = 330$			SS	FS	2-F	8-F	SS	FS	2-F	8-F	SS	FS	2-F	8-F	
$w = 0$	$\mu_2 = 0$	$\Psi_{x_0,n} - \Psi_{x_0}$	0.00	-0.06	0.00	0.00	0.00	-0.06	0.00	0.00	0.00	-0.05	0.00	0.00	
		$\Psi_{x_0} \in C_{0.95}$	0.96	0.45	0.88	0.88	0.95	0.50	0.89	0.85	0.94	0.59	0.92	0.91	
		$\theta_0 \in E_{0.95}$	0.94	0.21	0.85	0.79	0.93	0.22	0.86	0.77	0.59	0.30	0.82	0.76	
	$\mu_2 = 0.5$	$\Psi_{x_0,n} - \Psi_{x_0}$	0.01	-0.01	0.02	0.01	0.01	-0.01	0.01	0.01	0.06	-0.01	0.06	0.05	
		$\Psi_{x_0} \in C_{0.95}$	0.95	0.94	0.86	0.92	0.96	0.95	0.88	0.91	0.81	0.96	0.52	0.54	
		$\theta_0 \in E_{0.95}$	0.93	0.92	0.73	0.84	0.93	0.93	0.74	0.83	0.42	0.91	0.16	0.22	
	$\mu_2 = \tau$	$\Psi_{x_0,n} - \Psi_{x_0}$	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.00	0.06	0.01	0.06	0.05	
		$\Psi_{x_0} \in C_{0.95}$	0.94	0.95	0.94	0.95	0.94	0.96	0.91	0.92	0.71	0.96	0.26	0.30	
		$\theta_0 \in E_{0.95}$	0.92	0.92	0.91	0.94	0.92	0.92	0.91	0.92	0.56	0.92	0.10	0.10	
$w = 1$	$\mu_2 = 0.5$	$\Psi_{x_0,n} - \Psi_{x_0}$	0.01	-0.02	0.01	0.01	0.01	-0.02	0.01	0.01	0.03	-0.02	0.03	0.03	
		$\Psi_{x_0} \in C_{0.95}$	0.94	0.88	0.87	0.89	0.94	0.89	0.85	0.89	0.94	0.90	0.80	0.83	
		$\theta_0 \in E_{0.95}$	0.90	0.89	0.74	0.78	0.93	0.88	0.73	0.76	0.74	0.88	0.43	0.55	
	$\mu_2 = \tau$	$\Psi_{x_0,n} - \Psi_{x_0}$	0.00	-0.01	0.00	0.00	0.00	-0.01	0.00	0.00	0.02	0.00	0.02	0.02	
		$\Psi_{x_0} \in C_{0.95}$	0.94	0.94	0.96	0.93	0.95	0.96	0.92	0.92	0.93	0.97	0.85	0.87	
		$\theta_0 \in E_{0.95}$	0.94	0.92	0.92	0.91	0.94	0.93	0.92	0.93	0.91	0.93	0.78	0.77	
	$n = 2000$ $\mathbb{E}n_{20,x_I} = 330, \mathbb{E}n_{21,x_I} = 660$			SS	FS	2-F	8-F	SS	FS	2-F	8-F	SS	FS	2-F	8-F
	$w = 0$	$\mu_2 = 0$	$\Psi_{x_0,n} - \Psi_{x_0}$	0.00	-0.04	0.00	0.00	0.00	-0.04	0.00	0.00	0.00	-0.04	0.00	0.00
			$\Psi_{x_0} \in C_{0.95}$	0.95	0.41	0.90	0.89	0.96	0.44	0.92	0.88	0.94	0.51	0.96	0.91
$\theta_0 \in E_{0.95}$			0.94	0.10	0.88	0.80	0.95	0.11	0.89	0.78	0.63	0.18	0.88	0.72	
$\mu_2 = 0.5$		$\Psi_{x_0,n} - \Psi_{x_0}$	0.01	0.00	0.01	0.00	0.01	0.00	0.01	0.01	0.06	0.00	0.06	0.05	
		$\Psi_{x_0} \in C_{0.95}$	0.94	0.94	0.90	0.91	0.96	0.94	0.92	0.92	0.61	0.95	0.23	0.32	
		$\theta_0 \in E_{0.95}$	0.92	0.92	0.83	0.89	0.96	0.92	0.89	0.90	0.28	0.93	0.03	0.05	
$\mu_2 = \tau$		$\Psi_{x_0,n} - \Psi_{x_0}$	0.00	0.00	0.00	0.00	0.01	0.00	0.01	0.00	0.06	0.01	0.06	0.05	
		$\Psi_{x_0} \in C_{0.95}$	0.95	0.91	0.94	0.94	0.96	0.93	0.94	0.96	0.45	0.92	0.04	0.05	
		$\theta_0 \in E_{0.95}$	0.93	0.91	0.96	0.94	0.97	0.91	0.95	0.95	0.29	0.88	0.00	0.00	
$w = 1$	$\mu_2 = 0.5$	$\Psi_{x_0,n} - \Psi_{x_0}$	0.00	-0.02	0.01	0.00	0.00	-0.01	0.00	0.00	0.02	-0.01	0.02	0.02	
		$\Psi_{x_0} \in C_{0.95}$	0.94	0.91	0.90	0.91	0.95	0.91	0.94	0.92	0.89	0.91	0.76	0.83	
		$\theta_0 \in E_{0.95}$	0.93	0.87	0.82	0.84	0.96	0.87	0.86	0.88	0.71	0.91	0.40	0.52	
	$\mu_2 = \tau$	$\Psi_{x_0,n} - \Psi_{x_0}$	0.00	-0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.02	0.00	0.02	0.02	
		$\Psi_{x_0} \in C_{0.95}$	0.94	0.94	0.94	0.92	0.96	0.95	0.94	0.94	0.90	0.95	0.75	0.73	
		$\theta_0 \in E_{0.95}$	0.93	0.92	0.95	0.94	0.98	0.93	0.93	0.93	0.84	0.94	0.58	0.59	
	$n = 4000$ $\mathbb{E}n_{20,x_I} = 660, \mathbb{E}n_{21,x_I} = 1320$			SS	FS	2-F	8-F	SS	FS	2-F	8-F	SS	FS	2-F	8-F
	$w = 0$	$\mu_2 = 0$	$\Psi_{x_0,n} - \Psi_{x_0}$	0.00	-0.03	0.00	0.00	0.00	-0.03	0.00	0.00	0.00	-0.03	0.00	0.00
			$\Psi_{x_0} \in C_{0.95}$	0.95	0.32	0.91	0.86	0.94	0.36	0.92	0.86	0.96	0.47	0.95	0.92
$\theta_0 \in E_{0.95}$			0.95	0.04	0.87	0.80	0.96	0.07	0.89	0.82	0.64	0.14	0.86	0.79	
$\mu_2 = 0.5$		$\Psi_{x_0,n} - \Psi_{x_0}$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.05	0.00	0.05	0.04	
		$\Psi_{x_0} \in C_{0.95}$	0.94	0.94	0.94	0.95	0.95	0.95	0.93	0.94	0.52	0.95	0.14	0.19	
		$\theta_0 \in E_{0.95}$	0.94	0.97	0.91	0.93	0.97	0.97	0.92	0.94	0.18	0.97	0.02	0.01	
$\mu_2 = \tau$		$\Psi_{x_0,n} - \Psi_{x_0}$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.05	0.01	0.05	0.05	
		$\Psi_{x_0} \in C_{0.95}$	0.94	0.93	0.95	0.97	0.93	0.94	0.94	0.93	0.21	0.89	0.01	0.01	
		$\theta_0 \in E_{0.95}$	0.94	0.94	0.93	0.93	0.97	0.94	0.95	0.95	0.05	0.91	0.00	0.00	
$w = 1$	$\mu_2 = 0.5$	$\Psi_{x_0,n} - \Psi_{x_0}$	0.00	-0.01	0.00	0.00	0.00	-0.01	0.00	0.00	0.02	-0.01	0.02	0.02	
		$\Psi_{x_0} \in C_{0.95}$	0.94	0.91	0.92	0.92	0.95	0.92	0.89	0.90	0.88	0.93	0.70	0.77	
		$\theta_0 \in E_{0.95}$	0.92	0.93	0.88	0.87	0.94	0.92	0.87	0.90	0.70	0.95	0.35	0.44	
	$\mu_2 = \tau$	$\Psi_{x_0,n} - \Psi_{x_0}$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.02	0.00	0.02	0.02	
		$\Psi_{x_0} \in C_{0.95}$	0.96	0.94	0.93	0.93	0.93	0.95	0.94	0.94	0.82	0.96	0.52	0.50	
		$\theta_0 \in E_{0.95}$	0.95	0.95	0.93	0.94	0.96	0.95	0.94	0.94	0.67	0.95	0.25	0.22	

Table 6: Bias, overlap coverage, and joint overlaps and CTPs coverage for semi-deterministic full sample and K-fold cross-fitting estimators. Uniform kernel.

			$d_n = 0$				$d_n = (nh_n^{1.5})^{-1/2}$				$d_n = (nh_n^3)^{-1/2}$			
$n = 1000$			SS	FS	2-F	8-F	SS	FS	2-F	8-F	SS	FS	2-F	8-F
$\mathbb{E}n_{x_0, x_I} = 165, \mathbb{E}n_{x_1, x_I} = 330$														
$w = 0$	$\mu_2 = 0$	$\Psi_{x_0, n} - \Psi_{x_0}$	0.00	-0.05	0.00	0.00	0.00	-0.05	0.00	0.00	0.00	-0.04	0.00	0.00
		$\Psi_{x_0} \in C_{0.95}$	0.96	0.53	0.91	0.89	0.93	0.56	0.90	0.89	0.93	0.69	0.94	0.94
		$\theta_0 \in E_{0.95}$	0.95	0.34	0.85	0.83	0.95	0.35	0.87	0.80	0.75	0.53	0.87	0.81
	$\mu_2 = 0.5$	$\Psi_{x_0, n} - \Psi_{x_0}$	0.02	-0.01	0.02	0.01	0.02	-0.01	0.03	0.02	0.07	0.00	0.08	0.07
		$\Psi_{x_0} \in C_{0.95}$	0.94	0.93	0.81	0.87	0.93	0.95	0.81	0.87	0.70	0.97	0.21	0.32
		$\theta_0 \in E_{0.95}$	0.89	0.91	0.63	0.77	0.90	0.91	0.64	0.75	0.13	0.90	0.00	0.03
	$\mu_2 = \tau$	$\Psi_{x_0, n} - \Psi_{x_0}$	0.01	0.00	0.01	0.01	0.01	0.00	0.01	0.01	0.08	0.01	0.08	0.07
		$\Psi_{x_0} \in C_{0.95}$	0.94	0.95	0.93	0.94	0.93	0.98	0.89	0.92	0.52	0.95	0.10	0.13
		$\theta_0 \in E_{0.95}$	0.93	0.92	0.91	0.93	0.94	0.92	0.89	0.91	0.28	0.89	0.01	0.02
$w = 1$	$\mu_2 = 0.5$	$\Psi_{x_0, n} - \Psi_{x_0}$	0.01	-0.02	0.02	0.01	0.01	-0.02	0.02	0.01	0.04	-0.02	0.04	0.04
		$\Psi_{x_0} \in C_{0.95}$	0.91	0.85	0.84	0.83	0.94	0.88	0.83	0.87	0.87	0.91	0.69	0.75
		$\theta_0 \in E_{0.95}$	0.89	0.84	0.69	0.75	0.93	0.87	0.68	0.76	0.57	0.87	0.25	0.34
	$\mu_2 = \tau$	$\Psi_{x_0, n} - \Psi_{x_0}$	0.00	-0.01	0.01	0.00	0.00	-0.01	0.01	0.01	0.04	0.00	0.04	0.03
		$\Psi_{x_0} \in C_{0.95}$	0.95	0.94	0.91	0.92	0.96	0.95	0.90	0.91	0.89	0.96	0.65	0.73
		$\theta_0 \in E_{0.95}$	0.93	0.90	0.89	0.89	0.95	0.92	0.90	0.92	0.78	0.94	0.43	0.52
$n = 2000$			SS	FS	2-F	8-F	SS	FS	2-F	8-F	SS	FS	2-F	8-F
$\mathbb{E}n_{x_0, x_I} = 330, \mathbb{E}n_{x_1, x_I} = 660$														
$w = 0$	$\mu_2 = 0$	$\Psi_{x_0, n} - \Psi_{x_0}$	0.00	-0.04	0.00	0.00	0.00	-0.04	0.00	0.00	0.00	-0.03	0.00	0.00
		$\Psi_{x_0} \in C_{0.95}$	0.95	0.47	0.91	0.89	0.96	0.52	0.91	0.88	0.95	0.63	0.95	0.93
		$\theta_0 \in E_{0.95}$	0.92	0.23	0.90	0.84	0.96	0.23	0.90	0.84	0.82	0.36	0.91	0.82
	$\mu_2 = 0.5$	$\Psi_{x_0, n} - \Psi_{x_0}$	0.01	0.00	0.01	0.01	0.01	0.00	0.02	0.01	0.07	0.00	0.07	0.07
		$\Psi_{x_0} \in C_{0.95}$	0.93	0.94	0.87	0.89	0.96	0.94	0.84	0.89	0.44	0.94	0.04	0.08
		$\theta_0 \in E_{0.95}$	0.92	0.90	0.78	0.82	0.95	0.89	0.77	0.84	0.03	0.89	0.00	0.00
	$\mu_2 = \tau$	$\Psi_{x_0, n} - \Psi_{x_0}$	0.00	0.00	0.00	0.00	0.01	0.00	0.01	0.01	0.07	0.01	0.07	0.07
		$\Psi_{x_0} \in C_{0.95}$	0.95	0.92	0.93	0.94	0.95	0.92	0.92	0.94	0.28	0.89	0.01	0.02
		$\theta_0 \in E_{0.95}$	0.93	0.91	0.94	0.93	0.96	0.90	0.92	0.93	0.10	0.86	0.00	0.00
$w = 1$	$\mu_2 = 0.5$	$\Psi_{x_0, n} - \Psi_{x_0}$	0.01	-0.01	0.01	0.01	0.01	-0.01	0.01	0.01	0.04	-0.01	0.04	0.03
		$\Psi_{x_0} \in C_{0.95}$	0.93	0.88	0.84	0.89	0.95	0.90	0.89	0.91	0.81	0.92	0.57	0.65
		$\theta_0 \in E_{0.95}$	0.92	0.86	0.78	0.85	0.95	0.87	0.78	0.83	0.47	0.89	0.13	0.22
	$\mu_2 = \tau$	$\Psi_{x_0, n} - \Psi_{x_0}$	0.00	-0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.04	0.00	0.04	0.03
		$\Psi_{x_0} \in C_{0.95}$	0.95	0.94	0.90	0.93	0.96	0.96	0.94	0.93	0.79	0.96	0.52	0.56
		$\theta_0 \in E_{0.95}$	0.92	0.90	0.93	0.93	0.98	0.92	0.92	0.90	0.63	0.93	0.27	0.33
$n = 4000$			SS	FS	2-F	8-F	SS	FS	2-F	8-F	SS	FS	2-F	8-F
$\mathbb{E}n_{x_0, x_I} = 660, \mathbb{E}n_{x_1, x_I} = 1320$														
$w = 0$	$\mu_2 = 0$	$\Psi_{x_0, n} - \Psi_{x_0}$	0.00	-0.03	0.00	0.00	0.00	-0.03	0.00	0.00	0.00	-0.03	0.00	0.00
		$\Psi_{x_0} \in C_{0.95}$	0.95	0.43	0.92	0.88	0.94	0.47	0.91	0.90	0.97	0.58	0.95	0.93
		$\theta_0 \in E_{0.95}$	0.94	0.16	0.89	0.85	0.97	0.20	0.87	0.85	0.85	0.32	0.93	0.87
	$\mu_2 = 0.5$	$\Psi_{x_0, n} - \Psi_{x_0}$	0.00	0.00	0.01	0.00	0.01	0.00	0.01	0.01	0.06	0.01	0.07	0.06
		$\Psi_{x_0} \in C_{0.95}$	0.95	0.93	0.90	0.93	0.95	0.93	0.89	0.92	0.28	0.95	0.00	0.02
		$\theta_0 \in E_{0.95}$	0.94	0.96	0.84	0.88	0.94	0.96	0.83	0.89	0.01	0.96	0.00	0.00
	$\mu_2 = \tau$	$\Psi_{x_0, n} - \Psi_{x_0}$	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.00	0.06	0.01	0.07	0.06
		$\Psi_{x_0} \in C_{0.95}$	0.94	0.94	0.95	0.97	0.94	0.93	0.94	0.92	0.13	0.85	0.00	0.00
		$\theta_0 \in E_{0.95}$	0.93	0.94	0.92	0.93	0.96	0.94	0.92	0.94	0.01	0.87	0.00	0.00
$w = 1$	$\mu_2 = 0.5$	$\Psi_{x_0, n} - \Psi_{x_0}$	0.00	-0.01	0.00	0.00	0.01	-0.01	0.01	0.00	0.03	-0.01	0.03	0.03
		$\Psi_{x_0} \in C_{0.95}$	0.95	0.90	0.89	0.93	0.93	0.92	0.89	0.89	0.72	0.93	0.43	0.52
		$\theta_0 \in E_{0.95}$	0.94	0.90	0.84	0.86	0.96	0.91	0.81	0.88	0.38	0.94	0.06	0.13
	$\mu_2 = \tau$	$\Psi_{x_0, n} - \Psi_{x_0}$	0.00	-0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.03	0.00	0.03	0.03
		$\Psi_{x_0} \in C_{0.95}$	0.96	0.91	0.92	0.93	0.94	0.94	0.91	0.93	0.70	0.94	0.25	0.30
		$\theta_0 \in E_{0.95}$	0.94	0.92	0.93	0.91	0.95	0.95	0.91	0.91	0.45	0.95	0.06	0.06

Table 7: Bias, overlap coverage, and joint overlaps and CTPs coverage for semi-deterministic full sample and K-fold cross-fitting estimators. Epanechnikov kernel.

			$d_n = 0$				$d_n = (nh_n^{1.5})^{-1/2}$				$d_n = (nh_n^3)^{-1/2}$			
$n = 1000$			SS	FS	2-F	8-F	SS	FS	2-F	8-F	SS	FS	2-F	8-F
$\mathbb{E}n_{x_0, x_I} = 165, \mathbb{E}n_{x_1, x_I} = 330$														
$w = 0$	$\mu_2 = 0$	$\Psi_{x_0, n} - \Psi_{x_0}$	0.00	-0.07	0.00	0.00	0.00	-0.07	0.00	0.00	0.00	-0.06	0.00	0.00
		$\Psi_{x_0} \in C_{0.95}$	0.96	0.19	0.89	0.87	0.96	0.25	0.88	0.87	0.93	0.39	0.93	0.90
		$\theta_0 \in E_{0.95}$	0.94	0.02	0.87	0.82	0.95	0.03	0.86	0.77	0.77	0.09	0.86	0.78
	$\mu_2 = 0.5$	$\Psi_{x_0, n} - \Psi_{x_0}$	0.03	-0.02	0.03	0.02	0.02	-0.02	0.03	0.02	0.07	-0.01	0.08	0.07
		$\Psi_{x_0} \in C_{0.95}$	0.91	0.93	0.76	0.84	0.92	0.93	0.78	0.85	0.71	0.95	0.25	0.29
		$\theta_0 \in E_{0.95}$	0.88	0.88	0.59	0.73	0.87	0.90	0.61	0.72	0.17	0.93	0.01	0.03
	$\mu_2 = \tau$	$\Psi_{x_0, n} - \Psi_{x_0}$	0.01	0.00	0.01	0.01	0.01	0.00	0.01	0.01	0.08	0.01	0.09	0.08
		$\Psi_{x_0} \in C_{0.95}$	0.94	0.94	0.91	0.94	0.94	0.96	0.88	0.92	0.46	0.95	0.07	0.14
		$\theta_0 \in E_{0.95}$	0.92	0.92	0.89	0.94	0.93	0.92	0.88	0.91	0.25	0.90	0.01	0.01
$w = 1$	$\mu_2 = 0.5$	$\Psi_{x_0, n} - \Psi_{x_0}$	0.02	-0.03	0.02	0.01	0.01	-0.03	0.02	0.01	0.04	-0.03	0.04	0.04
		$\Psi_{x_0} \in C_{0.95}$	0.93	0.76	0.80	0.83	0.96	0.82	0.83	0.85	0.89	0.87	0.68	0.74
		$\theta_0 \in E_{0.95}$	0.88	0.71	0.69	0.74	0.90	0.76	0.67	0.73	0.59	0.84	0.26	0.32
	$\mu_2 = \tau$	$\Psi_{x_0, n} - \Psi_{x_0}$	0.01	-0.02	0.01	0.00	0.01	-0.01	0.01	0.01	0.04	-0.01	0.04	0.04
		$\Psi_{x_0} \in C_{0.95}$	0.92	0.89	0.92	0.91	0.94	0.93	0.89	0.91	0.85	0.95	0.61	0.68
		$\theta_0 \in E_{0.95}$	0.92	0.88	0.90	0.89	0.94	0.90	0.87	0.89	0.75	0.93	0.39	0.44
$n = 2000$			SS	FS	2-F	8-F	SS	FS	2-F	8-F	SS	FS	2-F	8-F
$\mathbb{E}n_{x_0, x_I} = 330, \mathbb{E}n_{x_1, x_I} = 660$														
$w = 0$	$\mu_2 = 0$	$\Psi_{x_0, n} - \Psi_{x_0}$	0.00	-0.05	0.00	0.00	0.00	-0.05	0.00	0.00	0.00	-0.05	0.00	0.00
		$\Psi_{x_0} \in C_{0.95}$	0.96	0.17	0.91	0.90	0.97	0.21	0.92	0.88	0.95	0.31	0.95	0.92
		$\theta_0 \in E_{0.95}$	0.95	0.00	0.89	0.80	0.98	0.00	0.90	0.83	0.81	0.03	0.90	0.76
	$\mu_2 = 0.5$	$\Psi_{x_0, n} - \Psi_{x_0}$	0.01	-0.01	0.01	0.01	0.02	-0.01	0.02	0.01	0.07	0.00	0.07	0.07
		$\Psi_{x_0} \in C_{0.95}$	0.90	0.91	0.86	0.90	0.94	0.92	0.80	0.88	0.46	0.95	0.06	0.10
		$\theta_0 \in E_{0.95}$	0.89	0.90	0.71	0.80	0.96	0.92	0.72	0.79	0.04	0.92	0.00	0.00
	$\mu_2 = \tau$	$\Psi_{x_0, n} - \Psi_{x_0}$	0.00	0.00	0.00	0.00	0.01	0.00	0.01	0.01	0.08	0.01	0.08	0.07
		$\Psi_{x_0} \in C_{0.95}$	0.95	0.91	0.93	0.93	0.95	0.94	0.90	0.92	0.26	0.90	0.01	0.02
		$\theta_0 \in E_{0.95}$	0.93	0.91	0.94	0.93	0.95	0.91	0.89	0.93	0.11	0.88	0.00	0.00
$w = 1$	$\mu_2 = 0.5$	$\Psi_{x_0, n} - \Psi_{x_0}$	0.01	-0.02	0.01	0.01	0.01	-0.02	0.01	0.01	0.04	-0.02	0.04	0.03
		$\Psi_{x_0} \in C_{0.95}$	0.92	0.79	0.84	0.89	0.95	0.83	0.86	0.91	0.80	0.86	0.60	0.64
		$\theta_0 \in E_{0.95}$	0.94	0.73	0.78	0.83	0.96	0.76	0.77	0.81	0.51	0.82	0.16	0.21
	$\mu_2 = \tau$	$\Psi_{x_0, n} - \Psi_{x_0}$	0.00	-0.01	0.00	0.00	0.00	-0.01	0.00	0.00	0.04	0.00	0.04	0.03
		$\Psi_{x_0} \in C_{0.95}$	0.95	0.87	0.91	0.90	0.97	0.93	0.92	0.94	0.77	0.96	0.50	0.56
		$\theta_0 \in E_{0.95}$	0.95	0.86	0.91	0.89	0.98	0.89	0.89	0.87	0.64	0.93	0.24	0.30
$n = 4000$			SS	FS	2-F	8-F	SS	FS	2-F	8-F	SS	FS	2-F	8-F
$\mathbb{E}n_{x_0, x_I} = 660, \mathbb{E}n_{x_1, x_I} = 1320$														
$w = 0$	$\mu_2 = 0$	$\Psi_{x_0, n} - \Psi_{x_0}$	0.00	-0.04	0.00	0.00	0.00	-0.04	0.00	0.00	0.00	-0.04	0.00	0.00
		$\Psi_{x_0} \in C_{0.95}$	0.97	0.10	0.90	0.86	0.95	0.13	0.91	0.88	0.97	0.21	0.94	0.90
		$\theta_0 \in E_{0.95}$	0.93	0.00	0.87	0.83	0.97	0.00	0.89	0.84	0.83	0.01	0.93	0.80
	$\mu_2 = 0.5$	$\Psi_{x_0, n} - \Psi_{x_0}$	0.00	0.00	0.01	0.00	0.01	0.00	0.01	0.01	0.06	0.00	0.07	0.06
		$\Psi_{x_0} \in C_{0.95}$	0.94	0.93	0.87	0.91	0.93	0.94	0.86	0.89	0.30	0.97	0.02	0.01
		$\theta_0 \in E_{0.95}$	0.93	0.96	0.80	0.87	0.95	0.97	0.79	0.88	0.01	0.98	0.00	0.00
	$\mu_2 = \tau$	$\Psi_{x_0, n} - \Psi_{x_0}$	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.00	0.06	0.01	0.07	0.06
		$\Psi_{x_0} \in C_{0.95}$	0.95	0.93	0.94	0.97	0.93	0.94	0.92	0.93	0.13	0.86	0.00	0.00
		$\theta_0 \in E_{0.95}$	0.95	0.95	0.92	0.93	0.95	0.94	0.90	0.93	0.01	0.87	0.00	0.00
$w = 1$	$\mu_2 = 0.5$	$\Psi_{x_0, n} - \Psi_{x_0}$	0.00	-0.02	0.01	0.00	0.01	-0.01	0.01	0.00	0.03	-0.01	0.03	0.03
		$\Psi_{x_0} \in C_{0.95}$	0.94	0.82	0.87	0.91	0.95	0.87	0.87	0.87	0.74	0.92	0.42	0.50
		$\theta_0 \in E_{0.95}$	0.93	0.81	0.83	0.83	0.94	0.84	0.81	0.89	0.37	0.91	0.07	0.11
	$\mu_2 = \tau$	$\Psi_{x_0, n} - \Psi_{x_0}$	0.00	-0.01	0.00	0.00	0.00	-0.01	0.00	0.00	0.03	0.00	0.03	0.03
		$\Psi_{x_0} \in C_{0.95}$	0.95	0.90	0.92	0.92	0.93	0.94	0.91	0.92	0.70	0.94	0.25	0.31
		$\theta_0 \in E_{0.95}$	0.95	0.88	0.94	0.93	0.95	0.95	0.92	0.90	0.43	0.96	0.04	0.07